UNDERSTANDING HOW AUSTRALIAN RESEARCHERS EXPERIENCE OPEN ACCESS AS PART OF THEIR INFORMATION LITERACY

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Abstract

Open access has become an essential component of spreading and sharing scholarly knowledge. It has become a significant alternative to traditional publishing options. This thesis reports on a qualitative study that explores how Australian researchers experience open access as part of their information literacy. The research was informed by the relational perspective of information literacy which emphasises the relationship between people and information, and the way people experience the world.

The qualitative data was collected from 31 researchers employed at Australian universities. Semi-structured interviews were employed and analysed using thematic analysis. The study revealed four themes that represent the researchers' experiences of open access as part of their information literacy. The themes revealed are: Understanding Open Access, Institutional Repositories, Publishing and Dissemination, and Finding Information. The themes signify that open access is shaping how Australian researchers experience information literacy, influencing how Australian researchers are, and can be, informed and informing researchers. For example, as informed and informing researchers, participants have their own individual understandings of what open access is, its benefits to them, and its benefits to the academy and society at large.

The study's findings create new knowledge for researchers to use and build upon in the field of open access and information literacy. The research contributes to existing knowledge by providing an empirical understanding of researchers' use of open access as part of their broader information world. The research also contributes to Australian universities by providing an evidence base to inform the development of information literacy and/or open access programs, resources and strategies offered by university libraries, thus helping to ensure the programs genuinely meet the lived needs of researchers.

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Statement of Original Authorship

The work contained in this thesis has not been previously submitted to meet requirements for an award at this or any other higher education institution. To the best of my knowledge and belief, the thesis contains no material previously published or written by another person except where due reference is made.

QUT Verified Signature

Signature: _____

Date: 05/04/2018

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Chapter 1: Introduction

1.1 Introduction

This chapter introduces the study. It begins by providing the background to the study in order to orientate the work within the broader context from which it originated and was inspired. This is followed by an outline of the study's aim and the research design employed. An overview of the study's key findings, including significance to both theory and practice, is provided. The chapter concludes by outlining the overall structure of the document.

1.2 Context

Access to high quality and relevant scholarly publications is important for the research process. Researchers have tended to predominantly rely on library collections and database subscriptions when securing access to journals. Recent advances in technology, however, are changing scholarly communication. Of particular note is the phenomenon of open access, which refers to publications that can be viewed and downloaded online, without copyright restriction and financial barrier (Willinsky, 2006). Open access has rapidly emerged as an alternative viable source to library collections and databases, for academics to find and share scholarly publications (Bernius, 2010).

The lower restrictions of open access compared with traditional publication methods of access may result in wide dissemination of a publication. Many studies have suggested that research published by open access avenues are more likely to be used or cited by other research (Willinsky, 2006). Research outputs published through open access are disseminated in two main ways: green and gold (Harnard et al., 2004). Lewis (2012) stated that green open access is free online access to articles through the Web other than via a publisher's website. It is a model that allows authors to publish their papers on personal websites or in institutional repositories (Lynch, 2003; Harnard et al., 2004; Westell, 2006). Articles published through gold open access are available only from a publisher's website. Gold open

access consists of journals making their peer-reviewed articles available for online reading without a fee or need to register (Crawford, 2011).

Despite growing interest in and the potential benefits of open access, there are a number of concerns for researchers. For example, unexpected article processing charges, researchers receiving unsolicited and potentially predatory emails from open access journals, and inconsistent and unreliable peer review (Osborne, 2015; Shen & Björk, 2015; Chang, 2017). Hence, it can be important for researchers to assess the quality of open access research articles they use for research, and to be cautious when selecting where to publish their research outputs (Zhao, 2014).

In this regard, other stakeholders may contribute to ensuring that research funded by public funds benefit the wider community. The establishment of an institutional repository (IR) as a digital outlet that stores and shares research outputs can be seen as part of a university's effort to extend the impact for their scholarly outputs (Kennan, 2011). Further, the provision of institutional repositories may help researchers to comply with funding bodies that require their research outputs are publicly available.

The discussion of open access may continue to evolve, but there is some evidence that shows research outputs that are disseminated through open access may not only provide academic impact (e.g., increased citation), but also an economic and societal impact (Tennant, et al., 2016).

Open access has an economic impact by challenging the production cost of digital and print publications that become the reason for publishers to charge a subscription fee (Tennant, et al., 2016). Readers do not need to pay a subscription to access open access journals. This arguably benefits institutions by reducing their subscription expenses. It may also benefit the wider community, for instance small businesses and other industries. The accessibility of recent research findings may be useful in developing innovations (Mazzucato, 2011), enhancing their return on financial investment (Beagrie & Houghton, 2014).

Open access has a social impact in that it may also support researchers from developing countries, who may have difficulties accessing scholarly articles, by providing quality journal articles that would be useful to conduct research (Tang, Bever, & Yu, 2017). This global dissemination of research outputs and the communication shared between researchers through open

access transcends traditional academic collaboration and supports lifelong learning (Tennant, et al., 2016).

Scholarly communication is impossible without information literacy (Hebrang Grgig, 2016). Researchers, regardless of discipline, need to know how to find, evaluate, produce and present information in appropriate and ethical ways. In addition, researchers need to know how to make their scholarly information more visible and accessible, how to improve its quality and impact. As open access is rapidly becoming a preferred form of scholarly communication for both scholars and funding agencies, it is becoming increasingly important for researchers to develop their open access information literacy skills to ensure that scholarly information in open access is valid, reliable, properly published and used. Very little however is known about researchers' experiences of open access as part of their information literacy. There are limited empirical studies investigating the interplay between open access and information literacy. It is within this broad context that the current study is situated.

1.3 Aim of the study

A growing body of research has begun to explore the open access phenomenon. Existing research has focussed on two main areas of enquiry: impact and user perspective. From the perspective of impact, research has investigated how frequently articles published in open access journals are cited compared to articles published in closed-access journals (Gaule & Maystre, 2011; Koler-Povh, Juznic, & Turk, 2013). From the user perspective, research has begun to explore research students' awareness, attitude, and engagement with open access generally, and with university repositories specifically (St Jean, Rieh, Yakel, & Markey, 2011; Stanton & Liew, 2011).

The aim of this study was to explore Australian researchers' open access experience as part of their information literacy. The concept of information literacy may be interpreted in a number of ways. In this study, a relational perspective of information literacy is adopted. The relational perspective of information literacy was introduced by Bruce (1997). It emphasis the relationship between people and information when learning in different contexts.

The main research question for this study was:

How do Australian researchers experience open access as part of their information literacy?

Emanating from this question, this study's objectives were identified as:

1) To identify the researchers' experience of open access as part of their information literacy
2) To provide descriptions of how researchers experience open access as part of their information literacy
3) To develop greater understanding of how researchers experience open access as part of their information literacy

1.4 Research design

This research aims to explore Australian researchers' experience of open access as part of their information literacy. The philosophical foundation that underlies this study is interpretivism. This research paradigm assumes that reality is socially constructed, as such there is no single observable reality but multiple interpretations (Merriam, 2009). Interpretivism believes that researchers do not find knowledge but they construct it (Merriam, 2009).

In keeping with the interpretivist paradigm, the study employed a qualitative research approach. A qualitative approach is used to study the interactions of people in naturally occurring situations (Litchman, 2006). It is designed to help researchers understand people, and the social and cultural contexts where they live. It allows the complexities and differences of people's experience of their world to be explored and represented (Philip, 1998). Similarly, the research is informed by the relational perspective of information literacy. The relational model does not seek to measure information literacy but instead emphasises the relationship between users and information, and the way users experience their information world (Bruce 1997).

The data in this study was collected through 31 semi-structured interviews with Australian researchers based in Australian universities who had used open access resources while conducting their research in the 12 months prior to their interview. Audio-recorded interviews were transcribed and

analysed using thematic analysis (Braun & Clarke, 2006). The analysis revealed four themes that describe the experience of the participants with respect to open access. The next section will give an overview of the identified themes. A full description is presented in Chapter 4.

1.5 Key findings

The analysis of the interview transcripts revealed how academics experience open access. This is presented through four themes:

- Understanding Open Access
- Institutional Repositories
- Publishing and Dissemination
- Finding Information

The participants' experience of open access as part of their information literacy relates to their roles as informed and informing researchers. The term "informed researcher" relates to the participants' role as information users, while the term "informing researcher" relates to the participants' role as information producers. Across the themes, it is through participants' roles as informed and informing researchers that their information literacy related to open access is enacted. The themes revealed in this study are:

Theme 1 - Understanding Open Access

Understanding Open Access describes the participants' understanding of what open access is and how it can benefit them as informed and informing researchers. Understanding what open access is, including the nature of open access publications, ways of accessing them, and the economic models that drive them, was an important aspect of researchers' information literacy. This understanding helped researchers navigate the practice of publishing and informed resource selection when they were looking at the research literature. In this study, participants understood "open access" to be "freely available".

Theme 2 – Institutional repositories

The participants engaged with institutional repositories as both informed and informing researchers. As informed researchers, they used information found in repositories as part of their research. Institutional

repositories enabled the participants to interact with other scholars in their field through the publications other researchers had deposited. As informing researchers, they deposited their work in the institutional repository to enable other academics, industry and society at large to access and benefit from their work. Participants reported on how Institutional repositories played a role in allowing them to communicate their research, and to build their profile as a researcher.

Theme 3 – Publishing and dissemination

As informed researchers, participants made decisions about where to publish their work (closed or open access sources), they also made judgements about using social media to disseminate their research. Each participant revealed their own preferences and insights on selecting where to publish, and the role that open access played in sharing research.

Theme 4 – Finding information

As informed researchers, participants found, evaluated, and used information as part of their research. For example, they found information for a literature review, and kept up to date with the latest research in their field. While they may not have actively sought out open access information, participants engaged with open access resources as part of their research.

1.6 Research significance

The research contributes a new empirically-derived understanding of researchers' experience of open access as part of their information world broadly, and of their information literacy specifically. The research provides, for the first time, empirically-derived evidence exploring how researchers experience open access as part of their information literacy. It is anticipated that this study will be of interest to the research domains of information literacy, library and information science, and higher education. More specifically, the findings may be used to inform both the policy and practice of Australian universities. The research supports Australian universities by providing an evidence base that can be used to inform the development of information literacy programs managed by university libraries, thus helping to ensure the programs meet the information literacy needs of the researchers

they serve. It also provides an evidence base that can inform Australian universities in their approaches to supporting and enabling researchers' access to open access information, including the design of institutional repositories to reduce barriers for use.

1.7 Key terms

A number of key concepts are of relevance to this study. This section will establish how the concepts which have informed this study have been interpreted.

Open access

Open access means a free and unrestricted online access to peer-reviewed research articles without financial and legal barriers (Budapest Open Access Initiative, 2002; Bailey, 2007; Willinsky, 2006; Suber, 2015). Authors disseminate their open access research in two main ways, gold and green (Harnard et al., 2004). The gold route means the papers are published by journals (Suber, 2004; Norris, Oppenheim, & Rowland, 2008), while the green route refers to an article, usually a pre-print version, made freely available through the Internet, for example via the author's personal website, or via an institutional repository (Bjork, Laakso, Welling, & Paetau, 2013).

Institutional repository

The term institutional repository refers to a digital collection of scholarly work created by the faculty, research staff and students of an institution, and accessible worldwide (Westell, 2006). It is a service provided by a university for the preservation and dissemination of the digital collection to the academic community (Lynch, 2003). Authors of the scholarly works are the key contributors to the institutional repository, however librarians are usually involved in the management and organization of the content, by ensuring the accuracy of the metadata and by respecting any copyright related to the content (Bjork, Laakso, Welling, & Paetau, 2013).

Open access journal

Open access journals refer to journals that peer review and publish articles, and disseminate them freely (Suber, 2012; Zhao, 2014). Some journals

are differentiated by their degree of openness and timing when the articles are made open. Some open access journals offer full and immediate access to their published articles, hybrid journals allow their published articles to be freely accessed, and delayed open access journals make the articles available only after a period of time (Zhao, 2014; Harnard, 2015).

Researcher

Researcher in this study refers to an academic who has expertise in research conceptualization, proposal, and conduct within one or across more than one discipline, and produces scholarly written works and research papers (Leckie, 2008). Researchers have a number of responsibilities, including maintaining awareness of critical issues related to their study interest, keeping up to date with scholarly publications, and ensuring their research is conducted according to accepted methodological standards (Leckie, 2008).

Information literacy

This study employs a relational perspective of information literacy. As such, it interprets the concept of information literacy as being "the complex of ways in which people experience using information when learning in different contexts" (Bruce, 1997; Bruce, 2008). In the relational perspective, the meaning of information literacy originates from the various ways in which people relate to information. It refers to the different ways in which people interact with elements of their information world. Consequently, the relational perspective does not seek to measure information literacy but instead emphasises the relationship between users and information, and the way users experience their information world (Bruce 1997).

Lived experience

The term lived experience denotes human experiences and choices, and how these factors influence individuals' perceptions (Boylorn, 2008). In this study, lived experience is understood to be the life-world as it is lived, felt, undergone, made sense of, and accomplished by a specific group of researchers (Schwandt 2001, 84). This study explored the lived experience of Australian researchers with open access as part of their information literacy.

The lived experience of participants was explored through qualitative interviews. Lived experience was the core unit of analysis for the study.

1.8 Overview of chapters

Following this introduction, Chapter two provides an overview of relevant literature in order define the scope of study, mapping the major research territories. Chapter three discusses the methodology that was applied in this study. It situates the study within the interpretivist research paradigm and introduces semi-structured interviews with thematic analysis as the selected research approach. Chapter four introduces the participants of this study. It acknowledges their role as co-creators of the findings that emerged from the study. Chapter five presents the study's findings. It describes the four themes that depict how the participants experience open access as part of their information literacy. Chapter six discusses and summarises the research outcomes. It examines the contributions of this study to open access and information literacy research. The study's limitations and suggested directions for future research are also outlined.

1.9 Conclusion

This chapter has provided an introduction to the study. The chapter outlined the background, articulated the aim and objectives, and discussed the research design, key findings and significance. Key terms used in this study were briefly presented. The next chapter explores and critically discusses the literature relevant to the study.

Chapter 2: Literature review

2.1 Introduction

This literature review discusses existing relevant research on open access and information literacy. The purpose of this review is twofold. Firstly, it seeks to define the scope of the study, mapping the major research territories. Secondly, it aims to synthesize the ideas and viewpoints of this study's exact territory, and thus prepare the way for responding to the research problem.

This literature review begins by providing a discussion on the role of, and approach to, the literature review within the context of the current study. This is followed by a brief background to open access. The background includes an overview of the different open access strategies currently available, a critical exploration of current research into open access and a discussion on the limitations of existing open access research. The chapter also discusses information literacy, with particular focus on exploring the four different information literacy perspectives used to guide current empirical work. The chapter also critically considers current studies exploring the information literacy experiences of researchers. The literature review concludes by articulating the key research gap that the study fills.

2.2 Literature review: role and approach

The literature review played a core role in the current study. It was used to define terms and constructs, identify previous research relevant to the study, reveal the existing gap in knowledge, identify the research problem to be addressed, and inform the research design (Bruce, 1994; Ridley, 2008). The literature review therefore served as the "driving force and jumping-off point" (Ridley, 2008, p. 2) for this study.

There are many different types of literature reviews, and each type is approached in a different way. This study used a traditional or narrative literature review. This type of review critiques and summarises a body of literature about a given topic (Cronin, Ryan, & Coughlan, 2008). This type of review consists of critical analysis of the literature published in books and

electronic or paper-based journal articles that are relevant to the topic being explored (Cronin, Ryan & Coughlan, 2008). The traditional or narrative literature review is typically selective in the material it uses, for example focussing on a specific geographic area, time coverage or theoretical perspective.

This review examines empirical literature that relates to the research question: *How do Australian researchers experience open access as part of their information literacy?* The literature discussed is limited to that published since 2000. Open access is a rapidly evolving field (i.e., due to advances in technology) and therefore currency of information was key. A time limitation was used to ensure only highly relevant empirical work was used to inform the study. The literature review was not limited to any particular geographical context due to the limited amount of research on this topic. This literature review focuses on how researchers experience open access as part of their information literacy. Very few studies have explored the *intersection* of open access and information literacy. For this reason, the literature review has been structured in three parts. The first part provides a critical discussion on literature exploring how researchers experience open access. The second part provides a critical discussion on literature exploring the information literacy experiences of researchers. As this research is focussed on researchers' lived experiences, studies on how libraries, librarians or publishers use, support or produce open access and/or information literacy are not included in the first two parts. Finally, the third part explores the limited studies that have explored researchers' experience of both of the two elements of the topic, open access *and* information literacy.

2.3 Open access

There are many definitions of open access. A frequently cited definition is provided by the Budapest Open Access Initiative (BOAI) which describes open access as literature that is freely available online (Budapest Open Access Initiative, 2002). The key concept in this definition is "free":

> There are many degrees and kinds of wider and easier access to this literature. By "open access" to this literature, we mean its free availability on the public internet, permitting any users to read,

> download, copy, distribute, print, search, or link to the full texts of these articles, crawl them for indexing, pass them as data to software, or use them for any other lawful purpose, without financial, legal, or technical barriers other than those inseparable from gaining access to the internet itself. The only constraint on reproduction and distribution, and the only role for copyright in this domain, should be to give authors control over the integrity of their work and the right to be properly acknowledged and cited.

Open access offers an alternative way for publication of research works. By distributing the works freely through the world wide web without copyright barrier, the argument is that the advancement of knowledge and science becomes faster than by publishing in non-open access journals. (Willinsky, 2006)

2.3.1 Open access strategies

Open access dissemination of research outputs can be done in two main ways. Most studies cite a colour scheme introduced by Steven Harnard: green and gold. Lewis (2012) observed that in green open access authors will self-archive articles on their personal website, or to an institutional or subject repository. Self-archiving as an approach to publishing is not new. Disciplines such as physics, economics, and computer science have been using electronic archives to distribute research articles since 1991 (Creaser et al., 2010). arXiv provides evidence of this. Established in 1991, arXiv is owned and operated by Cornell University, a private not-for-profit educational institution. It is a highly automated electronic archive and distribution server for research articles in the fields of physics, mathematics, computer science, quantitative biology, quantitative finance, and statistics. There are currently 1,296,952 items in arXiv.

When an author is self-archiving an article that has been published elsewhere, such as a non open access journal, they need to make sure they comply with the following two conditions (Bailey, 2006): (i) they have retained copyright; and (ii) they have transferred rights to the publisher, but the publisher permits authors to distribute the article's post print version under specified terms and conditions. Jenkin et al. (2007) established a four colour

classification scheme to help authors in their self archiving. The colours represent the different levels of self-archiving that is allowed with published works: (i) White: neither preprint nor post print versions can be self-archived; (ii) Yellow: only pre-prints can be self-archived; (iii) Blue: only post-print versions can be self-archived; and (iv) Green: both pre-prints and post-prints can be self-archived. In the context of this classification scheme a pre-print is a draft version of an article that has not undergone peer-review and modification; and a post-print is the final published version of the article.

The second approach for disseminating research outputs freely is by open access journals (Gold Open Access). An open access journal is a journal that provides immediate and free access to peer-reviewed articles. It consists of journals that make peer-reviewed articles available for online reading without a fee or a need to register (Crawford, 2011). However, the question of funding inevitably arises: who pays for the publication of scholarly materials via gold open access? Mukherjee (2010) explores possible answers to this question, noting that one of the most common funding models is author-pays. The journal publisher charges either the author or their sponsoring bodies, such as universities or the government, to publish. However, in his study, Mukherjee (2010) also said that in purely open access journals the author does not pay any fee to the publisher and the article is permanently made available to the journal website. The journals receive support by grant funding or by subsidy from the institution which pays the staff and provides the infrastructure.

Besides the green and gold scheme, other colours have been used in the open access publishing discourse, as revealed by Walt Crawford (2010), Bhaskar Mukherjee (2010), and the Association of College and Research Libraries (2004). They classify open access publishing as follows:

1. Open access journals (Green)

Green open access journals provide free access to all articles they publish and apply minimum restrictions on the use of the articles, for example the Creative Commons Attribution License. For example, *Information Research* is an international peer-reviewed scholarly journal that provides research results covering the field of information science, information management, and information systems. The license says that the readers are free to share,

which means free to copy and redistribute the material in any medium and format (Attribution-Non-commercial-NoDerivs 3.0 Unported).

2. Free access journal (Cyan)

Cyan open access journals provide free access to all articles and apply a variety of copyright statements but not a Creative Commons Attribution License or similar license. An example of this kind of journal is *D-Lib* magazine, a journal produced by the Corporation for National Research Initiative (CNRI). The access terms and conditions are clearly displayed on the D-Lib website: "material contained in D-Lib and D-Lib magazine are subject to copyright claims and other proprietary rights." However, permission is given for the material to be used for research purposes or general non-commercial purposes.

3. Embargoed access journals (Yellow)

Yellow open access journals have a specified embargo period and use a conventional copyright statement. An example is *Libri*, a journal of library and information services that investigates the functions of libraries and information services, and analyses the role of information in cultural developments. The articles are available online to their e-subscribers on publication of the printed issue, but are not available to non-subscribers until one year after publication. John Willinsky (2006) calls this delayed open access.

4. Partial access journals (Orange)

Orange open access journals offer free access to selected articles and use a conventional copyright statement. *The International Journal of Aging and Society* is an example of this kind of journal. It is a journal that publishes interdisciplinary research including psychology, economics, public health, and health education. Some of the articles are freely available, and some can be viewed and downloaded for a fee.

Open access publishing has increased the number of research articles now accessible and available. For example, *The Directory of Open Access Journals* indexes 9916 journals and has more than 2 million articles currently

available, and is growing (DOAJ, 2017). Open access is here to stay but at this point in time very little work has explored this emerging phenomenon. The next section provides a critical review of research into open access that has been conducted in the last sixteen years.

2.4 Research into open access

A number of studies have begun to explore the open access phenomenon. This research has tended to focus on the issues of impact and user perceptions. The following section reviews the key studies from these two areas. It should be noted that the literature review does not seek to be exhaustive, citing all works in the field, rather it provides critical commentary on the main or significant recent studies that have shaped and continue to shape scholarly discourse relevant to the phenomenon under investigation. As this research is focussed on researchers' lived experiences, studies on how libraries, librarians or publishers use, support or produce open access are not included.

2.4.1 Impact

Impact in the context of open access refers to how many times an article or item has been accessed, downloaded, cited, and to what extent the findings outlined in the article have been applied in other studies (Harnard et al., 2004). However, most of studies exploring the impact of open access have done so only from the perspective of citation count. For example, a study conducted by Koler-Povh, Juznic, and Turk (2013) investigated the impact of open access on the citation of scholarly publications in the field of civil engineering. They analysed 2,026 articles published in 14 non-open journals and in the author's own website or institutional repository. The study confirms that open access publications have a positive impact on the citations of articles, noting that open access articles were cited 43 times, whereas non open access articles were cited only 29 times. The work by Koler-Povh, Juznic and Turk (2013) suggests that publishing in open access journals increases the probability of being accessed and cited. Knowing that an article is cited in all likelihood indicates that the article was read and used to inform scholarly work. The study does not provide insights into how the researcher who cited

the article actually engaged with the article, nor does it reveal how the article cited fits within the researcher's overall information literacy experience.

The relationship between open access works and increased access and citation observed by Koler-Povh, Juznic and Turk (2013) was explored by Gaule and Maystre in a 2011 study investigating the role of author choice. Gaule and Maystre analysed a sample of 4388 biology papers published between May 2004 and March 2006 in *Proceedings of the National Academy of Sciences* (PNAS). They conclude that using open access alone does not increase citations, noting that authors of higher quality articles are more likely to choose open access publication options.

Gaule and Maystre (2011) note several reasons for authors choosing open access: a paper is more likely to be published in open access if the quality of the paper is high, if the present value of citations is high, if the cost of publishing is low and if the increase of the readership associated with open access is high. This study suggests that open access does not necessarily increase the number of citations, but publication quality and journal readership does. Gaule and Maystre note that an important limitation of their study was that it did not capture the "invisible readers" or those who read the article but did not cite it. This point has relevance to the current papers' focus on how researchers experience open access materials during their research.

Author choice was further explored by work undertaken by Chua et al. (2017). More specifically, the work explored academics' use of JIF in choosing where to publish. Many authors choose to publish in journals with high impact factors, believing this means their publications are more likely to be read and cited. This has implications for open access journals, which for the most part do not have high impact factors. However, because open access journals are freely available, open access articles are easy to find and the opportunity to read is high.

In total, 102 open access journals and 101 non-open access journals published in January 2011 were selected for a Chaua, Quereshi and Krishnan et al (2017) study. Five years of citations in open access and non-open access journals were compared. Source papers were obtained from PubMed, while citations were derived from Web of Science, Scopus, and Google Scholar

databases. The Thompson-Reuter's impact factor was also included as a tool for the analysis. The findings reveal that open access publications receive more citations compared with non-open access publications. The study suggests that the correlation of JIF to the open access publications is very small, but the correlation of JIF to the non-open access journals is high. The study concludes that JIF makes a small contribution to the citation count, but open access becomes a significant factor to gain citations. Based on the study, there is an opportunity to study academics' attitudes on selecting where to publish, including what factors influence their decision.

Further support for the high citations for open access articles over restricted access has been provided by Tang et al. (2017). They undertook a study exploring open access and non-open access journals published from 2009 to 2013 in the field of ecology and indexed in the Journal Citation Reports. Open access articles were paired with non-open access articles that were selected randomly from the same issue of the journal. 3534 journal articles were used and analysed for the study. The study revealed that open access articles received more citations than the non-open access articles every year from 2009 to 2013. The results also showed which continents the citations were coming from, indicating a citation advantage in North America, Asia, Oceania, Europe, and Africa.

Tang et al's study provides evidence that, in the ecology field, open access articles were disseminated, hence used, across many continents. However, the study focussed only on the ecology field. A further study from different areas of research may provide different insights. In addition, Tang et al. did not explore how open access articles were found, or reveal the contribution of the authors in disseminating the journal articles.

Tang's work supported and extended the 2015 study by Sotudeh, Gahsempour, and Yaghtin (2015) which explored the performance of open access and toll access papers published in author-pays open access journals. Publications in Springer and Elsevier's author-pays open access journals from 2007 to 2011 were reviewed. A total of 576 journals were identified. Together, Springer and Elsevier published 18,654 open access papers in the author-pays journals identified. This represented 4.7% of the 396,760 papers published

between 2007 and 2011. The study reveals that the number of open access papers has been growing more rapidly than toll access papers. In addition, it was noted that the impact values of open access papers were larger than those of the toll access papers for each year examined. There is need for further research to explore the awareness, attitudes, and decision making of academics when making decisions on whether to publish in open access or toll access journals.

Norris, Oppenheim and Rowland in 2008 investigated the citation advantages between journal articles that have been published by open access versus articles that were published by toll access (i.e., restricted access) within different disciplines. They selected four subjects to explore this issue: ecology, applied mathematics, sociology, and economics. Citations were counted using the Web of Science and the open access status of articles was determined by searching Google, Google scholar, OAIster, and OpenDOAR. A total of 4633 articles were examined. The study revealed that there was a citation advantage for those articles published by open access verses by toll access. This advantage however varied between the disciplines, for example sociology had the smallest number of open access articles but had the highest citation advantage. This study supports the previous work exploring citation advantage and open access, however the reason why there is a citation difference between open access and restricted access articles has not yet been investigated or explained.

Although open access articles are available for free online, this does not mean they are easy to find. Norris, Oppenheim, and Rowland (2008) used four popular search engines to locate the articles in their study. Two of the search engines were dedicated to retrieving open access articles, yet they noted challenges in locating the articles. The study shows the importance of ensuring easy access to articles published via open access channels if they are to be cited and used. The issue of ease of access is not new. Lawrence (2001) raised it over a decade ago, noting that “online availability of an article may not greatly improve access and impact without efficient and comprehensive search services; a substantial percentage of the literature needs to be indexed by these search services before scientists consider them useful” (p. 521). There is

a clear need for research exploring how researchers seek information via open access channels and how they use existing open access search platforms.

Extending the Norris, Oppenheim and Rowland study is work by Dorta-Gonzales and Santana-Jimenez in 2017 in which they analysed gold open access from 27 subject areas in order to investigate the citation impact of open access journals. The study excluded hybrid journals that allowed authors to pay the processing fee of publishing. The study used Scimago Lab for data gathering, which then sorted using the Scopus database and tagged as open access/non-open access by using the *Directory of Open Access Journals* (DOAJ). A total of 3,737 open access journals and 18,485 non-open access journals published in 2015 were included in the study.

The study findings revealed that open access journals from multidisciplinary and medicine subject areas showed a high citation number. In addition, in 18 of the 27 subject areas, the Scimago Journal Ranking (SJR) from non-open access journals was higher than those from open access journals. When they analysed journal impact, the findings showed that the advantage of open access was observed in only one journal out of 27.

The study concludes that there is no generalizable open access citation advantage. However, the study also considered the visibility of the journals had an impact on citations. Dorta-Gonzales and Santana-Jimenez stated that most of the open access journals included in their study were not on the top of rankings that measure impact, hence the journals were less visible than the non-open access journals that were in the top rankings. The study only focused on gold open access, it did not investigate the green model of access, for example, authors' personal websites or institutional repositories. This is the gap that the current study will fill.

Ottavian (2016) is one of the few to explore the citation advantage for articles deposited in institutional repositories as an open access channel. Ottaviani (2016) compared the citations of non-open access articles that become open access after the embargo period and non-open access articles that were available in the institutional repository of the University of Michigan. The non-open access articles were allowed to be deposited in the repository by the University, under a blanket licensing agreement between the

University and the publishers. A total of 3,850 peer-review articles published from 1990 to 2013 were selected for the study. Data was gathered from Thomson Reuters' Web of Science and Journal Citation Reports databases in order to get the citations and calculate the expected value of citations. The study findings suggest that good quality articles receive more citations when they are made open access. Additionally, a citation benefit was found for open access articles when compared with non-open access articles in the same issue of a journal.

Ottaviani's study confirmed the impact of open access articles, however it did not reveal how the articles in the institutional repository were found by the users. Ottaviani did not investigate the academic's contribution to making their publications available in the repository.

Knowing that articles published in open access channels received more citations rather than the scholarly works published in other channels, it can be argued that open access articles are likely to be used more by researchers than the non-open access articles. Following that argument, the question of usage was addressed by Chaudhuri and Thohira (2010). Specifically, their work explored the usage patterns of open access and hybrid open access journals in selected scholarly publications. They analysed eleven high impact science and medical journals for 2004, 2006, and 2008. Among the subjects investigated, biology, medical science and mathematics were the three highest users of open access journals, while biology and mathematics were the two subject areas that had the most significant increase of use. The study concluded that overall most of the journals have indicated increase in use over the years. However, given that the majority of the journals examined were journals with high impact factors and were of high standing in their fields, for example *Proceedings of the National Academy of Science* (PNAS) and *Nature*, this finding is not unexpected. As these non-open access journals already have strong profiles and high readership, it might be expected that as they become available as open access, the readership patterns will follow. Further research exploring how the users found the articles (e.g., via the journal website or via search engines) would provide insight into the information literacy of researchers and the implications for ensuring open access publications are easy to find. Chaudhuri and Thohira do not provide a working definition of "usage".

It is unclear in their study whether it means accessing, reading, or citing the articles. Again, the term "use" does not explain how the readers engage with information they gain from the articles they accessed.

2.4.2 User perspective

In the last six years a growing number of studies have begun to explore open access from the user perspective. These studies showed that the academics were aware of open access and willing to use open access sources in future, but open access awareness does not necessarily mean the academics disseminated research via open access channels. The findings shed light on academics' attitude towards open access in terms of publishing and dissemination. These studies did not investigate how open access contributed to the information need of the academics, in other words these studies did not explore how the academic engaged with open access in the context of their information world and associated information literacy experiences. That is the gap that the current study fills.

Research conducted by Stanton and Liew (2011) examined doctoral students' attitudes to open access forms of publication. Data was collected through interviews with eight doctoral students enrolled in a range of disciplines in a New Zealand university and a self-administered online questionnaire of 251 students. The students were asked to discuss their awareness of open access generally and the digital repository managed by the university specifically.

Data was also gathered on the students' willingness to comply with a university mandate to archive their thesis in a university repository, and most importantly their use of open access and institutional repositories for their information needs while conducting their doctoral studies. The students were drawn from the University's five colleges: Business, Humanities, Social Sciences, Education, and Sciences.

Stanton and Liew (2011) noted that while the mandatory submission policy had been applied by Massey University since 2007, the awareness of students of university repository was fairly low. The research students who took part in the study understood the benefits of open access, and valued the

institutional repository in helping to ensure their research outputs were disseminated widely. However, they raised concerns about copyright and the potential of plagiarism, and expressed confusion regarding journal publishers' copyright restrictions and the implications these restrictions had on making submissions to the university repository.

The work by Stanton and Liew (2011) was one of the first to shed light on how researchers actually engage with university repositories. Although the study provided new insights into the awareness and attitudes of doctoral students towards open access, it did not explore the way in which the articles retrieved contributed to the students' doctoral work. Stanton and Liew (2011) have shown that doctoral students value open access and think of the university repository as an important information resource for their study, but how do the students use the institutional repository for finding information while conducting research? How do articles obtained via the university repository fit within the students' information need? These are some of the questions that the current study will shed light on.

Another study to explore the perceptions and experiences of a group of institutional repository users was conducted by St Jean, Rieh, Yakel, and Markey in 2011. The participants were recruited through five university repositories. Twenty people participated in the study, including six undergraduate, four masters, and three doctoral students; five academics; one library staff member; and one museum staff member. Findings showed that the study's participants understood institutional repositories at different levels. Most of the participants were able to provide a basic definition of a university repository; this level of awareness was established by attending library workshops and via messages from their university on the obligation of archiving their thesis or dissertation in the repository, and from recommendation from professors.

The participants' reasons and methods for visiting the institutional repository were diverse. Participants visited the repository to obtain content for a course, to access raw data for a research project, to identify academics or researchers with similar research interests, for networking and collaborations, to find best practice examples, and to help locate information

for everyday life. The study found that the participants reached the repository through various paths such as clicking on a link from the university library's webpage or searching for the name of the repository on the university's homepage.

The study concluded that users were uncertain about the scope of institutional repositories and what they offered. There was confusion about the relationship between repositories and library databases. The participants indicated they were not sure if all materials in the repository had been through a peer review process. It was noted however that institutional repositories were considered more trustworthy than popular search engines such as Google or Google Scholar, because of the association with the library and the university. St Jean et al. (2011) concluded their work noting that "our study also suggests both the need for, and the potential value of, future studies involving institutional repository end-users" (p. 40). The work by St Jean et al. provided insights into how different stakeholders in the university setting were aware of and engaged with university repositories. Importantly this work began to consider how repositories were being used as an information source by users. In common with other studies exploring open access, the work by St Jean et al. (2011) did not focus on how researchers use the materials obtained via institutional repositories in their research processes.

More recently, Borrego (2017) explored academics' use of institutional repositories in comparison to ResearchGate. The work arose out of a recognition that despite the increasing number of institutional repositories worldwide, most were underpopulated. In contrast, academic social networking sites such as ResearchGate and Academia.edu appeared to be well used by academics in sharing their scholarly works. The availability of scholarly outputs in the institutional repositories of 13 top Spanish universities and in ResearchGate were compared. The findings revealed that the proportion of articles available by academics in the institutional repositories of the Spanish universities investigated was low. In contrast, the proportion of articles available in full text on ResearchGate was much higher than that of articles deposited in institutional repositories.

Two main reasons were identified for why an academic would upload their scholarly output on ResearchGate but not their institutional repository. The first reason was a lack of knowledge of the existence or operation of the institutional repository, for example some participants commented on not having received appropriate training and as a consequence not knowing how to upload an item. The second reason was the awareness of the advantages offered by ResearchGate, for example reaching a wider audience. The finding of Borrego's study indicated that academics prefered to use ResearchGate over institutional repositories. However, the study may also highlight the role of librarians in helping academics develop their awareness and capacity to use institutional repositories. Borrego's study sheds light on academic's open access preferences as it relates to two key types of repositories (e.g., institutional repositories and ResearchGate). However, the study does not explain the intentions or preferences of academics on using these types of repositories for finding information.

A number of studies in the last three years have explored the awareness of academics of open access and whether awareness leads to use or engagement. For instance, the study by Lwoga and Questier (2015) undertook research in Tanzanian health science universities that investigated academics' awareness, attitudes and use of open access, and the role of information professionals to support open access. Interviews were conducted with 16 librarians and 415 academics completed an online questionnaire. Participants were drawn from eight Tanzanian health sciences universities. The study found that although most of the academics who participated in the study were aware of open access, they did not disseminate their work through open access channels. The study also revealed that those academics who did use or engage with open access did so by publishing in open access journals and not by self-archiving in institutional repositories. The study's participants identified a number of barriers to open access usage, including awareness, skills, author pay model, copyright, and plagiarism concerns.

A study similar to the work of Lwoga and Questier (2015) was undertaken in Ethiopia by Mammo and Ngulube (2015). This study also explored the attitudes of academics to open access. Data was obtained via an online questionnaire and interviews. In total, 768 academics from seven universities

in Ethiopia participated in the study. Like the work by Lwoga and Questier (2015), this study found that most of the participants were aware of open access. The study also found that the participants had a generally positive attitude to open access journals indicating an interest in using them in the future. The participants also noted that they had been using or accessing open access journals as an alternative to traditional subscription-based journals. However, many of the academics in this study revealed that the open access articles they had found were through their information search without intentionally looking for open access articles. The participants also noted an expectation that librarians would promote and enhance the accessibility of open access journals in both their own university and in Ethiopia more broadly.

Exploring further the role of "attitude" with regard to academics' engagement with open access is the work by Massoro and Othman (2017). Sixty academics from the Agricultural Research Institutes in Tanzania completed an online questionnaire exploring the factors that predicted usage of open access journals, with particular attention to how attitude to open access could be influenced. The study's findings showed that colleagues, management, subordinates, and heads of departments played significant roles in influencing an academic's attitude to open access, and this in turn could influence that academic's intention to use open access journals. The study also found that predatory publishers were a strong determinant of the researchers avoiding using open access journals. Mistrust among the academics who participated in the study, about the content of open access journals negatively impacted on their use of open access journals.

Mistrust or concern by academics with regard to open access were also revealed in McDonald et al's 2016 study. A web-based survey was distributed to academics at two Canadian universities. A total of 346 academics participated in the survey. The findings reveal that academics in the science and health disciplines were the most positive about open access journal publishing. In contrast, the academics in the arts, humanities, and social sciences had mixed feelings about open access. Academics in business were the least positive. The academics who participated in the study had concerns focussed on open access journal quality and associated costs.

While the user perspective of open access was investigated by some early studies, Chang (2017) explored the difference between authors in the library and information science (LIS) discipline who published in open access journals in comparison to those who did not. Thirteen non-open access journals and twenty open access journals were identified. The authors' occupation and institutional affiliation were extracted from the full text of articles published between 2008 and 2013. The study's findings suggest that traditional scholarly communication has not faced obvious threats from open access publishing. While open access journals have provided additional options to disseminate their scholarly work, academics in the LIS discipline still prefer to publish in non-open access journals. Half of the authors who published in open access journals were LIS practitioners (e.g., librarians), and it was noted that practical discussions are more often found in open access journals and the latest research appears in non-open access journals. Chang's study sheds light on how different types of authors (e.g., academic, practitioner) use open access and non-open access journals. There would be considerable benefit in replicating this work in other disciplines.

2.4.3 Summary

There is growing body of research exploring open access. This research has focussed on two main areas of enquiry: impact and user perspective. Research exploring open access from the perspective of impact has investigated how frequently articles published in open access channels are cited compared to articles published in other channels (Gaule & Maystre, 2011; Koler-Povh, Juznic, & Turk, 2013). However, citation count does not consider the ways in which researchers *use* the articles obtained via open access. Research exploring open access from the perspective of user engagement has tried to fill this gap. For example, studies have explored how research students are aware of and use their university repository (St Jean, Rieh, Yakel, & Markey, 2011; Stanton & Liew, 2011), and academics' preferences when disseminating and factors influencing their decisions (Massoro & Othman, 2017; Borrego, 2017; Chang, 2017). While providing an interesting insight into the researchers' experience of open access, these studies have focussed only on the attitudes and perceptions of researchers mainly for publishing and disseminating research, the studies have not explored the researchers'

experience of open access as part of their broader information world and most importantly as part of their personal information literacy. Two constructs, open access experience and information literacy, will be explored further in this current study. The concept of information literacy is introduced and explored in the next section.

2.5 Information literacy

This section introduces the concept of information literacy. The term information literacy was first used in 1974 by Paul Zurkowski, in a proposal submitted to the United States National Commission on Libraries and Information Science. The proposal stated that an information literate individual was someone "who had learned to use a wide range of information sources in order to solve problems at work and in his or her daily life" (Grassian & Kaplowitzz, 2001, p. 4). Over time, the concept of information literacy has been described in numerous ways with many different attempts at a definition (Welsh & Wright, 2010). A frequently cited definition is one provided by the American Library Association in 1989: "To be information literate, a person must be able to recognise when information is needed, and have the ability to locate, evaluate, and use effectively the needed information" (p. 1).

The American Library Association (ALA) *Presidential Committee on Information Literacy: Final Report* (ALA, 1998) states information literate people are those who can find information when they need to complete tasks or to make decisions. From that perspective, information literacy can be seen as personal empowerment as it allows people to experience information searching and motivates them in pursuing knowledge.

Regardless of which definition of information literacy might be adopted, there is some agreement about its core elements. Harding (2008) noted that there was consensus in the existing literature that information literacy was "a problem solving activity that involves critical thinking and the ability to apply information to an individual's life" (p. 3). Similarly, Eisenberg, and Brown (1992), and Grassian and Kaplwitz (2001), noted that information literacy definitions all included a focus on lifelong learning, critical thinking, problem solving, and the capacity to use information in all aspects of life.

In addition to consensus regarding some of the key elements of information literacy, there is agreement emerging around the value or impact of information literacy. The *Australian Library and Information Association* (ALIA) Statement on Information Literacy for All Australians (ALIA, 2006), grounded in the *Alexandria Proclamation on Information Literacy and Lifelong Learning* (2005), argues that society will be advanced by all people being empowered to find, use, and create information effectively to achieve their personal, social, occupational, and educational goals. Similarly, Todd (2000) suggests that: "At the heart of information literacy are people being able to effectively engage with their information world: to connect with, interact with, and use information meaningfully and purposefully to get on with their lives. The outcome is not information literate persons per se, but people able to get the best out of, and contribute to living" (2000b, p. 29).

In academic settings, which is the context for the current research, students and academics are dealing with a vast amount of information. The ability to retrieve the needed information and understand the information found is very important, especially for academics engaged in empirical research. When conducting research, academics are required to have knowledge about how information is organised and to be able to apply the best ways to retrieve the information needed (Bruce, 1990). Along with that, as information is regarded as the core of research (Bruce, 1990), academics and students should be able to evaluate the quality of information and use it effectively to meet their scholarly needs.

2.6 Theoretical perspective of information literacy

Limberg, Sundin, and Talja (2012) noted that understandings of information literacy differ based upon the "theoretical lens from which it is approached" (p. 93). Over the years varying perspectives on information literacy have been identified and discussed (Markless & Streadfield, 2007; Bruce, 2008; Lloyd & Talja, 2010). The four main theoretical perspectives for information literacy that will be discussed in the following section are:

a. Behavioural
b. Relational
c. Sociocultural

d. Discourse analytic

A brief discussion of each follows.

2.6.1 Behavioural perspective

The behavioural perspective describes and understands information literacy as a collection of skills and behaviours which individuals possess to acquire and use information (Bruce, 1997). This perspective is closely associated with the work of Christina Doyle (1992) who conducted a study aimed at developing a consensual definition of information literacy and establishing outcome measures for the concept. The definition that emerged from the study was: "information literacy is the ability to access, evaluate, and use information from a variety of sources" (Doyle, 1992, p.2). The definition is followed by 10 indicative outcomes which state that an information literate person is an individual who understands the need for information, makes decisions based on accurate and complete information, develops questions based on information needs, locates sources of information, formulates search strategies, and is able to use information technology to find information, evaluate information, organise information, and synthesize and use information (Doyle, 1992).

The behavioural perspective has influenced information literacy standards that are mostly applied in educational contexts. An example is the United Kingdom Society of College, National and University Libraries (SCONUL) called Information Skills in Higher Education, established in 1999. The standard introduced the Seven Pillars of Information Skills Model. The model was then updated in 2011 and was presented as a generic core model for higher education (SCONUL, 2011). The model defines information literacy as the ability to gather, use, manage, combine as well as create information and data effectively.

Similarly, the Association of Colleges and Research Libraries (ACRL), a North American organisation, established the Information Literacy Competency Standards for Higher Education in 2000. These Standards define information literacy as a set of abilities individuals need to locate, evaluate, and use information effectively (ACRL, 2000). More recently, ACRL released the

Framework for Information Literacy for Higher Education (2016). Building upon the Standards, the Framework recognised that information literacy, was, or could be, "an educational reform movement" (ACRL, 2016) and as such it needed to be articulated via a richer, more complex set of core ideas. Drawing upon the concept of metaliteracy, the Framework is based upon, and provides, an expanded definition of information literacy that emphasizes dynamism, flexibility, individual growth, and community learning:

> Information literacy is the set of integrated abilities encompassing the reflective discovery of information, the understanding of how information is produced and valued, and the use of information in creating new knowledge and participating ethically in communities of learning. (ACRL, 2016)

The Framework is organized into six frames, each consisting of a concept central to information literacy, a set of knowledge practices, and a set of dispositions. The six concepts that anchor the frames are:

- Authority Is Constructed and Contextual
- Information Creation as a Process
- Information Has Value
- Research as Inquiry
- Scholarship as Conversation
- Searching as Strategic Exploration (ACRL, 2016)

There has been a growing criticism of the behavioural perspective to information literacy (Johnston & Webber, 2003). The behavioural perspective focuses on the process and the behaviour of individuals getting and using information, but it is not related to the assessment of how people use and engage with information in order to learn (Bruce, 2008). The standards derived from the behavioural perspective are a list of skills that are necessary for the information literate individual to possess. Johnston and Webber (2003) suggest that this is little more than a "tick the box" approach, which only "[reduces] a complex set of skills and knowledge to small, discrete units" (p. 337). The recent ARCL Framework represents a shift away from the behavioural perspective to one that also acknowledges the role of affect, cognition, and

attitudes in an individual's engagement with their information worlds. This point has been recognised in other information literacy perspectives.

2.6.2 Relational perspective

The relational perspective of information literacy was introduced by Bruce (1997) and is derived from a study of information literacy as experienced by university staff. The relational perspective of information literacy focuses on the various ways people relate to information and interact with elements of their informational worlds. Bruce's study used phenomenography. This is a method that explores differences or variations in how people experience particular aspects in the world. In Bruce's study the aspect being explored was how information literacy was experienced among people who use information. The people who took part in the study were academics, librarians, learning advisors, and educational developers. Bruce's study identified seven different ways in which participants experienced information literacy. Her work, *The Seven Faces of Information Literacy* is described in Table 1 (overleaf). As a result of her study, Bruce described the information literate person as being "one who experiences information literacy in a range of ways, and is able to determine the nature of experience as it is necessary to draw upon in new situations" (1997, p. 169). Bruce's research is significant as it is one of the early studies to move beyond skills and behaviour, and to show the complexity and nuanced elements of information literacy. Bruce offers an alternative to the skill-driven approach where information literacy is described in terms of competencies rather than skills. Andretta (2012) added this idea by mentioning that it is the competencies that are required to complete the task when people are learning.

Grounded in the results of this study, Bruce (1997) developed a "relational model of information literacy" which consists of three components: descriptions of information literacy, information literacy education, and information literacy research. This model is presented in Figure 1. The key aspect to this model is the focus on information users, not information experts. Equally important is that the model does not seek to measure information literacy per se, instead it focuses on the relationship between user and their experience of their information world.

No	Categories	
1	The information technology conception	IL is seen as using information technology for information retrieval and communication
2	The information sources conception	IL is seen as finding information
3	The information process conception	IL is seen as executing process
4	The information control perception	IL is seen as controlling information
5	The knowledge construction conception	IL is seen as building up a personal knowledge base in a new area of interest
6	The knowledge extension conception	IL is seen as working with knowledge and personal perspectives adopted in such a way that novel insights are gained
7	The wisdom conception	IL is seen as using information wisely for the benefit others

Table 1. Seven conceptions of information literacy (Bruce, 1997, p.110)

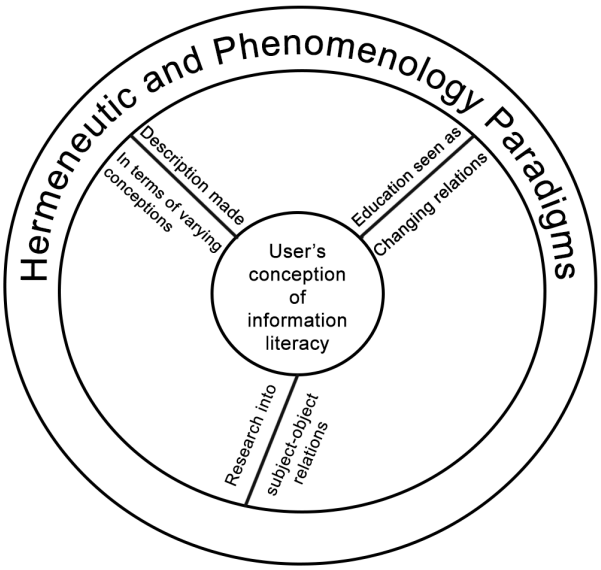


Figure 1. The Relational model of information literacy (Bruce, 1997)

More recently, Bruce (2008) renamed the relational model of information literacy. The new name, Informed Learning, was adopted to promote the model's focus on the "learner-centred, experiential, and reflective approaches to the information literacy agenda" (p. 5). In addition to a new name, Bruce also articulated a revised definition of information literacy: "experiencing different ways of using information to learn" (p. 5). The inclusion of the term "learn" is significant. Although this concept has been a part of many descriptions of information literacy (see 1974 statement by Paul Zurkwoski), by intentionally including the term 'learning' in the new name for the relational model of information literacy, Bruce was introducing a greater emphasis on the role of learning in the phenomenon.

Through the relational perspective, Bruce introduced an evaluation of people's experience of information literacy to clarify the connection between the relational perspective of information literacy and learning: "learning should be seen as qualitative change in a person's way of seeing, experiencing, understanding, conceptualising something in the real world – rather than a change in the amount of knowledge which someone possesses" (Marton & Ramsden, 1988, cited in Bruce, 1997, p.60).

Since 1997 there have been numerous studies conducted that have adopted the relational perspective to investigate people's experiences of information literacy. Recent examples in the context of higher education have included: Maybee (2006) exploring undergraduates students' experiences of information literacy; Boon, Johnston and Webber (1997) examining English academics' experiences of information literacy; Andretta (2012) investigating the experience of information literacy among postgraduate students studying Information Services Management; and Johnston (2014) investigating the information literacy experiences of English as a Foreign Language student.

The aim of the current study is to explore Australian researchers' open access experience as part of their information literacy. The relational perspective is adopted to uncover the experiential aspect of Australian researchers' engagement with open access during their research. Therefore, through this perspective, this study does not seek to measure information literacy but instead emphasises the relationship between people and

information, and the way people experience their information world (Bruce 1997).

In addition, the relational perspective is essential for information literacy research to investigate the way people learn and to reveal some important aspects that are beneficial to improve people's experience of using information (Bruce, 1997). It represents a scholarly approach that can be applied for personal development in the quest to be information literate individuals (Catts, 2005).

2.6.3 Sociocultural perspective

The sociocultural perspective to information literacy emphasises the relationship between individuals and communities. It focuses on where individuals engage with each other and have to communicate with each other (Limberg, Sundin, & Talja, 2012). In this perspective, information literacy is identified as a transformative agent that enables individuals to learn from others (Lloyd, 2005). The use of tools for communicating and for creating a relationship is emphasised (Limberg, Sundin, & Talja, 2012). This includes tools which enable people to find, work with, and use information, for instance scientific journals, databases, and search engines. James Wertsch (in Limberg, 2012) describes the sociocultural perspective as being primarily concerned with describing, interpreting, or explaining action. Using the word action emphasises that information literacy happens through activities of social character, information literacy is thus seen as integrated into different social practices (Limberg, Sundin, & Talja, 2012).

The sociocultural perspective is being used to inform a growing number of information literacy studies. For instance, Spiranec and Zorica's (2010) study attempts to re-conceptualized the definition of information literacy and identify the development of information literacy. Spiranec and Zorica analysed previous research focusing on the features of information literacy, information literacy 2.0 and library user education. They noted that while the conceptual core of information literacy was on abilities of individuals when seeking and using information, the theoretical foundation of information literacy has constantly evolved. A shifting focus of information literacy from traditional understanding towards information literacy 2.0 was identified. Information

literacy 2.0 was recognised as a concept that emphasized a social, economic, and cultural aspects of information literacy to reflect the idea of using information to learn in a social and cultural collaborative environment (Spiranec & Zorica, 2010).

Wang (2010) applied sociocultural theories to explore the experience of librarians and students in the curricular integration of information literacy. The purpose of this research was to reveal how information literacy can be systematically integrated into higher education curricula as well as to propose a model for curricular integration of information literacy. The research reveals that librarians, lecturers, course coordinators, head of faculties, and students are the key stakeholders for the success of the integration. By interviewing librarians and students from four universities, the researcher developed an information literacy integration model.

2.6.4 Discourse analytic

Although Bates (2005) has noted that discourse analysis is a theory which has influenced contemporary information behaviour research, it has only attracted limited consideration in relation to information literacy research with the notable exception of the work by Limberg, Sundin and Talja (2012). The primary tenant of discourse analysis is that language is central to social life. In discourse analysis, it is not only what is said or written which is important, but also the styles and strategies employed by the language user (Robson, 2002).

The discourse analytic perspective of information literacy looks at the socially and culturally shaped ways of understanding information competences and information practices (Limberg, Sundin & Talja, 2012). The focus of this perspective to information literacy is not on what people do with information; instead the focus is on the ‘interpretive repertoires’ (Talja, 1999) through which people give meaning to information competences and practices.

The discourse analytic perspective is influenced by the work of Michael Foucault (Bryman, 2012). Foucault regarded Discourse as more than talking and writing, but it is seen as complex relationships between individuals, texts, and ideas (van Leeuwen, 2005). The primary aim of the discourse analytic perspective is to bring into view the “take for granted understandings” and

background assumptions that underpin a specific way of discussing information literacy (Limberg, Sundin & Talja, 2012). Most discourse analytic studies published in library and information research have tended to focus on how texts depict the roles and competences of information users. For example, Haider and Bawden (2007) analysed discourse on information poverty and concluded that the "information poor" (Haider and Bawden, 2007, p. 111) are products of institutionalised professional library and information discourses, and that these discourses were tightly interconnected with general historical knowledge formation, such as the discourse on the evolution of the information age.

2.7 Research into information literacy and researchers

A growing number of studies have begun to explore the information literacy experiences of researchers. The majority of research has been conducted outside the Australian context. The following section provides a review of the key studies, internationally and from Australia. It should be noted that the literature review does not seek to be exhaustive, citing all works in the field, rather it provides critical commentary on the main or significant recent studies that have shaped and continue to shape scholarly discourse relevant to the topic. As this research is focussed on researchers' lived experiences, studies on how libraries, librarians or publishers use or support information literacy are not included.

2.7.1 International studies

The international studies fall into two broad groups. The first group includes three studies that explored the information literacy of higher degree students. The second group explored the information literacy of researchers or scientists. A brief overview of each group follows.

2.7.1.1 Information literacy of higher degree research students

Three studies have been conducted in the UK and Europe exploring the information literacy of higher degree research students. Taken together, these studies show that information literacy is regarded as a set of skills that are considered necessary for research students. The skills include how to use online databases and advanced research tools, and using information

technology applications for managing, recording, and presenting information. They explore how researchers identify different sources of information, how they value the information they find, and whether they also shared the information they found.

A study of the information literacy of doctorate students was undertaken by Vezzosi (2009). The study involved doctoral researchers in the field of biology from the University of Parma, Italy, with the aim of understanding their information needs and how they searched for information, with a view to suggesting improvements to the University's library service. Doctoral students were considered important for the library (and as consequence for the study) as they devoted great amounts of time to bibliographic research, and accessed digital collections and online databases. The need to keep themselves up to date with the most valuable contributions published in their research field and the importance of literature review for the doctoral dissertation makes this type of user very interested in all types of services offered by the academic library.

In-depth semi-structured interviews conducted with 18 participants investigated which information sources students used, the research strategies they employed, the role people adopted in their information finding process, and the attitude students had towards library services. Doctoral students in the field of biology were recognised as heavy users of digital libraries and those who preferred to access documents through the web. Data analysis revealed doctoral students prefer the online version of any document. They appreciate simple and easy research tools. However, as their study progresses over time, they become more aware of the importance of bibliographic databases and appreciative of seeking information through advanced searches.

The way doctoral students searched for and managed information was also investigated by Heading, Siminson, Purrel, and Pears (2010). Their study explored the experiences of doctorate students who attended information literacy skills and bibliographic software training. The training was held to respond to the Joint Skills Statement of the UK Research Councils which required every research student to be able "to identify and access appropriate bibliographical resources, archives, and other sources of relevant information"

(Heading, Siminson, Purrel, & Pears, 2010, p.206). Students needed to be introduced to the many information technology applications available for managing, recording, and presenting information. This suggests that information literacy in this study was understood as a set of skills and behaviours to acquire and use information (Bruce, 1997).

The doctoral students taking part in the study were invited to attend an information literacy workshop covering key topics including the use of bibliographic reference management software, library catalogues, database searches, and online repositories. By the end of the training, data on the doctoral students' experiences of the training was gathered by an online questionnaire followed by focus group discussions. Of the 349 students who took part in the training, 211 students responded to the questionnaire, providing positive feedback on the content of the training. Most of students participating in the training indicated that the session that provided the most benefit concerned how to undertake a search strategy and confidently use bibliographic management software. The focus group that followed with 21 students showed that research students were able to link the information they found and the bibliographic management software they used. In terms of using information technology, the participants through the training had learnt how to search for information via online databases.

Whitworth, McIndoe, and Whitworth (2011) describes common information problems that postgraduate researchers in the UK experience and the outcomes of an information literacy course specifically tailored to meet their information needs. For instance, moving from undergraduate study into postgraduate research work has been "recognised as a difficult transition in terms of information skills" (Beetham et al., 2009 in Whitworth, McIndoe, & Whitworth, 2011, p. 36). Whitworth, McIndoe, and Whitworth (2011) developed resources intended for postgraduate research students to be able to use technology for finding information. The resources are divided into seven themes: 1. Introduction; 2.Content: what is research, the role of information in research, where to look information; 3. Competency frame: information literacy procedure and standard; 4. Learning to learn frame: how information becomes knowledge; 5. Personal relevance frame: reflecting on information searching; 6. Social impact frame: how information is used in an ethical way;

7. Conclusion: objective and subjective value (Whitworth, McIndoe, & Whitworth, 2011).

The resources were pilot tested by eight postgraduate research students from a range of disciplines, through an information literacy course offered at the University of Manchester. On completing the course the students took part in a focus group exploring whether the resources delivered met with their information needs. A majority of the students confirmed that by participating in the course they had a better understanding of information literacy, why it is important for their lives and studies, and most importantly they knew how to find, use, and preserve information for future use (Whitworth, McIndoe, & Whitworth, 2011). However, as the course had a focus on information skills (e.g., how to find and use information), it could be argued that the research students' understanding of information literacy was inevitably limited to a behavioural understanding of information. They may have focussed, for example, on how quickly they could find information, or how many keywords and synonyms they were able to create when searching for information.

2.7.1.2 Information literacy of researchers or scientists

Five studies have been conducted in the UK and Europe exploring the information literacy of researchers (i.e., not research students). These studies reveal the information literacy of researchers, including how they find and store information during research, their implementation of search strategies when using search tools, their use of academic databases and search engines to retrieve information, and their use of journals to publish. By looking at the key findings reported by these studies, information literacy was seen as essential skills that researchers enacted during research. The studies did not investigate how the researchers related to their information world when conducting research. In addition, the studies did not explore how the researchers made decisions when they were selecting a database to retrieve information or how they made their selection to publish.

Alvarez et al. (2013) investigated the ability of Brazilian health researchers to find and use updated scientific information. In this study information literacy was understood to be skills and knowledge to find information based on students' academic need, to analyse and to use the

information in an effective way (Alvares et al., 2013). The six health researchers participating in the study were all employed at the University of Sao Paulo and the Oswaldo Cruz Foundation. The data collection was conducted in two phases, the first phase involved semi-structured interviews with a focus on exploring how the researchers searched for information and how they expected to conduct information searches in the following five years. The aim was to identify the information behaviours of the researcher. In the second phase, the researchers took part in a focus group that aimed to discover the way researchers managed information overload and to identify what skills they required to deal with information overload.

The study revealed the health researchers' experiences of disseminating information, their interactions with librarians, and their use of information and communication technology. The findings showed that all of the health researchers who participated in the study were familiar with information search strategies, but had limited knowledge of academic databases, for example refining search strategies and dealing with information overload. This study recommended that academic librarians needed to find new ways of working with health researchers to develop their information literacy skills. Even though this study only involved researchers from the discipline of health and from one location (e.g., the University Sao Paulo), the study highlighted researchers' preferences for information via commercial databases, including the difficulties they encountered. It also showed that the health researchers in this study relied on academic databases when searching for information.

Further exploration of medical researchers was undertaken by Roos (2015). More specifically, the study investigated how medical researchers sought, managed, shared, and used information during research. Eight medical researchers and four librarians from the Helsinki University Central Hospital and the University of Helsinki participated in interviews exploring information practices when undertaking research. The study revealed that medical researchers searched and used information mostly in the beginning and at the end of their research work. The researchers' main formal sources of information were journal articles. This was also where they chose to publish their own work. If needed, they found help and support from fellow or senior researchers. Ease of access and use were the most important conditions that

determined how medical researchers found information. The study concluded that to be successful, information services should be embedded in researchers' core activities. It was recommended that libraries should be designed to follow researchers' workflow and embed their tools and services in it. For example, libraries should link full text resources directly to the tools that researchers use and they should make library web pages part of the web resources researchers constantly use.

When investigating researchers' information literacy, medicine has not been the only discipline to be considered. Recent studies have been undertaken in the education, behaviour science and veterinary discipline settings. Zoellner, Hines, Keenan, and Samson (2015) undertook a study exploring how academics in the fields of education and behavioural science located, retrieved, and used information for research and writing, and how they published and stored their research materials. Interviews were conducted with 12 academics from the College of Education and Human Sciences, and the Psychology Department at the University of Montana. The findings reveal that the study's participants found information sources through active searching, and through professional affiliations. They made frequent use of library resources and services; they also relied on graduate students to compile bibliographies for research work being undertaken. E-resources were popular for their ease and accessibility, as a result many participants indicated that their reading patterns had changed with a preference for e-journal articles over books. The findings also show that academics advanced their research in strategic ways, for example gathering feedback on their initial work via conference presentations, then publishing in journals.

Veterinary science provided the discipline context for the work by Nel and Fourie (2016). The Faculty of Veterinary Science at the University of Pretoria provided the site for a case study exploring veterinary researchers' information needs, how they searched for information, and how they used information, with a particular focus on how well their needs were met by the University Library. Questionnaires, focus groups and citation analysis were used to gather data from the researchers, including academic staff, research students and three information specialists from the library. The study found that the information needs of researchers were influenced by the research

environment and expectations for research output. Context was therefore confirmed as having a significant impact on the information needs and the information behaviour of the researchers who participated in the study. The findings also revealed that most of the researchers' needs for information and support services were being met by the library. The findings however also revealed that the researches who participated in the study were frequently unaware of the skills and expertise of the information specialists available to support them (e.g., research data management and systematic reviews).

While Nel and Fourie's (2016) finding provides insights about information needs of researchers, the discussion of scholarly communication development that may influence the way researchers search for information was not investigated. The transition from print to electronic scholarly communication and its effect on academics' information behaviours was explored by Borrego and Anglada (2016). An online survey was distributed among academics affiliated to the member universities of the Consortium of University Services in Catalonia. A total of 2230 responses were received. Findings revealed that journals were the information resource frequently used for research and teaching purposes. Databases were the preferred starting point for bibliographic searches, although a significant proportion of the respondents indicated a reliance on Internet search engines. The main source for gaining access to documents was the library, followed by free materials available online. Scholarly journals were the preferred mechanism for disseminating research, with open access being a factor of marginal interest in determining where to publish.

Investigation of researchers' information literacy was also conducted by Nicholas et al. (2017), who investigated the information searching, use, sharing, and publication behaviours of early career researchers (ECRs). The study involved in-depth interviews with 116 researchers from seven countries (UK, USA, China, France, Malaysia, Poland, and Spain). The study's findings confirm the popularity of Google and Google Scholar by ECRs in finding scholarly information. The study also revealed that library platforms and web-scale discovery services were largely unmentioned by the participants. Social media was identified as a popular source for finding scholarly information and the study's participants acknowledged the benefits of open access in

delivering free access to papers. The study also identified that smartphones were becoming an increasingly popular mechanism to perform quick searches for scholarly information, but were rarely used for reading full text.

2.7.1.3 Summary

In summary, the eight studies outlined here have helped to expand current understanding of the information literacy of researchers, including doctoral students. However, two important aspects need to be highlighted. Firstly, the studies did not investigate the researchers' information literacy from any perspective other than the behavioural one. Thus, there is an opportunity to explore how researchers relate to their information sources when they conduct research. Secondly, the studies did not mention open access sources as alternative information resources when addressing researchers' information literacy. While it is considered important for researchers to look for a wide range of information sources when conducting research, it may be possible that researchers have found open access articles when they searched for information, hence further research investigating the information literacy of researchers in the open access context is considered important.

2.7.2 Australian studies

To date only two Australian studies have investigated the information literacy of researchers. Williamson, Bernath, Wright, and Sullivan (2007) explored the reliance of research students on electronic information resources. Using Grounded Theory, interviews were conducted with 15 research students (5 masters and 10 PhD) from the Faculty of Information Technology, Monash University. The interview questions explored three main areas: (i) use of information sources; (ii) when to stop finding information; and (iii) collecting and handling the information found. The findings indicated that most of the participants used Google for finding information as well as the databases provided by the university library. The reason for using these two avenues for obtaining information included familiarity, speed, and quality of results. However, even though the participants mentioned that one of the reasons to use Google was because of the quality of its results, this study did not explore how the research students evaluated and analysed the information

they found. Another key resource for the participants was their supervisors. The research students regarded their supervisor as helping them developing a framework for their study.

The study also explored the ability of the research students to recognise that they have enough information for their research and to know when to stop searching for information. Findings revealed that the students who took part in the study did not know how and when to stop finding information. For example, one participant said that the literature review did not finish until they submitted their final thesis, therefore it was necessary to keep looking for information relating to their research until the study was done. Another thing that related to managing information, the majority of research students were using software tools to manage their references. However, one student stated that it was easier for them to read the printed copy rather than the electronic form, even though they encountered difficulties organising the readings when they were printed.

This study provides insight that online information was considered to be an essential information source by students. Google was seen as a starting point for looking at information, followed by academic online databases. However, although mentioning Google and academic online databases as information resources, this study did not explain how the research students used the information found. It was clearly stated that they preferred to use the electronic version as it was easy and simple to use but it did not indicate how the information helped them to learn about their research topic.

Further investigation of doctoral researchers' engagement with information was conducted by Green and Macauley (2007). The study focused on observations drawn from literature and published research, and semi-structured interviews with American and Australian students enrolled in doctoral research degrees, to find out how they learned to gather literature and how these doctoral students learned through the literature process. Data was also collected from an interview with academic librarians who were interacting with the doctoral students, to provide comparative data.

By employing purposive sampling techniques, the study involved four doctoral students and two academic librarians each from America and

Australia. From the interviews conducted with the students, the study revealed that to fulfil their information needs, students were confident in their ability to conduct literature searches using a variety of search engines and subject databases, to retrieve scholarly sources. When the students engaged with information and information systems, they indicated that they were seeking efficiency, effectiveness, appropriateness, and customization to their personal learning styles and research interests. The study also reported that the doctoral students' engagement with information was affected by the experiences and knowledge that they brought to the situation. For example, they mostly appreciated information distributed by others (colleagues or librarians) only if it related to their study.

Green and Macauley (2007) provide insight into American and Australian doctorate students' perspectives on finding information and how they value information. Their study reveals that the participants saw information literacy as an activity when they were doing a literature review, as they sought and used information to build their understanding about their research. One way they sought information was through Google Scholar. This provided insight and opportunity for further study, to investigate why these groups of students preferred to use the free online search engine rather than the academic database or digital repository managed by the university.

2.7.3 Summary

There is a growing body of research exploring information literacy and researchers. To date the majority of studies have emanated out of Europe and the UK. Only two studies have been conducted in Australia, both undertaken ten years ago. The studies critically considered in this chapter have provided new insights into the information use of researchers in different countries, and in different disciplines. The studies however have tended to adopt a behavioural perspective of information literacy (e.g., as skills and knowledge to find, analyse and use information). In addition, none of the studies overtly investigated the researchers' experiences using institutional repositories. Nor did these studies explore how the researchers used open access information for their research.

2.8 Open access and information literacy: The intersection

Two studies have explored the relationship or intersection between open access and information literacy: Zuccala (2010) and Grgic (2016). A discussion of these studies and the implications for the current research follows.

Zuccala (2010) undertook the first study aimed at exploring the intersection between open access and information literacy. More specifically, the study explored what citizens of The Netherlands understood with regard to open access. The study was undertaken to inform Dutch policymakers on potential new approaches for supporting civic scientific information literacy. Twenty-three Dutch citizens participated in a series of four focus groups. A questionnaire was administered prior to the group sessions to obtain demographic information as well as a general assessment of each individual's level of digital and information literacy.

Findings revealed that open access literature could be useful for personal decision making, although there was some concern raised about the cognitive accessibility of scientific research. The study provides evidence that Dutch citizens were becoming familiar with the different types of information available on the Web, but did not necessarily know whether or not their search skills were effective, and could not always decide upon what to believe or not believe. To reap the benefits of open access Zuccalla (2010) recommends that the Dutch government focuses on developing programmes in schools to teach young citizens to develop an appreciation of scientific research and open access, and to develop their capacity to make decisions as online information consumers. The study provides invaluable insights from the perspective of the citizen, and their experience with open access and information consumers. This study is important as it is the first work to acknowledge and empirically explore the intersection between open access and information literacy. Further work is needed to provide perspectives of other user groups, for example researchers. This is a gap the current study will help fill.

The second study exploring open access and information literacy does so in the context of academic libraries. Recognising that academic libraries should play an important role in educating their users about open access, Grgic (2016) undertook a study to find out if, and how, user education about open

access was organised and planned in Croatian libraries. All Croatian academic libraries were invited to have a representative complete an online questionnaire. Forty-nine libraries completed the questionnaire, providing a response rate of 57%. The study revealed that all libraries were involved in open access publishing, either through an institutional repository or through open access journals. However, almost one quarter (22%) of the libraries did not educate their users about open access. The study also revealed that academic libraries recognised that open access literacy should be considered as an aspect of information literacy. In particular, the study showed that the two most important information literacy skills in the context of open access were finding open access information and understanding the benefits of open access citation. This was the first study to overtly discuss the important intersection between open access and information literacy. The study provides invaluable insights from the perspective of the librarian. Further work is needed to provide the perspectives of the users, for example researchers. This is the gap the current study will fill.

The studies by Grgic (2016) and Zuccala (2010) have acknowledged, and begun to articulate, the intersection between open access and information literacy. They have shed initial light on a new phenomenon that needs to be further explored. It is important to note that both studies adopted a behavioural perspective to information literacy. Information literacy was seen as skills for searching, evaluating and presenting information within the context of open access. The current study will build upon and extend the work by Grgic (2016) and Zuccala (2010). It will add a new group or community to explore the phenomenon (e.g., Australian researchers); and it will adopt an alternative theoretical perspective to information literacy by using the relational perspective as outlined by Bruce (1997).

2.9 The Research gap

This literature review has revealed there is a growing body of work exploring open access, especially in the areas of impact and user perspective. Research exploring open access from the perspective of impact has investigated how frequently articles published in open access channels were cited compared to articles published in other channels. However, citation does

not explain how researchers actually use and engage with the articles they obtain via open access. Research exploring open access from the user perspective has tried to fill this gap. However, these studies have focussed only on the attitudes and perceptions of researchers to institutional repositories, the studies have not explored open access in the broader context, for example the green and gold models of open access. In addition, this literature review has also revealed there is a body of scholarly work emerging that explores researchers' information literacy. This work has focused on understanding the information needs of researchers including how they find, access and use information as part of their research work. While open access has emerged as a topic of consideration in some of these studies it was not a primary focus. This work has also been mostly undertaken in the UK and Europe.

The current research addresses the gaps in previous studies by (i) undertaking a study in Australia; (ii) adopting a relational perspective to information literacy; and (iii) investigating both information literacy and open access. By doing this, the research adds to the growing body of knowledge on researchers and their experience.

2.10 Conclusion

This chapter has presented a discussion of literature concerning open access and information literacy and the intersection of both constructs in the context of researchers' experience. The next chapter describes the research methodology that was used for this study. More specifically, the chapter outlines the qualitative approach which involved the use of semi-structured interviews and thematic analysis that enabled an exploration of how Australian researchers experience open access as part of their information literacy.

Chapter 3: Research methodology

3.1 Introduction

The main purpose of this study is to explore how Australian researchers experience open access as part of their information literacy. This chapter describes the research methodology that was used for this study. It begins with an overview of the research paradigm that underpins the study. This is followed by the rationale for the appropriateness of qualitative method as the research approach. Details of the research design and implementation, including sample and recruitment, data collection techniques, and data analysis processes are outlined. The final part of the chapter discusses the ethical considerations that were adhered to in conducting the study.

3.2 The interpretive paradigm

Researchers have different beliefs and ways of viewing and interacting with their surroundings. As a result, the way researchers conduct their studies can vary greatly (Kinash, 2010). There are certain models and rules that guide a researcher's actions and these standards can be referred to as a paradigm. Creswell (2013, p. 6) noted that a research paradigm is a general philosophical orientation about the world and the nature of research that a researcher brings to a study. It is a set of beliefs and feelings about a specific context that should be understood and studied (Lincoln & Guba, 1985). These beliefs will lead to embracing a specific research design (Creswell, 2013). A research paradigm informs the selection of research procedures, including the method of data collection, analysis, and interpretation (Creswell, 2014).

This study is positioned within an interpretivist paradigm, a research paradigm that assumes the existence of a gap between human beings and their world, things, and processes that happen in their world (Humphrey, 2013.) The interpretivist starts from the premise that the way people experience their world is the representation of their consciousness about themselves, others, and the world they experience (Humphrey, 2013).

Interpretivism deals with subjective experiences, perceptions, and meanings that people assign to their world (Denzin & Lincoln, 2005). Its complexity is reflected by the interaction of each individual with themselves, society, and culture (Black, 2006).

This philosophical view has implications for methods and methodology for research. The most common research method in the interpretivist paradigm is the interview, guided by a topic chosen by the researcher but conveyed by open-ended questions in order to allow themes based on the participants' experience to emerge from the conversation (Silverman, 2011; Humphrey, 2013). The interview helps the researcher to develop their understanding of the life-story of the people being interviewed (Silverman, 2011).

This study relies on analysing and interpreting the participants' experience of open access as part of their information literacy, making it an interpretivist study focussed on obtaining a deeper understanding of the topic. Thus, this study employs a qualitative method to explore academics' opinions and perspectives about their experience with open access as part of their information literacy. The aim of the study is not to examine research hypotheses nor to look for causal relations. Therefore, the interpretivist paradigm was considered appropriate for the study.

3.3 Research method

The research method is a strategy of inquiry which moves from the underlying assumptions to research design and data collection (Myers, 2011). The most common classification of research methods is qualitative and quantitative (Creswell, 2013).

Quantitative research methods were originally developed in the natural sciences to study phenomena (Bryman, 2012). While qualitative research methods were developed in the social sciences to study social and cultural phenomena (Creswell, 2013). Quantitative work is expressed in numbers and in statistical models, whereas qualitative work is expressed in natural language. Quantitative work is based on large samples, whereas qualitative work employs small samples. According to Myers (2009), qualitative research is designed to

help researchers understand people, and the social and cultural contexts where they live. Qualitative studies allow the complexities and differences of people's experience of their world to be explored and represented (Philip, 1998).

Qualitative research is naturalistic (Denzin & Lincoln, 2003). It studies the everyday life of different groups of people and communities in their natural settings. Qualitative research involves an interpretive, naturalistic approach to its subject matter. It attempts to make sense of, or to interpret, phenomena in terms of the meaning people bring to them (Denzin & Lincoln, 2003). In addition, qualitative research aims to explore and discover issues about the problem on hand, when very little is known about the problem. There is usually uncertainty about the dimensions and characteristics of the problem (Domegan & Flemming, 2007). According to Myers (2009), qualitative research is designed to help researchers understand people and the social contexts within which they live.

Conversely, quantitative research is more theory driven and aims to generalise from the sample being studied (Riazi, 2016). Quantitative methods use numerical or measurable data with accompanying statistical analysis to answer the research question. Objective reality amenable to cause and effect relationships is investigated for explanation. This is usually achieved through hypothesis testing using quantification and statistical analysis of the data, to make inferences about the validity and generalizability of the explanations reached at the sample level (Riazi, 2016).

For data collection, quantitative research uses surveys and questionnaires where the participants are asked questions in the same order. The questions usually are closed questions. Qualitative methods are more flexible because they offer spontaneity and an interaction between the researcher and the participant, therefore in a qualitative approach the questions asked are open-ended and are not necessarily asked in the same way for each participant (Mack, Woodsong, Macqueen, Guest, & Namey, 2005). Hence, semi-structured methods such as in-depth interviews, focus groups, and participant observation are common for qualitative research (Myers, 2009).

In a qualitative method, the participants have the opportunity to respond more elaborately and in more detail than in the quantitative method. This means the researchers have the opportunity to interact directly with the participants, through probe questions responding to information the participants have provided (Mack, Woodsong, Macqueen, Guest, & Namey, 2005).

3.4 Rationale for choosing qualitative method

In selecting a research method Lincoln & Guba (1985) suggest choosing an approach that is most relevant to the phenomenon being studied. The purpose of this study is to explore and understand the ways people experience a particular phenomenon (i.e., open access as part of information literacy), consequently, based on the commentary above, a qualitative method was employed.

A qualitative research method is used when the results are intended to provide detailed description about context, activities, and processes (Speziale & Carpenter, 2007). It is a method used to uncover the meaning of a phenomenon from the perspective of those who are involved in the phenomenon. Therefore, qualitative researchers are interested in "understanding how people interpret their experiences, how people construct their world, and what meaning they attribute to their experiences" (Merriam, 2009, p.5). In addition, given the lack of research on the topic to date, a qualitative approach was appropriate for addressing the study's stated research question (Taylor & Bogdan, 2015).

Qualitative research methods involve qualitative data collection (e.g., interviews) to understand and explain social phenomena (Myers, 2011). The researcher might go into the 'field', for example interviewing people in their professional or social setting. This approach to data collection provides the researcher with the opportunity to interact directly with the participants by having a direct conversation, asking questions, and asking probe questions responding to information the participants have provided (Mack, Woodsong, Macqueen, Guest, & Namey, 2005). In this sense, the qualitative method is more 'flexible' because it offers spontaneity and an interaction between the investigator and the participants (Mack, Woodsong, Macqueen, Guest, &

Namey, 2005). The strength of the qualitative approach is that it illuminates issues and reveals possible explanations (Gillham, 2000). Furthermore, Gillham (2000, p.11) notes that a qualitative approach enables the researcher "to get 'under the skin' of the participants" to find out what is really happening.

The purpose of analysis in qualitative research is to inquire deeply into the meaning of different situations, and different participants' experiences and understandings of the phenomenon under study (Creswell, 2013). For this reason, Braun and Clarke's (2006) thematic analysis was used for data analysis. This is important in the current study as the unit of analysis is the lived experience of the participants.

This study is about academics' experience of open access as part of their information literacy. Very little empirical research has currently explored this phenomenon. This study did not have a hypothesis to begin the research, rather it employed inductive data analysis in order to explore participants' experience of open access as part of their information literacy. The focus of the study was on the participants' experiences, perceptions and meanings of their engagement with open access sources as part of their information literacy, consequently qualitative approaches were most appropriate to describe the interrelationship of the participants' experience of open access sources as part of their information literacy.

3.5 Research design

This section focuses on the design of the study. It provides details on participant selection, sampling approach, and the use of semi-structured interviews for data collection.

3.5.1 The Research Phenomenon

The aim of this study was to explore Australian researchers' open access experience as part of their information literacy. Hence the phenomenon being explored in this study is *open access and information literacy*.

In this study open access refers to free and unrestricted online access to peer-reviewed research articles without financial and legal barrier

(Budapest Open Access Initiative, 2002; Bailey, 2007; Willinsky, 2006; Suber, 2015). Researchers using open access disseminate their research in two main ways, gold and green (Harnard et al., 2004). The gold route means the papers are published by journals (Suber, 2004; Norris, Oppenheim, & Rowland, 2008), while the green route refers to an article, usually a pre-print version, made freely available through the Internet, for example via the author's personal website, or an institutional repository (Bjork, Laakso, Welling, & Paetau, 2013).

In this study, a relational perspective of information literacy is adopted (Bruce, 1997). This is the perspective of information literacy that focuses on the various ways people relate to information and interact with elements of their information worlds. The relational perspective moves away from focussing solely on skills and behaviours. By including a focus on experience, for example, attitudes, beliefs and emotions, the relational perspective reveals intricate and nuanced elements of information literacy. Through this perspective the study does not seek to measure information literacy but instead seeks to explore and understand the complex (i.e., behavioural and experiential) relationship between people and information and their information worlds.

3.5.2 The participants

The population of the study were researchers who worked in Australian universities. In accordance with the aim of the study, two criteria were considered for the recruitment process of study participants.

The first concerned their employment status; only individuals who were academics currently employed (sessional, contract or full time) were approached for the study. Full time research students were not approached. The focus of the recruitment was academic staff with the consideration that they had already completed an adequate number of research projects, and therefore it was considered that the academic staff would have an extensive experience of research, which lead to experience in finding and publishing their research.

The second criterion concerned researchers' experience with open access, thus only researchers who had engaged with open access in the previous twelve months were recruited. As this study was exploring the lived experience of researchers using open access as part of their information literacy, it was important that the participants had some degree of engagement with or use of open access sources. This engagement or use could be selecting, evaluating, or publishing articles in open access sources. A 12-month time frame was determined to ensure that the participants could speak in a meaningful way about their experiences.

This study population was not limited to specific disciplines, because the aim of the study was not to explore the nature or application of open access within information literacy experiences for specific groups of researchers (i.e., health, history, or business). Therefore, for the purpose of this study, researchers from all disciplines were targeted for data collection.

3.5.3 Sample and recruitment

There are many different approaches to sampling that can be used in research involving interviews. Two of the more common are purposive and snowballing which were used in the current study.

Purposive sampling is a non-probabilistic sampling technique that requires the researcher to select sections of a population that show some aspect that is required for the study (Cohen et al., 2007; Silverman, 2005). Purposive sampling is widely used in qualitative study to engage participants that it is expected can provide detailed information related to the phenomenon (Patton, 2001). The process involves identifying and selecting individuals that are knowledgeable about and experienced with the phenomenon being studied. Bernard (2002) mentions that in addition to knowledge and experience, availability and willingness to participate in the study are also important. The strengths of the purposive approach are that it is inexpensive, convenient and timely. However, the weakness lies in the subjective nature of the approach (Neuman, 2012). Purposive sampling was employed in this qualitative research to achieve the required depth of understanding of the phenomenon being experienced by the participants.

Snowball sampling is when the researcher approaches other participants based on people the study's existing participants have referred to. Snowball sampling was appropriate for this study because the technique is designed to enhance access to participants from a population that may be hard to approach due to low numbers of potential participants (Berg, 1988).

There are no set rules about the actual sample size required for qualitative data collection, that is, how many participants are sufficient to guarantee the validity and generalizability of findings from interviews (Morse, 2000). Determining an appropriate sample size can be influenced by the research methods being employed, as well as other factors such as the scope of the study, nature of the topic, and quality and amount of data obtained from each participant (Morse, 2000). The sample size is perceived to be appropriate and adequate for a qualitative study if it adequately responds to the research question (Marshall, 1996). In other words, sampling must be continued until complete explanation of the phenomenon or saturation of the concepts is achieved. Generally, a reasonable sample for interviews is between 20 and 50 participants, to ensure a rich and complex data pool has been gathered (Morse, 2000). Douglas (1985) contends that interviews with at least 25 people are necessary before reaching saturation point, while Bertaux and Laso (cited in Guest et al., 2006, p. 61) note that "fifteen is the smallest acceptable sample size in qualitative research". With this in mind, there was no specific target sample size for this study, however it was expected that between 15 and 50 participants would be required. It was eventually determined that sufficient data had been collected to address the research question after 31 participants were interviewed. Profiles of the participants who took part in the study are provided in Chapter 4.

Two key strategies were used to recruit participants. Firstly, Heads of School[1] from 39 Australian universities were contacted about the study. The Heads of School's email addresses were collected from university websites. They were found under the faculty and school web pages. The Heads of School were introduced to the project and provided with a recruitment flyer to be

[1] Australian universities are made up of Faculties and Schools, and the Head of School is responsible to the Dean of the Faculties for the academic leadership and management of the school.

distributed to academics in their school (see Appendix A). The Heads of School were selected as the first point of contact because of their management role which could provide the required permission for participant recruitment within their schools.

The recruitment materials invited academics who were interested in the study to contact the lead researcher about their willingness to participate and to arrange a suitable time and place for the interviews. Participants who lived in Brisbane, Australia were interviewed in person, while those who did not live in Brisbane were interviewed through Skype or phone.

In total, 710 Heads of School were emailed to ask for permission for their school to be part of the study, and to distribute the recruitment flyer to academic staff. The 39 Universities that were contacted during the study are provided in Table 2. All Universities were members of Universities Australia, the national peak body for higher education institutions.

No	States	University
1	New South Wales	Charles Sturt University
2		Macquarie University
3		The University of Newcastle
4		The University of New England
5		The University of New South Wales
6		The University of Sydney
7		University of Technology Sydney
8		University of Wollongong
9		Western Sydney University
10		Southern Cross University
11	Queensland	Queensland University of Technology
12		University of Queensland
13		Griffith University
14		University of Southern Queensland
15		Central Queensland University
16		James Cook University
17		University of Sunshine Coast
18		Bond University
19	Victoria	Monash University
20		Deakin University
21		Swinburne University
22		Victoria University
23		La Trobe University
24		RMIT
25		The University of Melbourne
26		Federation University Australia
27	South Australia	Flinders University
28		University of South Australia
29		The University of Adelaide
30	Western Australia	Curtin University
31		University of Western Australia
32		Edith Cowan University
33		Murdoch University
34		University of Notre Dame
35	ACT	Australian National University
36		University of Canberra
37		Australian Catholic University
38	Tasmania	University of Tasmania
39	Northern Territory	Charles Darwin University

Table 2. List of Australian universities contacted for recruitment

The main difficulty encountered in recruiting participants was the low response from academics. The Heads of School generally responded positively to the initial contact and were willing to distribute the recruitment flyer to the academics in their schools. However, only 13 academics took up the offer to participate in the study. An additional challenge was ensuring that the academics who agreed to participate fully met the selection criteria, especially the requirement to have engaged with open access in the previous twelve months.

As a result of the low participation rate, a second recruitment strategy was used. The second strategy involved approaching Australian academics who had published in open access journals in the previous twelve months. *PLoS One* was used as it was frequently mentioned by the participants who had already participated in the study. In addition, suitable open access journals were identified through the Elsevier's site of open access journals (https://www.elsevier.com/about/open-science/open-access/open-access-journals). The Elsevier site was used because it included open access journals from a broad range of fields, including natural sciences, medicine, engineering, and the related social sciences and humanities. Their open access journal site was then selected and identified for Australian authors, mostly for papers authored by Australian academics that had been published in the 12 months prior to the interview. The list of open access journals for the Australian authors identified is not provided in this report to respect the anonymity of the participants, however it may be provided upon request.

Forty Australian authors were identified. A recruitment email and flyer was sent to the authors. Seven authors agreed to be interviewed for the study. One of these participants was interviewed face to face, and the other six participants were interviewed via Skype.

3.5.4 Interviews

The purpose of the interview is to explore the views, experiences, and motivations of people on specific matters (Gill, Stewart, Treasure, & Chadwick, 2008; Kvale, 2007). Through the interview, the researcher has a "unique opportunity to uncover rich and complex information" (Cavana, Delahaye, & Sekaran, 2001, p. 138). This is because the research participants

are able to tell their own story in their own words. Rubin and Rubin (1995) suggest that an interview is "a great adventure... it brings new information and opens windows into the experiences of the people you meet" (p. 1).

Interviews were an appropriate choice for this study because of their suitability in obtaining information about participants' lived experiences (Arskey & Knight, 1999). The researcher was able to acquire a deep understanding of the phenomena through this "powerful method for capturing the experiences and lived meaning of the participants' everyday world" (Gill, Stewart, Treasure, & Chadwick, 2008; Kvale, 2007, p. 11). Interviews, however, have both advantages and disadvantages. Whilst they provide an invaluable insight into people's everyday lives, they can be limited by the small number of respondents that participate, the limitations on generalizability to a larger population, and the bias of the interviewers' influence and interests (Kvale, 2007). Every effort was made to strengthen the advantages and to limit the disadvantages of the interview approach used in this study.

Patton (2002) defines interviews as having three different types: informal conversational interview, interview guide approach, and standardized open-ended interview. Each of these has a particular meaning as well as its strengths and weaknesses.

The first type of interview is the informal conversational interview where questions are asked in a natural setting without any predetermined questions. The questions usually emerge from the immediate context. As the interviews proceed from observations, they can be matched to the participant's circumstances. However, the drawback of this type of interview is the interview is less systematic and comprehensive, hence it may cause difficulties in data organization (Patton, 1987).

The second interview type is the interview guide approach, where the questions have been prepared and outlined in a sequence. This kind of semi-structured interview offers more comprehensiveness and any logical gaps can be anticipated (Patton, 1987). However, Patton suggests that by being structured an important topic may be missed, especially if the interviewer only follows the sequence of questions that has been prepared. Therefore, it

might be necessary to use probe questions to pursue emergent topics that may come from the interviewee (Gillham, 2000).

Patton (2002) suggests another type of interview, which is a standardized open-ended interview. This interview uses a list of questions that have been prepared in a systematic sequence, and the interviewer does not clarify or engage in follow-up or probing of the answers provided by the participants. The participants are asked the same the questions in the same order. However, it makes it difficult to standardise such interviews with all the participants because each might have a different context (Preece, Rogers, & Sharp, 2002).

This study used the interview guide approach, because of the flexibility in combining the formal and informal aspects of the conversation. The questions were designed and structured prior to the interview, and there was also a chance for the interviewer to follow up responses given in order to keep the conversation focussed on the topic being discussed. The informal conversation that rises during the interview is useful to anticipate any gaps for data collection and to clarify different perceptions that come from the participant. Since the purpose of the study was to explore the participants' experience of open access as part of their information literacy, planned questions were considered necessary to manage the interview process and to obtain a deep and broad range of information (Marshall & Rossmann, 2011).

3.5.5 Developing the interview guide

The development of the interview guide is an essential part of conducting the interview guide approach for data collection. The interview guide directs the interview process, ensures that all important topic areas are covered, and helps to manage time effectively during the interview (Marschan-Piekkari & Welch, 2004). The full interview guide for this study is provided in Appendix D. The questions were revised and modified after conducting a pilot study (see Appendix F). The questions used in the main data collection were:

1. What research have you been working on over the last 12-month?
2. Can you tell me about how you find information for your research?

3. Can you tell me about a time when you used open access sources to find information for your research?
4. Can you tell me about a time when you published in open access?
5. How would you define open access?

The purpose of Question 1 was to provide the participants with the opportunity to describe their current research work. This question helped to establish a conversational tone (Magnusson, 2015) where the participant was invited to begin by discussing a topic they would feel comfortable to discuss.

Through Questions 2 and 3 participants revealed their information literacy experiences as relevant to their research work. Through these questions participants revealed how their accessed, evaluated, and used information relevant to their research, as well as how they shared their work with others. Additionally, as this research was informed by the relational perspective of information literacy which emphasises the relationship between people and information, and the way people experience the world, the questions were designed to reveal both the behavioural and experiential aspects of participant's information literacy experience. To tap into the behavioural aspects, participants were asked about the actions or activities they undertook. To tap into the experiential aspects, the participants were asked about their preferences and attitudes to their information literacy experiences.

Question 4 revealed the participant's understandings of open access, while Questions 5 and 6 served the purpose of bring the two parts of the phenomenon together by focussing the conversation on the participants' experiences, perceptions, and understandings of open access as part of their information world.

Adding to the main questions, follow-up questions were asked during the interviews to clarify the story that the participants told. Follow-up questions are useful to "fill out" participants' stories (Magnusson, 2015, p. 54) so that the interviewer receives clear and complete information about the topic being discussed. Some follow-up questions that were used during the interviews were:

1. Can you tell me more about that?
2. What did that make you think about that?
3. What do you mean by that?

3.5.6 Conducting the interviews

In a qualitative study, the interview guide approach is a particular kind of conversation between the interviewer and the interviewee that requires active listening and asking specific questions and follow-up questions that relate to the topic being discussed (Hesse-Biber & Leavy, 2011). The topic of this study's interviews was the experiences of the participants in using open access as part of their information literacy. Before the interview, participants received a Participant Information Sheet (Appendix B) that introduced the study and the purpose of the interview. This is an important part of the research process as it provides the participant with a clear explanation of the topic being explored (Galletta, 2013). On the day of the interview, the researcher also reads an introduction to the interview, that explains the intent of the study, outlines how the interview would be conducted, and provides ethics considerations for the participants, for example, the anonymity of responses provided and their right to withdraw from the study at any time. The participants are also told that they may take as long as they feel is necessary to tell their story (Staples, 2012). Before formally starting the interview, participants were invited to ask if they had any questions or needed any further clarification. The participants also received a Consent Form to complete.

The participants were given a choice of day and time most convenient for them to undertake the interview. The interviews were conducted in a variety of ways. Participants who were located outside of Brisbane where interviewed through Skype or phone. Twelve participants were interviewed face to face, six participants were interviewed over the phone, and thirteen interviews were conducted through Skype. All interviews were audio recorded and then transcribed. This is commonly used in interviews as it contributes to a more relaxed atmosphere because the interviewer is freed from the distraction of note-taking and can concentrate on asking questions and actively listening (DiCicco-Bloom & Crabtree, 2006). The interviews ranged from 35 and 50 minutes in duration.

A reflective memo was written by the researcher for each interview. Writing a memo has the benefit of "capturing moments of insight" (Blumenreich, 2016, p. 69). The reflection was intended to improve the quality of the subsequent interviews and served as initial phase of analysis. Ely (1997) suggests that the reflective memo is useful to help the researcher shift thinking during the data collection process. The memo was written after each interview about what had transpired, including the noteworthy features about the content, or participants' expressions or emotions on some issues. For example, the researcher noted some comments from several participants who mentioned their use of social networks for sharing research, or comments from some participants about how they understood open access.

There are some essential steps that need to be considered to conduct an effective interview guide approach. A fundamental principle of the interview guide approach is to listen. It is about trying to hear the meaning of what participants are saying, understanding where there is a subtext that needs to be explored and hearing the nuances in the participants' accounts (Galletta, 2013).

Follow-up questions were used for the interviews to explore participants' responses. Follow-up questions were only asked following the interviewee's answer (DiCicco-Bloom & Crabtree, 2003). The researcher used a pre-determined list of thematic probe questions to help decide what kind of follow-up question might be asked related to the participant's answer. Listening carefully to the participants' responses was also helpful to remember points that needed to be followed up at a later stage. One response from an interviewee may trigger two or three points to probe (Magnusson & Marecek, 2015). A swift decision was taken several times with regard to the immediate issue that needed to be followed up. A short note was also written to return to the other issues raised, either once the conversation dealt with the immediate issue or later in the interview when the interview dealt with a relevant topic (Magnusson & Marecek, 2015).

3.6 Data analysis

The purpose of this section is to describe general principles that underpin the data analysis conducted, and to outline the process undertaken for this study.

3.6.1 Data analysis: overview

The purpose of analysis in qualitative research is to inquire deeply into the meaning of different situations and different participants' understanding of the phenomenon (Creswell, 2013). For this reason, thematic analysis was used in this study (Braun & Clarke, 2006). Thematic analysis is a method for identifying and analysing patterns or themes within the data, that are considered to be important to the descriptions of the phenomenon being studied (Braun & Clarke, 2006; Fereday & Muir-Cochrane, 2006). This is important in the current study because the unit of study is the lived experience of the participants regarding a particular phenomenon (i.e., open access as part of information literacy).

3.6.2 Data analysis: process followed

The analysis was implemented in three phases. These phases included familiarisation with the data, generating codes, and theme identification (Braun & Clarke, 2006). The remainder of this section provides a discussion of each of these phases.

Phase one: familiarisation with the data

In this first phase of analysis the primary aim was to reach a basic level of familiarisation with the transcripts and to begin the process of immersion in the data. To achieve this, all 31 audio recordings from the interviews were transcribed verbatim to ensure that all information was included. A verbatim transcription is useful for data analysis as it helps the researcher understand the data better (Halcomb & Davidson, 2006). Transcription of the interviews can be an effective way to start familiarizing yourself with the data (Riesmann in Braun & Clarke, 2005). The participant's privacy was assured through the use of pseudonyms, for example, Participant 1. Any other persons or locations

mentioned by name were removed and replaced with text such as “University” or “Institutional Repository”.

The transcripts were read and re-read a couple of times to obtain a thorough understanding of the data (Bluemenreich, 2016). Braun and Clarke (2006, p. 87) suggest that “this phase will provide the bedrock for the rest of the analysis”. A summary paragraph was written after reading each transcript, containing the reflections of the researcher about the data contained in the transcript.

Phase two: generating codes

The second phase in the thematic analysis is generating coding for the data. Codes identify a feature of the data that appears relevant to the phenomenon (Boyatzis, 1998). To ensure the process of generating coding focuses on the phenomenon under the study, the research question (i.e., how researchers experience open access as part of their information literacy) was written on a piece of paper and kept visible while the process of generating codes was taking place. This act was a reminder to the researcher that the objective of the coding was to focus on the phenomenon under study. The phenomenon in the current study was participants’ experience of open access as part of their information literacy. As this research was informed by the relational perspective of information literacy (Bruce, 1997), which emphasises the relationship between people and information and the way people experience the world, the data analysis was undertaken with the intent of revealing both the behavioural (i.e., skills and knowledge) and experiential aspects (i.e., preferences, attitudes, feelings). The process of coding involved initial coding, and reviewing and modifying the codes (Braun & Clarke, 2006). A brief description of each will follow.

Initial coding

Initial coding refers to the process of generating codes from the data, that are relevant to the phenomenon being studied (Braun & Clarke, 2006). This process involved reading the interview transcripts. Reading the transcripts thoroughly reveals possible patterns within the data (Braun & Clarke, 2006). The initial coding was done in MS Word. MS Word was chosen as a tool for

doing the initial coding as the interview transcripts were already in MS Word, hence there wasn't any process to import or export the data. The availability of MS Word in most computers allowed the researcher to work anywhere, for example at the researcher's office or at the library.

The researcher used an inductive approach with the purpose of condensing and organising the data into patterns of meaning. The inductive approach to coding refers to generating codes that "derive in what is in the data" (Braun & Clarke, 2012, p. 58). However, during the coding process it is possible that the existing concepts and knowledge of researcher influence the researcher when generating the codes. Braun & Clarke (2012) suggest that it is almost impossible to do a pure inductive approach when generating codes because researchers may bring something to the data when they analyse the data.

In this initial coding stage, in-vivo codes were generated where possible, this refers to the act of generating codes from the actual words used by the participants, as such this ensures the codes generated remain close to the data and can provide the key elements that are important to the phenomenon being studied (King, 2008). As the coding progresses, and when some key concepts related to the topic of the study emerged, the data were reviewed more thoroughly.

After initial codes were developed for each transcript, the transcripts were shared with the researchers' supervisors. The initial codes were revised based on feedback and questions received from the supervisors. This process helped ensure consistency and reliability in the initial coding process. Once the initial codes were finalised a codebook was developed. The codebook was an important element for the analysis in this study, as it was used to organise the codes following the interpretations (Crabtree & Miller, 1992). The codebook was also useful to assist the analysis of the coding and counting the frequency for each code in order to identify patterns related to the phenomenon being studied (Crabtree & Miller, 1992). Table 3 (overleaf) shows an excerpt of the code book from the initial coding.

Codes	Total number of references	Number of interviews used in	Participant 1		Participant 2	
			Number of references	Line number	Number of references	Line number
Acknowledge the authorship	1	1				
Academic research should be accessible	1	1				
Access difficulties	1	1	1	240-245		
Accessible for lay person	1	1				
Depositing open access copy	1	1			1	172-179
Invited to publish non-open access	1	1				

Table 3. Excerpt of codebook documenting the initial codes

Following the initial coding, codes that had similar features were grouped into higher order categories. For example, it was noticed that several codes such as allocating funds and paying author processing charges could be moved to a broader category such as paying publication fees. The identified broad categories or groups were named according to their content. Table 4 shows groups of codes organised by their content.

Groups	Some codes included	Description
Commitment to open access	• Academic research should be accessible • Accessible for lay person • Breaking down research dissemination barriers • Depositing open access copy • Important of open access for poorer countries • Making information accessible • Moral imperative • Regret to publish to non-open access journal	The quote indicates the participant is committed to open access as a cause, or is committed to publishing open access
Defining open access	• Completely free publications • Creative Commons License • Different level of open access • Download ready • Easy access • Formal open access • Free scientific publication	Participants' personal understanding of open access

Deposit processes	• Aware of repository reminder • Difficult to use repository • Depositing legality • Extracting publications automatically • Prefer not to use repository • Research support assistance • Too many process for updating profile	Data that refers to aspects of the process of depositing in an institutional repository generally

Table 4. The excerpt of groups of code

Reviewing the codes

The initial processes of coding and organisation of codes resulted in creating a hierarchy of coding groups. In this phase, the groups created and the codes allocated to each group were reviewed to determine whether the codes were located in the right group. Several codes were moved to new places or merged with other similar codes, irrelevant codes were deleted, and overlaps were removed. At the end of this phase, 43 groups were identified. A sample of the groups is provided below and the complete list of the groups is provided in Appendix F.

A sample of the coding groups:

- Commitment to open access
- Finding information resources
- Finding open access materials
- Funder requirement for open access
- Identifying institutional repository benefit
- Identifying open access benefit
- Identifying open access journals
- Keeping up-to-date
- Making work accessible
- Monitoring citations of own work
- Motivations for depositing
- Motivations for not publishing open access
- Motivations for publishing open access
- Navigating copyright issues

- Paying publication fees

During this step, memos were written to record personal reflections about the findings of the study as well as to document the process of data analysis.

Phase three: theme identification

The third stage in thematic analysis deals with identifying key themes of the study. Themes refer to patterns in the data that explain and organise aspects from the phenomenon derived from the data (Boyatzis, 1998). The codes and groups that had emerged in the previous phase were reviewed and examined multiple times against the main research question: "*How do Australian researchers experience open access as part of their information literacy?*", to determine the themes.

Tentative themes were identified based on similarities and differences observed between the groups and codes. Groups that appeared to have similar focus or meaning were collated together to create tentative themes. During this process of identifying tentative themes the transcripts were continually reviewed to ensure that the data referenced in the groups and codes formed a coherent pattern within the tentative theme. Each tentative theme was also compared to the other themes to discover similarities, differences, and relationships between them. Once the full list of tentative themes was identified, each theme was weighted according to their significance, relevance to the research question, and distinctiveness to be included as one of the themes of the study. At the end of this process four themes were identified:

1. Understanding Open Access
2. Institutional Repositories
3. Publishing and Dissemination
4. Finding information

Finally, using the memos recorded during the analysis, relationships between the themes were determined resulting in a thematic map, a conceptual map of the themes identified in the study (Braun & Clarke, 2006). A complete story of the findings is presented and discussed in Chapter 5.

3.7 Maintaining quality

This study uses a qualitative approach that seeks to understand phenomena in a context-specific setting. The researcher uses their interpretations during the process of the study that hence may cause bias (Sandberg, 2005). In response, it is important that the researcher acknowledges this potential bias to ensure the trustworthiness of the research (Creswell, 2009).

To maintain trustworthiness, this study applied some steps suggested by Lincoln and Guba (1985), thus establishing the study's credibility, transferability, dependability, and confirmability.

Credibility relates to the research effort to link the study's findings with reality in order to validate the truth of the findings. To ensure the credibility of this study's findings, the researcher worked together with the supervisory team during the process of data analysis. By involving the supervisory team, the influence of the researcher's subjectivity and bias was reduced.

To ensure transferability of the study, Given (2008) suggests that the study's findings should be applied across all environments related to the context being studied. Given (2008) further states that to increase the transferability of the study findings, the research should focus on two considerations: first, the closeness of the relationship of the participants with the context being studied; second, the alignment of the research question with the context being studied, ensuring that the research question is appropriately answered.

To respond to the first consideration, the participants of the study were selected through criteria that closely related to the context being studied. The participants were currently employed in Australian universities and had used open access sources in the preceding 12 months while conducting research. To respond to the second consideration, the research question was informed by both the existing knowledge base nationally and internationally regarding the phenomenon of the study, and the current Australian higher education context and research agenda and priorities.

Dependability refers to showing that the findings are consistent and can be repeated. This study followed the suggestion by Lincoln and Guba (1985) to have an external audit to evaluate the accuracy of the findings and also to evaluate the analysis process. However, instead of having an external body do the audit, the analysis process and the findings were both closely supervised and assessed by the supervisory team. Both members of the supervisory team were able to guide the accuracy of the study by having discussions during the process of data analysis.

Similarly, in regard to confirmability, which is to show the degree of the neutrality of the findings, this study followed the recommendation by Lincoln and Guba (1985) to have a confirmability audit focused on the analysis process. Instead of having an external body do the audit, the members of the supervisory team were able to provide advice during the analysis process to ensure that the findings were formed by the participants of the study and not by researcher interests.

In addition, Miles and Huberman (1984) suggest a technique to maintain the reliability of the analysis and the interpretation of a qualitative study is by double coding. In an instance, two or more of the research team have a discussion during the process of the study. Following this, during the process of data analysis, the researcher and the supervisory team engaged in ongoing discussions. By having the supervisory team and the researcher read the same interview transcripts, they were able to check the interpretation of the data. Second, during the process of research there were several opportunities when the research was presented (outlined below), to check the communicability of the study and its results.

1. Confirmation Seminar, Queensland University of Technology, Brisbane, March 2015
2. Presentation of Pilot Study, 8th International Evidence Base Library and Information Practice (EBLIP) Conference, Brisbane, Australia 6-8 July 2015
3. Presentation of the Study, QUT Information Systems School Doctoral Consortium, November 2015

4. Presentation of the Study, QUT Information Systems School Doctoral Consortium, November 2016

By delivering the study to these wider audiences, the researcher was able to receive feedback and ideas about the study being undertaken.

3.8 Research ethics statement

Full ethical clearance was obtained from the QUT Human Ethics Committee (QUT Ethics Approval Number 1500000045). The following ethical considerations were adhered to in the study:

- Voluntary participation was respected at all levels of the study
- A consent form was provided to every participant
- Participants could withdraw from the study at any point in time without penalty
- Any potentially identifying information was removed from the transcripts
- Audio recordings were destroyed once the research was completed.

3.9 Conclusion

This chapter has presented a qualitative method as the approach that was selected for the study. It has outlined the key features of semi-structured interviews and thematic analysis. The following chapter, Chapter 4, provides profiles on the 31 participants who took part in the study and contributed to the co-creation of the study's findings.

Chapter 4: Introducing the Participants

4.1 Introduction

The previous chapter, which discussed the research approach, detailed the strategies used for the selection of participants and the questions asked at the interviews. This chapter introduces the participants. Thirty-one Australian researchers were interviewed for this study. The findings incorporate their descriptions and lived experiences. In this way, the participants are co-creators of meaning (Smith, 2008). Each of the researchers will be introduced with their profile as an acknowledgment of their participation in relating the stories they told to frame the findings for this study.

4.2 An overview of the participants

The participants of this study are Australian researchers employed (sessional, contract or full-time) by Australian universities. The total number of 31 participants who took part in this study come from different states across Australia. Below is the state they come from and number of participants who participated in the study. Following that, the participants' discipline area is presented in Table 5 (overleaf).

- New South Wales: 8 participants
- Queensland: 14 participants
- Victoria: 4 participants
- South Australia: 2 participants
- Western Australia: 2 participants
- Australian Capital Territory: 1 participant

Participant	Discipline area
1	Technology
2	Science
3	Science
4	Science
5	Health and Medicine
6	Science
7	Social Science
8	Health and Medicine
9	Social Science
10	Health and Medicine
11	Social Science
12	Health and Medicine
13	Social Science
14	Arts and Humanities
15	Social Science
16	Health and Medicine
17	Health and Medicine
18	Social Science
19	Science
20	Health and Medicine
21	Health and Medicine
22	Science
23	Social Science
24	Social Science
25	Technology
26	Arts and Humanities
27	Health and Medicine
28	Health and Medicine
29	Science
30	Health and Medicine
31	Health and Medicine

Table 5. Participants' discipline areas

4.3 The profiles

Participant 1 spoke about an open access source that they used a lot when they undertook research. This open access source provided high quality information. They also used the institutional repository for finding information that related to their projects. Further, they mentioned that the repository was a useful tool that helped them to collaborate with other researchers. It also allowed the participant to find colleague's publications. Participant 1 deposited their research into the institutional repository in part because they realised their work was easy to find if stored in the institutional repository.

Participant 2 made two of their publications open access with the library support. That was the extent of the participant's engagement with open access publishing. However, Participant 2 was very concerned with journals that kept sending them invitations to publish, particularly journals that offered a low price to publish, which they identified as predatory journals. They believed that these journals were damaging the reputation of open access journals in general. They also said that all of their publications were deposited into the institutional repository. Visibility and easy access were the main reasons they were willing to put their research in the repository.

Participant 3 had editorial roles for both open access and non open access journals. They noted that researchers did not send their best papers to the open access journal they had an editorial role for because it had a low impact factor. In terms of the participant's own use of open access, they used the university repository to upload their publications, even though they said that the process of uploading was time consuming.

Participant 4 had published 30 research papers, at the time of interview, however they had not yet published in an open access journal. Nevertheless, they highlighted that everything they published was available in the institutional repository. They also had established a personal website where they listed their projects.

Participant 5 mentioned that it was important to have multiple sources of information. They used a number of databases to find information, including

PubMed and MedLine, and also used a lot of grey literature, which is a type of document produced by government and organisations that are not published by commercial publishing. Open access sources were very important for the participant as they allowed them to have ready access to information, which was important when direction shifts during a project as it allowed them to act on changes quickly. They were concerned about the journal copyright attached to their publications, which meant they were not allowed to upload their publications to the repository. In terms of defining open access, the participant saw open access as a particular journal from which people can access journal articles without having a subscription.

Participant 6 had published a number of open access papers. Visibility was the main reason for publishing as open access, however they stated that publishing in an open access journal was not cheap. They always chose the best journals in the field to publish and as such they had published in the same journals for the last three years. These journals now have an open access option available for a fee. Participant 6 was willing to take the open access option only when there were funds to cover the cost.

Participant 7 relied on Google Scholar for information searching because it helped them to determine whether the search terms were right. When they were unable to access a resource via Google Scholar, they would use the library catalogue to access the full text. They never used the institutional repository when looking for information, nor to upload their publications. They did not see that the institutional repository offered any benefits to them, but understood it helped the university to track the outputs of their academics. The participant had published in an open access journal. Impact factor and relevance were two criteria they had in mind when selecting the open access journal as a suitable publication venue, however, they mentioned that they charged an expensive fee to publish. They saw open access as "a big part of our world and does make something easy in some ways but in another way it makes things more expensive".

Participant 8 used Medline and Google to find information for their research. All the information that they obtained from the government websites was free and easy to get. They were willing to share their research,

and they listed all of their publications in the institutional repository and also in ResearchGate. They were concerned about complying with copyright conditions set by publishers and as such they had made the decision to only list the publications in both the institutional repository and ResearchGate without uploading the actual papers.

Participant 9 research was very specific and quite unique. They did not use the institutional repository, finding it was not helpful to search their institution's repository for information related to their research because it was so specialised.

Participant 10 understood open access to be a resource that is available without having to have an institutional subscription or pay a fee. They saw open access as any resource that anyone can access.

Participant 11 uploaded all of their papers into the institutional repository. They were motivated to do this because they knew that it would help them when the university evaluated their academic performance.

Participant 12 saw impact factor as very important when it came to publishing. Two of their publications were published in open access journals with high impact factors. The other factor that influenced them to publish in an open access journal was university financial support for open access. They wanted their paper to be easily accessible and available for free download by anyone who was interested in the paper. They saw open access as advantageous for early career researchers because they helped the research to be more widely disseminated.

Participant 13 publications were all listed in the institutional repository because the university required it. They noted that uploading their publications to the institutional repository also helped the university with reporting on research output. They noted that the deposit process was handled by a staff member working in the university's research office, rather than by individual academics.

Participant 14 wanted their publications to be as widely available as possible, and to achieve that, they put copies of their work at academia.edu. They also uploaded copies of their work into the institutional repository, both

to make the work accessible and to comply with the university requirement to deposit publications, which allowed the university to track publications for reporting purposes.

Participant 15 had only published one paper in an open access journal. The open access journal where they published was one that accepted a submission for an interdisciplinary study, which was the type of research they did. The fact that the journal was open access did not really factor into their decision about publishing there. They made their work available in the institutional repository, a practice that was prompted by reminders from the university to do so. They talked about the fact that making publications available in the institutional repository may help researchers from other countries who have limited access to resources.

Participant 16 noted that the university encouraged them to publish in open access journals and that they had done so. Participant 16's desire to have their research applied also motivated them to publish open access. They wanted to contribute to their field. In the interview, the participant said "you get more views, people reading it, which is definitely what you want anyway", and this motivated them to make their publications openly available. Participant 16 commented they received support from the library in depositing their work into the institutional repository.

Participant 17 used the library databases for their information needs. When they used open access articles, it was often because they found them when conducting a search using the library databases. They had also published in an open access journal. They chose to publish open access because of the quick turnaround time offered by the open access journal. Turnaround time was important when they needed to get something published quickly. They understood open access to mean that the publication is free for the reader to access, but not for the author to publish. They stated that the author pays for the publications to make them free for the readers.

Participant 18 had a well-established publication record. They typically found information through library databases and was therefore not always aware when they had found open access articles, but noted that they were sure that they had used open access articles for their research. They

mentioned that they used one particular journal that was an open access journal. They listed all of their publications in the institutional repository, however, not every listing in the repository was accompanied by the full text because they were concerned about copyright. They had also published in an open access journal. In this instance, they got a fee reduction to publish in this journal, otherwise they would not have published there.

Participant 19 had published three times in the same open access journal. They selected the journal as a publication venue because it was open access and there was no fee to publish there. Publication fees were a concern for them, believing publication fees created inequality through being free to access a publication but costly to publish. They did not use the institutional repository for either finding information or depositing. However, they commented that the publications they published had been deposited in the institutional repository by the university staff.

Participant 20 chose journals with a high impact factor when publishing and also considered whether the journal would reach the desired audience for the publication. Whether or not the journal was open access was not a consideration. They noted that all of their publications were available in the institutional repository, meaning that everyone could access the papers. They preferred the formatted, published version of a paper to be uploaded in the institutional repository, with tables and graphs in situ, however noted that this was not always possible.

Participant 21's usage of open access usually started with information searching using the MedLine database. They identified articles as open access when they were able to access them from home without a subscription login. They defined open access journals as journals that were available in the public domain.

Participant 22 had published in an open access journal that they noted was "a tough journal in the field". However, they were also concerned about the number of journals from which they received spam invitations to publish. They were concerned that this practice may negatively influence people's attitudes about open access journals.

Participant 23 was concerned about predatory open access journals. They noted particularly that some of these journals had a title that was similar to reputable journals but they accepted any article for payment and the quality of articles they published was below standard. They noted that legitimate open access journals were ones that had rigorous review processes.

Participant 24 believed that research should be broadly accessible and widely disseminated. On a number of occasions, they had made their publications open access through different channels, including the institutional repository.

Participant 25 noted that green and gold open access were very useful for their research. In particular, they talked about being able to access a green open access or author version of a paper when they could not get access to full text of an article they wanted to read because their institution did not subscribe. They understood open access to be publications that are freely available to read but that must be cited appropriately when used as part of their research.

Participant 26 had a long list of publications. They noted that all of the publications were registered in the institutional repository. They noted that the university used publication listings in the repository to assist with academics' performance management.

Participant 27 deposited their own publications in the institutional repository to comply with university regulations. They had also published in some open access journals. They noted that speed of publication was a factor that motivated them to publish in open access journals. They also noted that the most important thing they considered when selecting where to publish was whether the journal had a good peer review system.

Participant 28 used the institutional repository when looking for theses, but for other types of information, they relied on PubMed, which they accessed using their university account. When they did a formal systematic review, they said that they would use all the relevant databases to ensure that they found studies that related to their research.

Participant 29 used open access a lot for their research. They mentioned the institutional repository for finding information and also talked about open access journals where they tried to publish. The reason they chose to publish in open access journals was because the projects they worked on up until recently had been publicly funded, hence they felt obliged to make their publications openly available.

Participant 30 relied on Google Scholar for much of their own research because it was easy to use and retrieved results relevant to their research. Most importantly, they valued the ability to get a direct link to the full text of the publication from the search results. Additionally, they had not yet used the institutional repository for their publications.

Participant 31 chose open access journals because they identified that there would not be any other journal that accepted such a long and detailed protocol. They saw open access as journals that are essentially PDFs or the complete articles available to anybody that looked for them, as opposed to being available through paid subscriptions to the journals.

4.4 Conclusion

This chapter has presented the profiles of the participants. The profiles highlight the diversity of the participants in their areas of research interest, and their understandings of and engagement with open access. The study's findings will be presented in the following chapter.

Chapter 5: Findings

5.1 Introduction

This chapter presents the findings of the study. The chapter is divided into two parts. The first part of the chapter outlines the four themes that describe how the participants experienced open access as part of their information literacy. The second part of this chapter presents a discussion of the relationships between the four themes.

5.2 The four themes

Using thematic analysis, and following the steps presented in Chapter 4, the data was examined, coded, and reviewed to explore the participants' experiences of open access as part of their information literacy. For this study, four themes emerged. A theme refers to patterns in the data that explain and organize aspects of the phenomenon derived from the data (Boyatzis, 1998). Each theme in the data related to a research question (Braun & Clarke, 2006).

The themes revealed in this study are:

1. Understanding Open Access
2. Institutional Repositories
3. Publishing and Dissemination
4. Finding Information

The participants' experience of open access as part of their information literacy relates to their roles as informed and informing researchers. The term "informed researcher" relates to the participants' roles as information users, while the term "informing researcher" relates to the participants' roles as information producers. Across the themes presented in this chapter, it is through participants' roles as informed and informing researchers that their information literacy related to open access is enacted; their information literacy experiences with open access relate either to their role as users or producers of information, or both. Importantly, as a relational perspective to information literacy was adopted, both behavioural (i.e., skills, knowledge)

and experiential (i.e., preferences, attitudes, feelings) aspects are present in both roles.

As informed and informing researchers, participants have their own individual understandings of what open access is, its benefits to them, and its benefits to the academy and society at large. These understandings are explored in the theme "Understanding Open Access".

Participants engage with institutional repositories as both informed and informing researchers. As informed researchers, they use information found in repositories as part of their research. As informing researchers, they deposit their work in the institutional repository to enable other academics, industry and society at large to access and benefit from their work. Participants' engagement with the institutional repository as informed and informing researchers is explored in the theme "Institutional Repositories".

As informing researchers, participants make decisions about where to publish their work, publish their work in closed and open access sources, and use social media to disseminate the research. The participants' role as informing researchers, as enacted through their publishing and dissemination practices, is explored in detail in the theme "Publishing and Dissemination".

As informed researchers, participants find, evaluate and use information as part of their research. This includes, for example, finding information as part of a literature review, and the practice of keeping up-to-date with the latest research in their field. While they may not actively seek out open access information, participants do engage with open access resources as part of their research. Participants' engagement with open access information as informed researchers is explored in the theme "Finding Information".

A detailed description and discussion of each theme is provided in the following sections. Each theme is supported by a selection of quotations from the interviews.

5.2.1 Theme 1 - Understanding Open Access

This theme elucidates the participants' understanding of what open access is and how it can benefit them as informed and informing researchers. Understanding what open access is, including the nature of open access publications, ways of accessing them, and the economic models that drive them, is an important aspect of a researcher's information literacy. This understanding helps them to navigate the practice of publishing and informs resource selection when they are looking at the research literature. In this study, participants understood: open access as "freely available", the global benefits of "freely available", the benefits "freely available" has for the participant, open access as a "last resort" when publishing, and open access quality.

5.2.1.1 Understanding open access as "freely available"

Most participants defined open access as journals making articles freely available, however, freely available was defined in different ways. For many participants, it referred to not having a financial obligation to the journal to access an article – either at the individual or institutional level. This view is illustrated by the following quotations:

> I guess "open access" is a particular journal that people can access journal articles without having a subscription to that journal [Participant 5]

> It means that you can either access it by going directly to the place, or you can access it through things like Google like if you do a search, it comes up and when you click on it, the actual article opens up for you. And you're not asked to pay for stuff, and you're not asked to join things, and so it's there, it's free for everyone to look at [Participant 9]

The previous quote also illustrates that for some participants, freely available meant freely available with minimal effort. For some participants, freely available meant that the level of effort required to obtain full text is low, as illustrated by the following quotations:

> I'd like to think that it's one click away almost, like something that's one click. It's not, but something that is one click away because I'm a bit lazy sometimes and I want to be able to click it and have it open up and be available and not challenging to download [Participant 30]

> I would define it as full text available for the articles. That's sort of what I, how I understand, so that, yeah, that everything published in those journals is available in full text [Participant 28]

Other participants acknowledged that freely available did not necessarily mean free to publish. These participants understood open access to be about the author paying to publish their work to allow free access to others, as evidenced by the following quotations:

> The open access is you pay the publishers to make your publications free and accessible, you just have to pay lot of money to make your publications available for free. [Participant 2]

> They can download it for free. The payment to the journal has been made by the person submitting the article. [Participant 17]

Many participants acknowledged that open access was not black and white – that journals are not simply open or closed - with many closed access journals providing open access options for a fee:

> I think these days as well, some journals are completely open access so that their entire content is open access. But then there are some journals that you've always been able to just pay some extra money to make your article open access. Whereas other articles that the journal publishes is not open access. And that's always been the case. I think that there's a growing number who just provide their articles as open access these days even though the journal itself isn't necessarily identified as an open access journal. [Participant 21]

However, one participant questioned this view or approach to open access, calling it predatory – a strong term that is usually reserved for journals that are completely open but generally considered to be disreputable, and which actively solicit submissions from unsuspecting researchers. This strong language illustrates the depth of feeling that this participant had about this practice:

> True open access means that it's freely available and I know that some people also call the version where you can make your work open access by paying like $2,000 still call that open access, for me that's not really open access. If you have to pay $2,000 to make people be able to read your work, then that's just another part of the predatory journal publishing industry. It's not really open access at all. [Participant 14]

For several participants, freely available was about being accessible by everyone following a quality peer review process. This point is illustrated by the following two quotations:

> My understanding is that it is a journal that still uses some peer review processes but then the outcome is available to anyone. So it's a journal that can be read by anyone, yet still uses some peer review process [Participant 13]

> The difference between having one scam open access publisher and a reputable open access publisher is when I ask to review an article I said no I don't like this thing, it's not acceptable for publication, they will not publish it until the author change what I wanted to be changed, while for the scam they will publish it anyway because they want their money [Participant 3]

Interestingly, only one participant explicitly spoke about open licensing when discussing their understanding of open access. For this participant, freely available goes hand-in-hand with open access licensing.

> I define Open Access, so it is an article that is freely available, but also freely licensed, as well as being not just

> readable for free, but under an Open Access license, as well [Participant 11]

A small number of participants recognised that open access was not only about making research findings via publications freely available but also research data, as illustrated by the following two quotations:

> It's formal open access through journals and then there's [institutional repository] which is a semi formal version of it, and then they giving your own data away essentially [Participant 4]

> Certainly, it means making publications and other outputs freely available, but it can also mean making data available, and I think that's really important that I think open access should extend to data as well as outputs. That's kind of the end of the spectrum [Participant 29]

5.2.1.2 Understanding the global benefits of "freely available"

When commenting on their understanding of open access, many participants spoke of a strong commitment to embracing open access as part of their information literacy practices, and many also commented on the broader benefits this would have on industry and society. In this way, their understanding of open access went beyond freely available; they also understood the benefits of making their work freely accessible. This point is illustrated in the following quotations:

> I think it's good, it's great, and it's served the academic community and also non-academic community. I think it's really important that journalists and people who can't afford library subscriptions have access to quality information and I think it's very important to make open access information as much as we can. It's efficient access to high quality information, good for everybody, fast for everybody, better information [Participant 1]

> From the ideological point of view that not only access to information should be free or as free as possible, but also not incur cost it contributes knowledge as well. You're not only receiving but giving the information should be as free as possible. [Participant 22]

The freely available benefit identified encouraged some participants to share the open access information with others, which could be with their students or industry partners who might not have access to the closed-access publications.

> In terms of that kind of model which I think it would apply to the journal as well being able to access anywhere, being able to give it to my students and say here have a look at it remove all those barriers having to say putting in request or if you're off site you can't look at this so in this sense this kind of open access model I see it's been usually beneficial to researchers generally whether be open data which something I'm passionate about or open journal or anything like that just the actual freedom it gives you to use it and show to other people and to show it to industry partners, so everyone knows what everybody is doing, I think that is usually beneficial to other in general. [Participant 4]

Similarly, the following participant's commitment to open access was born out of their engagement and conversations with doctoral students, and a growing understanding of the divide between developing and developed countries in being able to access scholarly works. The participant statement also reveals their evolving understanding of the implications of the current economic structures of publishing:

> I have a lot of PhD students who come here from developing countries. They've talked to me about how difficult it is to access their materials. I think that means that we, as researchers, have a moral imperative, a duty, to publish in Open Access databases. I think that something very bad has happened in the past, where academics have given control of

> publishing through commercial publishing houses. These commercial publishing houses are parasites on the academic process. They are locking up knowledge that is paid for by the public. I think that that is a terrible situation [Participant 11]

While the following participant acknowledged similar sentiment to supporting access to scholarly work for all, especially those in developing countries, they also recognised that they tended to follow pragmatic practices that will ensure the best exposure of their work which may or may not result in using open access:

> I do like the fact that they have better access, so more people can read my work and quicker and it's free, the ones that I happen to choose. People outside of institutions where they might not have a good set of subscriptions they can read my work as well. Now we should be thinking more globally and thinking about all sorts of issues including equity and developing countries and all of this, poor people in general. If you are wanting to do research or interested in research and you're in some poor country, and you might not even be at a university then it is the right thing to do to provide these people with information as well. It shouldn't just be expensive. At an ideological level, I very highly support open information. In practice, when I'm submitting an article, what I do there is actually, I think I actually honestly forget about that ideology and focus pretty much on what journal is best for this publication and myself for that matter. Is it a good journal? Will it give my paper exposure? Will it get citations? To be honest that's a little bit greedy but they are my three top questions. That guides me to choose the journal and in some cases, the journal happens to be open access, and sometimes it doesn't. Whether it's open access or not, it's nice if it is but that's not what informs my decision. [Participant 22]

In fact, while many participants espoused a belief in the principles of open access, when questioned about the criteria they used to determine

where to publish, whether or not the publication is open access does not necessarily factor into their decision. This will be explored in detail later.

5.2.1.3 Understanding the benefits "freely available" has for the participant

Several participants noted that they were not always committed to open access. Their view on open access has changed significantly over the years. This change was a result of seeing first hand how open access can have a direct and tangible benefit for them as scholars, for example increased citations counts. This is evidenced by the following participant statement:

> Five or seven years ago I knew about open access but I didn't worry about it, or I didn't think much about it, I always just thought that if I'm doing good research and I publish in an international journal doesn't matter it's open access or not my research will get out there, it will have impact, but I think in the last couple of years I re-thought that, and from talking with colleagues, in my field, which is [research field], [research field], a lot of my colleagues now are thinking a lot more about can we make our papers open access, because I think a little bit of data I've seen out there that certainly if you publish on open access you're going to get your work accessible lot of people will see your research, and more importantly they might cite your research and as an academic that's very important when you're writing a research grant to say that you have a highly cited publication and also when you get a promoting round you need to have a good h-index, an increasing h-index, good citation numbers…these sort of things, so I want to get my research out there more, and I think that open access, I am convinced that open access is the way to go [Participant 6]

5.2.1.4 Understanding open access as a "last resort" when publishing

A significant number of participants admitted to being sceptical of open access information channels, and in particular, they spoke about open access publications as a last resort when seeking to publish. The following participant statement illustrates this:

> I don't think the quality of the papers in open access journals is as good. Generally speaking, I kind of feel like it's a last resort if people have not been able to get a publication in other journals. Or sometimes I know people use it as a strategy to publish something in an open access journal so it gets lots of citations. So then obviously that helps with their standing as a researcher, I guess, to show that they have publications with lots of citations. [Participant 13]

Building on this point, the following participant admitted that they and their research group would on occasion take advantage of the last resort options of some open access channels:

> Sometimes I know within our group, some of the papers that we wouldn't necessarily consider as strong have been sent there after it has been rejected from one or two other journals because we generally know they accept most things. [Participant 20]

Further, this idea that open access publications are a last resort was underpinned for some participants by the idea that publishing outlets were just trying to make money:

> When I think of open access journals I think of payment but I also think of predatorily journal emails that they're just trying to make money at. Deliberately there's no academic rigor or anything going on that makes you think that these journals are robust journals at all. If you have a look into them click on any, but I'll go and search those journals through Google and will then find it's only done one volume, most of the publications

were people from China or India or are in high continent, high volume continent, have a hard time in publishing their work they might have money to pay for it. [Participant 18]

5.2.1.5 Understanding open access quality

Being open access can have a negative impact on the perceived quality of a journal. During the period this research was conducted, journals were ranked as part of the research quality and evaluation framework Excellence in Research for Australia (ERA). Several participants noted their concern that open access journals were ranked low in ERA. The following participant comment illustrates this point:

> I did wonder at the time, and there was a lot of speculation at the time, whether the ERA ranking system had ranked it so poorly because it was in the electronic form only open access. [Participant 22]

Participants however also recognised the difference between perception and reality when it comes to the assumption that open access equalled poor or low quality. As illustrated by the following quotation:

> So if you have a [journal name] publication of any sort, then your publication is considered to be of the higher quality, whether or not that's true is another thing. But, it's considered to be higher quality. And [journal name] publications are not open access. So therefore, by definition, the higher quality journals are not open access. So, open access journals are considered a lower quality. But then again, I don't think that's necessarily true. That's just by reputation. That's not necessarily an indicator of the quality of the actual research if that makes sense. [Participant 21]

Many participants spoke about receiving emails from open access journals. The invitation to publish from open access publishers influenced some participants to have a negative perspective about open access journals.

> I get emails that would promote these open access journals. That in itself is a negative thing. Most established journals don't promote themselves, they don't have to. [Participant 22]

Another source of negative sentiment regarding open access journals was the issue of payment. Some participants questioned the quality of open access journals because authors paid to publish in them:

> When I think of open access journals I think of payment but I also think of predatorily journal emails that they're just trying to make money. Deliberately there's no academic rigor or anything going on that makes you think that these journals are robust journals at all. [Participant 18]

Likewise, the quality of the journal peer review process was questioned by some participants, as money was involved with the journal publications process. Some participants showed their concern whether the payment would be the main consideration for the acceptance rather than the paper quality. As illustrated by the following quotation:

> For some of the other ones, and one open access journal I am thinking of in particular the review process I don't find very rigorous. Part of me wonders, and it is a little bit of a deterrent for our publication because part of me wonders if they just want more publications for more money. [Participant 20]

In contrast, some participants believed that the concern of quality and reliability did not relate to the fact that the journal was open access. However, it was linked to the credentials of the people who managed the journal, as illustrated by the following quotation:

> I think that you have to adjudicate reliability emphasized based on authorship credentials of people managing the site, so many different reasons to evaluate reliability…. I don't think open access is something that would affect reliability judgment either way, not just because you find open access, I have no suspicion about the reliability of open access [Participant 1]

5.2.1.6 Conclusion

Participants' understanding of open access can be grouped into two main perspectives, from the user perspective and the author perspective. Information users would see open access as a "freely available" journal article, while most of the authors see the open access publication as "pay to publish". The changing directions of some subscription-based journals, that offered open access options with some fee, have also enriched the discourse of open access publications among researchers. The identification of global open access benefit has encouraged researchers to consider selecting the open access model to disseminate their scholarly work, with some consideration of quality assurance from the journal publishers. This understanding forms part of the researcher's information literacy, as their understanding of open access is a foundation for their experience of open access as informed and informing researchers.

5.2.2 Theme 2 - Institutional repositories

This theme depicts an aspect of the participants' experience of open access as part of their information literacy that was associated with using institutional repositories. Researchers' information literacy now encompasses engagement with the institutional repository across the spectrum of research activities, from initiating collaborations with other researchers, to finding the work of specific researchers, to depositing a copy of accepted publications. Participants report on how institutional repositories play a role in allowing them to communicate their research, and to build their profile as a researcher. Institutional repositories also enable the participant to interact with scholars in their field through the publications other researchers have deposited. Significant aspects of the participant perspective on institutional repositories revealed in this study were: motivations for depositing in the institutional repository, using the institutional repository to find information, issues experienced when depositing publications, and institutional repositories versus social network repositories.

5.2.2.1 Motivations for depositing in the institutional repository

Participants were motivated to deposit their publications in the institutional repository for a number of reasons. These motivations formed part of participants' information literacy related to open access in that they represented an understanding of what the institutional repository could do for them as informed and informing researchers.

Several participants mentioned that their motivation to deposit their research in the institutional repository was to ensure their research would be easy to find and by consequence available for other researchers to use. The following participant statement illustrates this point:

> It gets your research out there. I want people to find my research, I want people to read it and think about it, and use it. [Participant 10]

Depositing in the repository not only surfaced their work to people searching the repository, it may also surface it for people searching the Internet more broadly. In this way, depositing in the repository maked research findable to a wider group of people. Some participants clearly understood the relationship between institutional repositories and major Internet search engines such as Google. Being able to ensure their work could be easily found by taking advantage of this relationship was a strong motivator for depositing items in an institutional repository:

> When you search my name, you just get pages of all my work and presentations, and I'm fairly sure our institutional repository is in the top 10 Google results. [Participant 10]

For some researchers, the repository was a central location for storing and profiling the work of research groups. Making work available via the institutional repository also helped build research collaborations and networks. For one participant this was a strong motivator for using their institutional repository:

> We also want to be able to profile our work to the public and to external collaborators. I'm always looking for ways we can do that more efficiently. [Participant 11]

Related to this, participants also revealed how they used the institutional repository to gather information to monitor and promote the performance of their research group. This next participant's statement illustrates this point by making reference to how they used their institutional repository to generate a listing of current publications by group members:

> I use institutional repository to look at the research performance of my research group. I'm able to create searches within institutional repository that will return an RSS feed of every article published by anyone that I'm associated with here in the faculty. Then we can use that RSS feed to publish on a website, a list of new articles published by us. [Participant 11]

Some participants revealed their motivation to deposit in the institutional repository was to keep track of the impact of their research. The following participant's comment illustrates how they intentionally used the institutional repository to demonstrate their impact as a researcher:

> When I get publication I will certainly point to the repository because they keep the track of download which means like Altmetric which is very important for researchers reliability, you can say you have impact based on number of download from our repository. [Participant 1]

Some participants were cognisant of the fact that their institution used the repository to assess research output and evaluate the performance of researchers, and this motivated them to deposit their work. The following participant statements illustrate this point:

> The motivator is to register it, so that university knows what I…what my output is. So that they can see that I'm doing my work properly. That's my motivation. It's just a list of all my publications. If you go into our institutional repository, and

> you say search for and put my name, it will come up with all my publications in a list. [Participant 26]

> It's mandated as part of our institutional metrics. None of my research gets counted unless it's in our institutional repository. I'm about to go to my performance meeting tomorrow, and with my Head of School. My Head of School will look at my institutional repository record to evaluate my research. So everything has to go in the institutional repository. [Participant 7]

The institutional repository was an important platform for the participants in relation to their academic career in that it was the source to assess their performance. In fact, some universities *only* counted research when it was available in the repository. This is evidenced by the following quotation:

> We are required to put copies of our publications in there. So we have to put the final published version of record in that's for Excellence in Research Australia and other purposes. But that's not made open access unless it is legally right to do it. [Participant 25]

In addition, the desire and/or need to be recognized as an active researcher was highlighted by several participants: "the university can track our records too, and they know what's popular and who's working hard" [Participant 30]. In this way, they knew the university was watching their performance based on what they deposited and this motivated them to deposit, so that they could be seen to be doing their job and to be recognized as "research active".

The participants acknowledged that the institutional repository also served as a tool for reporting internally within the university and externally to key stakeholders. For some participants, knowledge of these reporting requirements and processes motivated them to deposit their work. As the previous quote illustrates, some participants were aware of how external reporting worked and who the university reported to. Other participants were less clear about external agencies, but were cognisant of the fact that the

university was reporting on their research output and that it did that by drawing on their data from the institutional repository.

> I think we have to do a lot of reporting. We report in schools, in faculties, and other internal groups. We also have to report to a lot of external groups, research funders, and things like that. [Participant 11]

Related to the idea that universities evaluate researchers by looking at publications that are available in the repository, it was clear from the data that universities used reminders to encourage researchers to deposit and that participants are motivated by these reminders. A number of participants commented that they were frequently reminded by the university to upload their work to the repository especially when it comes to the Higher Education Research Data Collection (HERDC) time where the universities send their research statistics to the Department of Training and Education. These participants understood it was their responsibility to ensure their records in the institutional repository were up-to-date. The motivations of complying to a university's evaluation processes are illustrated in the following quotation:

> I would think that every academic researcher at this university knows about the researcher performance evaluation and the institutional repository. It's part of our performance review documents process. At least in the Law Faculty, each year, in the annual performance review the instructions are to make sure that your institutional repository is up-to-date, and to bring a copy of your institutional repository, or your institutional repository profile, to your performance review. They are the things that get counted by your academic supervisor. We get additional reminders at the HERDC data collection round, when the assistance of research has to send off [the institution's] performance metrics, or the faculty's performance metrics to the government. We get constantly reminded to make sure that we update our institutional repository. [Participant 11]

As further illustration, the following comment reveals the role a university reminder can have:

> I only do that when they ask me to do it. Occasionally, once a year or so, they come and say, "I've noticed you've got these papers and we would like to have that in [the institutional repository]. Can you send me the pre-publication version of the paper?" Then, I do that. [Participant 15]

Participants in this study were also motivated to deposit in their institutional repository by a sense of obligation to "the greater good". This included both a motivation to serve the academic community and a motivation to serve the general public. One participant highlighted a sense of responsibility to make research more visible as part of their obligation to their academic community, as illustrated by the following quotation:

> I also think it's my obligation as a staff member of this university to make available my work to other academics so they can also find out who I am and what I am doing. [Participant 1]

Another participant spoke about making their publications openly accessible through the institutional repository as a way to help other academics by providing access to scholarly works that they might not normally have access to. The following quotation illustrates this point:

> Not every university can afford or every individual can afford to subscribe to all the relevant journals in a particular field. It's on the record that even Harvard University which is the best funded university in the world can't afford all the subscriptions and all the researchers it might require. The reason we do research is so that people will read it. If you make it available, then people are much more likely to read it. [Participant 25]

Beyond benefiting the academic community, participants understood that depositing in the institutional repository would make their work accessible to the general public and were motivated to deposit as a result. The

following quotation illustrates this idea of being motivated to deposit in institutional repositories by a desire to contribute to the public good:

> It is about making it freely accessible, part of it's about making accessible in terms of language and … not making it too dense and using lots of academic terms. There should be special ways of dissemination research, so that the average lay person, can really engage with what's being done and dispel that ivory tower sort of issue. I'm proud of that. if you're not a member of the university, there's no way you'd be able to afford to access most of those articles, so publisher is talking about $30, $40, $50 or more to access articles if you don't have a subscription…..I think it's a huge issue and realistically research is supposed to be done for the public good, but if it's not disseminated to the public, then I don't know if it really meets those sort of…Maybe we might need to break down a few barriers, and I think open access is probably a pathway to doing that….It's just like really difficult balancing act of publishing in the journal the university wants you to publish in. [Participant 24]

5.2.2.2 Using the institutional repository to find information

All participants commented on using institutional repositories as a means for finding information. Participants used institutional repositories to locate different types of information, for a range of reasons. Several participants noted they frequently used the institutional repository to obtain details about colleagues, for example their research interest, current scholarly work, or areas of expertise. They also found institutional repositories useful when searching for particular types of information, including theses. They did not, however, find searching institutional repositories useful when they were looking for information on a particular subject.

The following two comments depict the act of using an institutional repository to find recent publications of colleagues. The comments illustrate that participants were aware that their colleagues' publications would be

available in the repository and that the repository would be the best place to locate this type of information:

> I still know certain researchers who are here, and if I want to find out what they're doing I just go on our repository. [Participant 10]

> If I want to find information, more publications by particular author I'll go to [the institutional repository] if it is authored by [participant's institution]. [Participant 1]

In addition, the following comment illustrates how one participant used an institutional repository to not only learn more about a colleague's work, but to initiate research collaborations:

> I use [our repository] all the time. It's incredibly important to me for collaboration across the university as a tool. So for me, I always try to read up about an academic's work before I meet them if I'm going to try to suggest them to work on a project together and so I rely on a lot…much easier to find out what's going on then through a database. [Participant 1]

While most participants acknowledged the useful role of institutional repositories in finding information about colleagues, one participant observed that usefulness was dependent upon colleagues keeping their details up to date:

> Yesterday I checked the profile of a researcher from my department…not a lot of information is updated, and then I go to Google Scholar they actually have newer papers. [Participant 3]

Overwhelmingly, participants noted that institutional repositories were most useful when searching for a particular author. The following quotation illustrates that this type of search is what institutional repositories were really designed for:

> University repositories tend to be focused on the creators of the material, the researchers. They're focused on you being

> able to search for a person's work. That's what they're interested in. They're interested in their researchers' work being used. It's how they show value. [Participant 29]

In addition to author searches, a small number of participants indicated that they would use an institutional repository when searching for specific types of information sources, for example theses:

> I don't tend to use research repositories unless I'm looking for something like a thesis or something that is not a journal. [Participant 12]

One participant noted the role of the institutional repository in guiding and informing their own writing process. The following comment illustrates how one participant used the items found in an institutional repository to learn how to approach a piece of academic writing:

> I mainly used institutional repository to look up the structure of writing something, so I have to do a theoretical chapter, I will look up other repository, other PhDs and stuff, and see how they wrote their theoretical chapter, or see how someone that's doing qualitative interviews, and see how they wrote up their methodology. So that's what institutional repository helped me with that. [Participant 9]

While there were some instances in which searching an institutional repository was useful, very few participants described using institutional repositories to search for information on a particular subject area or topic. This was because the participant would "have to hit too many universities, it's ineffective, it's time consuming" [Participant 8]. A number of participants noted that institutional repositories were not designed for searching by subject, as illustrated by the following quotations:

> When you look up sexual consent, any time you look up "consent" you're going to get the "age of consent" and I'm not looking at that, and then it brings up all sorts of random things that I'm not interested in. I find that on the

> [institutional repository], when you try and look up through "subject", it's hard to find. [Participant 9]

> Our repository is not on very good software. It's not very good and it's hard to search and it also has a funny layout for how it puts in the association of publications with each report. [Participant 25]

5.2.2.3 Issues experienced when depositing publications

Many participants spoke about the issues they experienced when depositing publications in an institutional repository. These issues significantly impacted on the participants' willingness and ability to engage with institutional repositories as an information repository specifically, and as part of their information literacy broadly. To successfully work with an institutional repository, researchers had to have an understanding of publication versions and their rights with regard to depositing copies of publications in institutional repositories. They must also be able to navigate the complexities of the deposit process related to system functionality.

Almost all participants commented on the extremely long time it took to upload items to an institutional repository, as illustrated by the following quotation:

> [I] have published 11 papers which are not so many to upload, but it took me half a day to upload them to our repository. [Participant 3]

The process of uploading papers and adding the information manually to the institutional repository was an issue for many participants. One participant suggested that the institutional repository system should be able to gather the publications from other sources. This point is illustrated by the interview excerpt below:

> It's an additional process and I think that's the challenge. One of the suggestions… is why can't we [get publications into the system automatically]. ResearchGate and Google Scholar are getting their information automatically. Surely, we can build

> something that gets it automatically to or extracts it from ResearchGate. [Participant 30]

A number of participants noted that depositing in the institutional repository consisted of too many steps and was too complex. Some participants asked staff or their personal assistant (PA) to upload their publications for them, however not all participants could afford to employ staff:

> When I can afford a research assistant, I will get them to do it for me. If I'm working on a paper and I've got a research assistant who is sitting around, then I get them to do it because it's a pain. So many processes, usually I don't have a research assistant, so I do it myself. [Participant 11]

> I do have a PA, so my PA does those jobs for me, which may be slightly easier. I do understand why some of our academic staff, they complain to me, [the institutional repository] is very difficult to use. [Participant 27]

Several participants noted that they were confused by the concept of publication versions (e.g., pre print, post print, publisher versions) and that this impacted on their ability or willingness to deposit publications. The participants did not fully understand which version of a publication they should or could deposit in a repository. The following participant's comment illustrates how confusion over publication versions resulted in the participant only depositing items when directly asked to by university staff:

> Sometime they come back with you and say we do like these pre-publication copies, and you have to put them together for them, which I find is quite painful because it's not clear which version it is. You need to clean them. You need to format them yourself…. I don't normally do by default. [Participant 15]

The issue of copyright was raised by a number of participants. The following two comments illustrates how confusion over copyright resulted in participants choosing to not upload items to avoid potential copyright breaches:

> Everything I publish at the university it's the list referenced in the [institutional repository] but it's not necessarily available because … say I have published the [open access journal]…I'm not allowed to release the finished article in a finished form for public access because they want twelve months of ownership. On the [institutional repository] all of my publications are listed but they are not many publications attached to those…I don't want to break any copyright…and I certainly know that I can put it up pre-draft copies but it's all too complicated for me to know which one I can and which one I can't put out or when I can put out or put it up. [Participant 18]

> I won't upload it because I think the whole copyright matter can be just too complicated, I just don't, well you know about an embargo period, the author's version, that sort of thing, you just need to put more attention when you going to put your articles up to the university repository. [Participant 8]

One participant noted the role of the university staff in lessening confusion with regard to copyright. As one participant noted:

> I'm always confused when you put things in the repository. I'm really glad they review it before they put it publicly available, because I'll look at it and with the license and say "I've got no idea! Put it in and you can investigate…!" Because you never know, is it supposed to be embargoed? Is it supposed to be available to anyone? But because they have that whole, you click the button, it contacts the researcher and says "can I have the paper?", so I figure out at least that's there. You put it there, and the repository staff will decide the availability and terms of can they just release it instantly or whatever. [Participant 10]

Additionally, some participants noted that the version of the publication (e.g., the pre-print or author version) that is uploaded in the institutional repository was not their preferred version. The following

participant's statement illustrates this point by making reference to how they would prefer to not upload the item to the repository and instead email their preferred version of the item to people who contact them directly. The statement also reveals the participant's uncertainty about the copyright implications of this approach to sharing their research outputs:

> It is not best looking version because you have got all your figures and tables at the back of the document in its formatted Word version. I'd rather someone if they wanted that paper to just contact me. I have the PDF and I am not sure it works with the copyright thing or whatever but I would rather someone see the final formatted version because it looks the way you intend it to look. I'd rather with the repositories that we were just allowed to upload our final accepted PDF version, the one that is published on the journal's website because that is the final formatted version where all the graphs and tables are in their correct spots and it looks much better and it is easier to follow. I'd much rather be able to upload those. [Participant 20]

5.2.2.4 Institutional repositories versus social network repositories

Some participants spoke about the limitations of the IR as a personal archive of their scholarly content. In particular, one participant explicitly commented on the value of having an archive of their scholarly content that would move with them as they built a career. Social networking repositories, unlike institutional repositories, allow the participant to curate their scholarly content independent of their employment status, and as such their content is more likely to remain up to date:

> I think [institutional repositories] are a good idea in principal but in practice in my stage of my career not as good as it sounds. When you're a professor you got tenure and very surely you'll stay in one place for the rest of your life….at this stage the institutional repository doesn't work as well as the personal repository like Research Gate or Academia.edu because those I would keep with me…I prefer [a repository]

> that goes with me rather than the one that stay with institutions, because I can modify it for the rest of my life as long as I pay for the hosting. [Participant 3]

The differences in the process for maintaining and keeping current details in institutional repositories as compared to social networking repositories were noted by several participants:

> ResearchGate and Google Scholar are getting their information automatically. Surely, we can build something that gets it automatically too or extracts it from ResearchGate. [Participant 30]

5.2.2.5 Conclusion

Engaging with the institutional repository is not merely about complying with the university mandate. Participants were motivated by many factors to deposit their publications in the university repository. Their motivations, and the understandings that underpin them, form part of their information literacy as informing researchers. It is their understanding of the benefits of the repository and its role in promoting their research and informing the university's evaluation of their performance that drive participants to deposit. Further, participants' experience of the deposit process is part of their information literacy, and these experiences were largely plagued by issues and concerns with the deposit process and related concerns with copyright. Depositing work is time consuming and often not straightforward. In this way, participants' information literacy experience related to open access and specifically related to institutional repositories is not always positive.

5.2.3 Theme 3 - Publishing and dissemination

This theme denotes an aspect of participants' experience of open access as part of their information literacy related to publishing and disseminating their research outputs. This theme elucidates participants' preferences and insights on selecting where to publish, and the role that open access played in sharing research. Aspects of the participant experience revealed in this study are: criteria used when selecting where to publish, why

participants opt to publish in open access journals, paying to publish, peer review, and social media.

5.2.3.1 Criteria used when selecting where to publish

When selecting where to publish, participants used a variety of criteria to help them decide where to submit their work. These criteria included journal scope, impact factor, familiarity with the journal, and where the most notable researchers in their field publish. Notably, whether or not a journal was open access did not necessarily factor into the decision-making process for many participants.

A number of participants indicated that the key determining factor of where to publish their research was focus or coverage of the journal. The following participant statement illustrates this point. The statement also reveals that for this participant journal relevance was a greater factor in decision-making than publication fees:

> Probably relevance, I think…[open access journal]…is very relevant to the research we've been doing in [research topic] so that's why we ideally would like to get in to that journal, and it's just unfortunate it came with a fee. [Participant 7]

Not surprisingly, a number of participants noted the strong influence of the impact factor in their decision-making relating to publishing in open access outlets, as evidenced by the following quotation:

> When I'm choosing where to publish, I'll generally look for journals that are read by my target, then look at what sort of work they publish and decide for myself whether they are a good fit and decent quality, and then once I've made a short list, I'll look at their ranking or journal impact factor to help make my final decision as to where to publish. [Participant 10]

Building on this point, two participants noted that the extent to which impact factor determined where to publish was dependent on where a researcher was in their academic career. As early career academics, the participants needed

to focus on publishing in journals with high impact factors to secure their ongoing research career:

> Impact factor determines the prestige of your publication and the effect to your career as a young researcher I cannot afford to publish to a journal…I cannot afford to send my paper to there because it waste of my time, I cannot spend a few months' work on a paper or in a manuscript then I read it somewhere without impact factor that something that senior researcher can do but not me I need to have my papers published in best journals and that's in practice mean journal with impact factor…the highest as possible…and it's not my goal to publish to this [open access] journal…but that's not because it is open access it is because it is less prestigious. It doesn't look good in my CV. [Participant 3]

> Philosophically I like open access for my job…in term of trying to get permanent position it's really important that I get publication in journal that have very high ranking… and at the moment in my discipline the highest ranking journal often not open access…at this stage in my career is very tough and because of [my University's repository] it means it's easier for me to publish to a subscription journal and have a copy of the PDF available and accessible…so at this stage I have to choose the publishing model that most likely can get me that job. [Participant 1]

Several participants noted that familiarity with a journal was a key element in their decision-making about where to publish. The following two comments illustrate how familiarity can influence a participant's actions. The participants revealed that having knowledge of the quality of the articles previously published in a journal created trust and respect for the journal. This in turn resulted in the participants choosing the journal as a means for sharing their own research:

> I know them very well. I trust the research that published there. They got a wonderful reputation as a journal, and I can

> see when I publish there I do get a good citation. It certainly, these journals are well advertised and people in the field know it's a very good quality journal. [Participant 6]

> I like the work they've published already over the years. I've read articles published in those journals that I thought were good articles and good pieces of research, and they tend to produce mostly really good quality work. I'm not across the board on all topics that they published, but I certainly read really interesting articles in those journals. [Participant 12]

Similarly, the following quotation illustrates how the increasing number of open access journals making "cold calls" and the lack of familiarity with these journals not only deterred one participant to publish, but produced feelings of scepticism and uncertainty:

> I receive emails probably every day from a new open access journal asking me to submit an article, but because I don't know this journal, I don't know it's impact factor, it can be a new journal, it doesn't have impact factor, I don't know their reputation, I'm just feel very cautious. [Participant 6]

Another participant focused their selection on where to publish based on the overall professional standing of the journal and the quality of the peer review process:

> I choose to publish in journals that have the best peer review. And publishers might be in subject areas that are of interest to me and my readers. But also that it is of high quality. As an academic that is not publishing in journals of high quality, you lose a lot of respect. [Participant 19]

A number of participants noted that their publishing choice was influenced by their academic peers. For example, publishing in a journal with other notable authors working in the same field could increase your chances of having your work found and read:

> I would look to see where my dominant authors have published, and try and publish there. Anywhere where it can

> sit alongside people that have researched in the same areas because if I publish in random places, then you're not going to find them as easily as if they're all blocked together. [Participant 9]

Similarly, the following participant noted that they opted to publish their work in an unfamiliar open access journal solely based on their knowledge of, and respect for, the editorial team:

> I wasn't familiar with the journal. But I was familiar with the people who were doing editing that special issue, and I respected them. So, then I looked into the journal, and actually it was a new journal. It didn't yet have an impact factor but it's part of a series, and they expect it soon will, and it's a very respected series. So, I thought it's worth it. [Participant 10]

When asked about how they decide where to publish, many participants indicated that a journal being open access was not in itself a key factor in deciding whether to publish there or not:

> this has nothing to do with open access…for example if a journal, they are willing to accept my publications, they are efficient, they have a reasonably good peer review system, and they don't charge a very high fee, publication fee I mean, then I would submit. Whether it's open access or not, I don't really care. [Participant 27]

> Is it a good journal? Will it give my paper exposure? Will it get citations? To be honest that's a little bit greedy but they are my three top questions. That guides me to choose the journal and in some cases the journal happens to be open access, and sometimes it doesn't. Whether it's open access or not, it's nice if it is but that's not what informs my decision. [Participant 22]

5.2.3.2 Why participants opt to publish in open access journals

When participants do select open access publications, their reasons for doing so vary. They are motivated to publish open access because: it makes their work visible to a broader audience, it has the potential to increase citations, it benefits their academic career by helping them to establish a reputation, the publication process may be more timely, they want or need to comply with university or funding agency policy, they are motivated to do so because of a philosophical belief in the value of open access, or they want to retain the copyright for their work.

For some participants, publishing their papers with an open access journal means their research would be disseminated widely and consequently it is a way to make their work visible to a wider audience. A number of participants recognized that closed access publishing limited their ability to reach key communities and individuals relevant to their scholarly work. The following statements illustrate this point, and reveal the sophisticated understanding of participants regarding the academic publishing process generally, and digital publishing and audience reach specifically. The statements also reveal the participants' awareness of the divide between the academy and industry, and the need to cater to both when sharing research outputs:

> Basically, because it was a paper that we wanted to be really accessible to the professions via [via Australian Policy Online]. It was research that was more relevant to organizations working in the field, and they wouldn't necessarily access the information via a journal. [Participant 24]

> I wanted to publish within the open access, and I thought that if I published in a free journal then my target reader who are people working in local government, not just academics. I wanted those people to be able to get the paper without having to pay because they're not coming through a university. [Participant 26]

Further, some participants recognised that publishing open access would make their work visible to those who had participated in their research:

> It doesn't make sense for me to be writing up that knowledge and then putting it up somewhere where that the people who shared it with me can't access easily. [Participant 14]

Many participants also revealed that the possibility of their work being cited more as a result of being published in an open access journal motivates them to publish open access, as illustrated below:

> When you have spent $3000 to get your paper published you like to see that you get more download and more read to your article, and hope that will translate to citation, I guess in terms of motivation I hope it will lead to better citation and more exposure of my research. [Participant 6]

Open access was a key strategy for many participants in building their reputation as researchers. By using open access as part of their overall dissemination strategy participants ensured that their work was more visible and available. The following participant statement illustrates this point:

> I hope that I'm publishing better science than I was 5 years ago and maybe that's why I'm getting high citation, I've got more reputation in science now, which you do when you've spent more time in your field, but maybe I think the open access as well also having an impact and number of my colleagues have convinced that also the case for them so I think we're talking among ourselves thinking about how best to get our research out there, and I think most of people have embrace a strategy about open access. [Participant 6]

Open access is an important dissemination tool and/or platform for the participants in relation to their academic career. The following quotations reveal the important role of open access for early career researchers, ensuring their work is available to others, and in turn laying foundations to grow and evolve their research activities:

> It's positive for early career researchers, it helps get your stuff out there, get your research flourishing. Helps get your research out there to a broader community. [Participant 12]

> I would say very important, like I'm an early career, like almost five years since I finished my PhD. At this stage, I need people to be seeing what I'm doing, I need to try and establish this profile, and I think it's really crucial for me. [Participant 10]

A number of participants observed that they opted to publish open access because of the speed with which articles submitted would be published:

> [T]hey tend to turn around the article more quickly than, say, a quarterly journal or a monthly journal. From my understanding, they appear to be publishing on a rolling basis, so I've found in the past that the turnaround may be a couple of months, whereas with the subscription journal, you might be waiting up to a year to get your article published, so that would be one of my main motivations… [Participant 28]

Many participants indicated that they made their research open access because they wanted to make their work available straight away without going through the long process of publishing that many closed journals required. This point is illustrated in the following quotations, which reveal how two participants used non-traditional dissemination activities to complement and add value to traditional academic publishing via journals:

> Prior to submitting our journal article for publication in a journal, we have working paper series or discussion paper series. We put the preliminary findings in the working paper series, which is again Open Access. It's not peer review, so it's not considered to be a publication. It doesn't have a publication ID. It's the same number, so it's not a published outcome, still you offer your findings before sometime lengthy publication process, so people can look at what you did, without having to wait two years until a sociology journal

> makes a decision and eventually publishes your work. [Participant 15]

> I mean the other thing that we've been doing a current project right now kind of open access that kind of thing it's that we've been writing a report as it going through the project for our industry partners and we've just put them up on the Internet straight away on our project page. [Participant 4]

The frustrations of overly long peer review and processing times for journal publications led one participant to establish and use a personal website as part of their dissemination strategy:

> I'm trying to write in a journal at the moment which is drawn from those reports so we've got some kind of submission for that stuff it's been good in terms of journal under review for about five months, six months and while we've been waiting for that time, because the journal article is still out the on the review cycle so I have to finish off doing the review comments that sort of things after I've gone but if we hadn't put them out by ourselves no one will see any of the results because it's so long, the procedure so drawn out. It can take a year starting from writing a paper until see it in print by that time the industry move on no one cares what you have done any more, you just miss your opportunity. [Participant 4]

A number of participants noted that they published open access in accordance to the requirements of funding agencies:

> The National Health of Medical Research Council…want you to make your article open access, if they fund your research, they want your research to be out there in the public domain to benefit science, to benefit medicine, it's a good will thing, because it's taxpayers' money they want to see results. [Participant 6]

Similarly, other participants noted the need to choose publication outlets to comply with university priorities and policies. In one participant's case this was to *not* publish in open access since not many open access journals were top-ranked by the Scimago Journal & Country rank, an online portal that ranks journals based on information derived from Scopus, which is used by the university as an indicator for their academics when publishing their work. That way, the university encourages their academics to publish in journals that are top-ranked by the Scimago:

> I prefer to publish in Open Access models because it's more accessible but we're under very strong pressure as a university, to publish only in highly-ranked journals, so the message we are getting very loud through official university channels, is that we should be aiming to publish predominantly, in Scimago Q1 journals. Few of those Scimago Q1's that are relevant to me, are actually open access. [Participant 11]

For another participant open access was encouraged:

> With the…article that I submitted with a PhD student earlier this year, our university has a policy where they really encourage open access publishing, because of that reason, that it is easier to be able to access those articles for the general population. [Participant 12]

One participant noted a philosophical imperative to publish open access for research made possible by public funds:

> Philosophically I think it's right that the results of research which have been funded by taxpayers should be freely available. Where possible I have deliberately chosen to publish in open access. [Participant 29]

A desire to retain copyright control over their work was a motivating factor for one participant to publish their works in open access outlets:

> I have recalled another one of the motivations for choosing open access, and that was to keep the permission that we got.

> I wanted to use in other settings and I wanted that not to be owned, copyrighted by the journal itself. We chose to make it open access so that we keep the copyright for that particular one. That was another motivation. [Participant 30]

5.2.3.3 Paying to publish

Across the data, there was a strong thread of discussion related to paying to publish in open access journals and the impact this has on publishing patterns. Publishing open access may involve a fee and can be very expensive. Several participants noted that cost or publishing fees was a key element in their decision-making. The following two comments illustrate this point:

> It really depends on what it is that we're trying to publish and the aims of that particular paper. Predominantly, I can pretty much guarantee you that it won't be open access just because funding is limited. [Participant 31]

> I don't put any money aside for open access publication. Unless I get a particular grant or a particular funding to publish in a certain journal, the general policy in my research area is to go for non-paid journal. [Participant 28]

Building on this one participant spoke of the complex relationship of fees and funding that comes with publishing open access. The following participant's statement illustrates this point by making reference to how they would like to publish open access but can only do so rarely, given their financial constraints. The statement also reveals the power relationships involved in deciding where to publish:

> Sometimes it depends who's paying for it, who's going to pay for it to be published but sometimes I may look for journals which there's no fee for publication. Other times, if somebody I'm working for is prepared to pay, then I'll look for the best-rated journal, for a highly-rated journal so that I can try to improve my research profile. Some of my papers I'm co-author on, so I'm right down the bottom of the authorship so they haven't been 100% my decision. I think for some of my papers,

> in particular the model I created for my PhD, I wanted it out there for everyone to use because I wanted to make a change, in the way that my model created a measure of accessibility and I wanted it out there. I think it depends on what the research is as to whether or not you'd go open access. For me, somebody paid for my PhD. I was on full scholarship. I thought it was the right thing to do to publish it open access. You get a bit confined by who's paying for your research a little bit. They may often determine which journals you publish in and that are what happened with the big projects that I've worked on. [Participant 16]

Few participants in this study were prepared, or more importantly able, to pay a fee for publishing open access. This point significantly impacted on the participants' willingness and ability to engage with open access through publication. As illustrated by the following quotation:

> Generally in the [research topic] they do not really go for that model yet, they likely stick to the traditional journal in a way because it's expensive, if you looking at sometime they charging $2000 to publish and generally for my group I'd like to publish at least 10 papers a year. [Participant 2]

A number of participants noted that the only way they could pay fees attached with open access journals was if someone else covered the cost:

> It's a fair bit of money. The last one that I did was $3,000. I'm just lucky. If you can get somebody else to pay for it, it's good. Otherwise it's really hard. [Participant 16]

Several participants indicated that they were supported by the university library in the costs associated with open access publishing:

> I've got…two open access tokens from the library, and that's [open access journal] publications so they have this token to use to contact the publisher to make it open access, I've done that twice. [Participant 2]

> The library will pay some of these fees, so there's a chance that I can make one my article open access. [Participant 1]

One participant noted that reducing library budgets was impacting on their ability to cover open access fees:

> I noticed they changed just after I arrived here, now there's no more money for publishing in open access journals so I received email from the library I think in September and they told me they have cut the open access thing so. [Participant 3]

Similarly, two participants noted the tension between funding agencies wanting research to be made publicly available via open access outlets, but not providing the funds necessary to make this possible:

> Most of my funding sources don't allow for a publication cost, so this is NHMRC, ARC, don't allow but they want you to publish in open access. If they're funding you to do research, they want your research available to the public, but they will not allow you to use your money to pay to put it in public space. [Participant 18]

> The grants we get from the government want us to publish in open access but they don't allow us to use any of the money we get to do that. [Participant 19]

Acknowledging that their university was unable to meet the costs associated with open access publication, one participant indicated they would still continue to pursue this form of research dissemination:

> I know that some universities will subsidize for publication fees but unfortunately [Participant's institution] does not do that, I've asked them about it, it will be a great idea, but I think they worry about the expense as well, but I'm not going to stop publishing, I'm still submitting manuscript and getting reviewed and do the revisions, but certainly would get [Participant's institution] name out more if it is open access,

> but they said they cannot afford it at the moment. [Participant 6]

Several participants questioned the use of funds to pay for open access, noting that the funds might be better spent in other ways:

> I'm not going to pay from my own pocket and if I have received a grant I won't allocate my grant for open access but if the university pay for it then all good, but if I have small grant for example $30,000, do I really want to spend [$1000 or $2000] to spend for open access when I can spend that amount of money for my research? [Participant 3]

> I weigh it up in two ways. I'm trying to make it easier for people to access especially in those places that don't have the option to access these papers. Yet, I could be employing one... somebody as a research assistant for 12 months to do anything…. I find it a waste of money, and at the same time, I have to do it that way. [Participant 30]

One participant noted that they had learnt the skills of negotiating fees in order to afford publishing in an open access journal:

> I've negotiated with [journal's staff], a reduction in their costs. I think I went through $1,500 to $800. [Participant 18]

The future of open access publishing given the unsustainable nature of the fees was questioned by one of the study's participants:

> I think the fees feel exorbitant, if I was going to give my opinion on it. I feel like the fees are really really expensive, especially for my area of research where it's really hard to get competitive funding, and in the current fiscal environment I don't know how sustainable that is. [Participant 12]

Another participant raised concerns about the ideological issues associated with publication fees:

> The prices are quite expensive. It's like $1,000 or $2,000, $3,000 even. That's something that is very exotic for me. It

> also feels very wrong, especially from the ideological point of view that not only access to information should be free or as free as possible, but also you should also not incur cost it contributes knowledge as well. You're not only receiving but giving the information should be as free as possible. To my mind, this is bad. I think within my specific research community, this is seen as bad as well. [Participant 22]

Researchers' information literacy relating to open access has a strong financial literacy component, with researchers now requiring an understanding of how the publishing industry works and of the financial models associated with publication.

5.2.3.4 Peer review

Participants always considered the journal's publication process, especially the peer review procedure, when publishing. This significantly impacted on the participants' willingness to engage with open access channels as an information resource specifically, and as part of their information literacy broadly. A number of participants noted concerns about the peer review process of open access journals:

> The review process probably wasn't as rigorous as I would have liked. It was a very basic review process. It was only one reviewer and the review was only about a paragraph long. I don't think there was even any changes that needed to be made. [Participant 13]

Some participants identified that it was difficult to determine the quality of the review process for some open access journals, especially journals that are not published by major publishing companies. This view is illustrated by the following quotation:

> I think that the legitimate open access ones is good and they have rigorous review processes. I think it's at lower 50% of the open access journals, the ones that are not hosted by say [publisher] or [publisher] and those sort of legitimate companies. As you go further down the list then it becomes I

> think a bit murky and more difficult to determine the quality of the review process. [Participant 23]

> Deliberately there's no academic rigor or anything going on that makes you think that these journals are robust journals at all. If you have a look into them click on any, but I'll go and search those journals through Google and will then find it's only done one volume, most of the publications were people from China or India or are in high continent, high volume continent, have a hard time in publishing their work they might have money to pay for it. Often the information that the journal article sell might be quite rigorous, it might be really good research but I don't know how robust the peer review process of the work was. [Participant 18]

A number of participants recognized that the peer review provided by open access journals was not always consistent - some open access journals provided good peer review processes, while others did not. The challenge was distinguishing the open access journals that provided appropriate peer review amongst the many journals available. The following participant's statements illustrate this point:

> For some of the better journals, it's good. For some of the other ones, and one open access journal I am thinking of in particular the review process I don't find very rigorous. Part of me wonders, and it is a little bit of a deterrent for our publication because part of me wonders if they just want more publications for more money. This also goes … Sometimes I know within our group, some of the papers that we wouldn't necessarily consider as strong have been sent there after it has been rejected from one or two other journals because we generally know they accept most things. I found in the case where I have reviewed for that journal too, I had rejected a paper for that journal as a reviewer yet it had still gone ahead and accepted them. Part of me thinks if the

> motivation is because every paper they accept they get x amount of dollars. [Participant 20]

> They supposed to have the same process, but it's unclear to me having read one of the open access journals and looking at the quality, some of those are really quite substandard. The more legitimate open access ones which had the appropriate peer review base, do have high standards. [Participant 23]

Participants have mixed experiences in having papers peer reviewed for open access journals. One participant had a positive experience:

> It's all been really easy. It's all been really, really easy. It's just like publishing in any other journal really. You send it in, it gets reviewed, you get some comments, you change it. It's no different to any other. [Participant 29]

Conversely, another participant did not have a positive experience:

> I actually found the process to be incredibly frustrating. I thought that given it was open access and we were paying for the pleasure of publishing in these journals that it would be a little bit easier. Not easier to get through, because I understand that it needs to be peer-reviewed, and I completely agree with all of that. But the time frame, it took close to a year to get our protocol paper published…so they held onto it because they had questions, but they didn't email us with the questions or get back to us, we had to follow that up. Then they took ages to get back to us. It delayed the publication process for something that really didn't need to be delayed, which was really frustrating. Then given we had to then pay them, I thought that was quite annoying. [Participant 31]

5.2.3.5 Social media

Social media played a significant role in the dissemination activities of many of the participants. The following comment illustrates how one participant intentionally used social media to ensure their scholarly work was

seen by the audience they wanted to connect with. It also reveals the shifting boundaries between personal and professional spaces, as they acknowledged Facebook is becoming more a part of their professional presence:

> I never used to be involved in Facebook with professional things, but there are so many of the community organizations that really have strong Facebook presence, so I found it's a good way to make sure they find out about the latest research. [Participant 10]

Several participants observed direct and tangible impact in using social media to make their research visible and available. This point is illustrated in the following two quotations. The quotations also reveal the slightly different experiences participants have in promoting their work with social media. While the latter participant had their research work listed on a popular research social networking site, they did not proactively promote this fact. In contrast, the former participant used social media in a more proactive and intentional manner, combining social media with other systems, in this case their email client, to increase visibility of their work.

> I saw a big uptake in people reading, like downloading off [institutional repository] once I started doing adding that link in my signature. I notice whenever I tweet about an article that I've published, I see a spike. [Participant 10]

> I have had a lot of, I am a member of Researchgate, and I have a lot of requests come through Researchgate for publications that aren't available on open access and I send through copies. But I don't actively promote. [Participant 31]

5.2.3.6 Conclusion

Selecting where to publish is no longer just about choosing the "best" journal in the field. Deciding where to publish is a nuanced decision, with researchers needing to balance a complex array of criteria and this process is part of a researcher's information literacy broadly. Arguably, open access has made the process of selecting where to publish more complex. In the age of open access, making a decision about where to publish requires researchers to

have an understanding of research impact, the publishing industry, and financial models related to publishing. These understandings form part of their information literacy, as does the process of using these understandings to make a decision about where to publish.

5.2.4 Theme 4 - Finding information

This theme portrays an aspect of the participants' experience of open access as part of their information literacy that was related to finding open access information. As informed researchers, participants found, evaluated and used information as part of their research, for example finding information as part of a literature review and the practice of keeping up to date with the latest research in their field. While they may not actively seek out open access information, participants did engage with open access resources as part of their research. The following comment illustrates how one participant used different open and closed access sources to locate the information they needed. The comment reveals the participant's awareness that different types of information, required different ways of searching, via different information channels:

> I search in many different ways it depends on the information I want to find, if I want to find information…publications by particular author I'll go to [institutional repository] if it is author [is from] [participant's institution], if it's different kind of authors I usually try to find their personal website from the university where their at, because they often list their publication, they often have extra information about their publications which is very useful, so probably wouldn't normally use a database for an author search, but that's not always the case. I've used [University library catalogue] to do a quick author search. It depends on whether I want to find little bit information about somebody or whether I'm really trying to get particular result. [Participant 1]

The ways participants found information are discussed below.

This theme also reveals that, when searching for information, participants may not need, and may not be able, to differentiate between open access and closed access information resources. As such, participants' experience of finding open access information was often a part of their broader experience of finding information, and they may not ever actively seek out open access information in particular. In this way, when it came to finding information, participants' experience of open access as part of their information literacy was often not specific to open access. Aspects of the participant experience revealed in this study are: Google as the preferred search tool, using social networking sites, current awareness, finding open access information, distinguishing between open and closed access, assessing quality, and the role of open access information.

5.2.4.1 Google as the preferred search tool

Google was one of the most common tools used by participants to find information. Several participants noted that Google Scholar was their preferred first step when looking for information. The following quotation illustrates this point and reveals the participants' understanding of the mechanics of searching for information via databases:

> My first step when I'm searching literature is usually to search Google Scholar and all of the databases that it indexes. I found that to be a really good approach in my research compared to some of the other proprietary databases. For me particularly, this has been useful because Google Scholar does a much better job at indexing all of the interdisciplinary literature that I study. I study not only, like from law, but I also draw on a lot of interdisciplinary work in media and communications and in other areas of research. For me, using one unified database like Google Scholar is better than going to subject matter databases, like Lexis for Law, or in other social science databases. [Participant 11]

Participants appreciated that Google – and particularly Google Scholar – offered them the opportunity to search a bigger database than any single

proprietary database. They also valued that it gave them access to a range of literature that was not indexed in academic databases:

> You can get to grey literature quite easily through Google Scholar, which is hard to do through other means. That's really important in terms of reports and things like that…Google Scholar is a good start. [Participant 29]

Building on this, the following statement shows how one participant relied on Google to locate information they were unable to locate in any other way:

> [I]f I'm looking for something specific and you pretty sure that it's out there and you haven't found it in any other way maybe I'll start through Google to make it come up. [Participant 8]

Another participant noted that Google Scholar was a means to access content available via institutional repositories without having to go to each repository:

> Generally speaking, I wouldn't deliberately go to [institutional repository] as a starting point, but I would end up using [institutional repository] copies of articles as a source of things so if I found articles on Google Scholar that weren't available without paying, so try to find them on [institutional repository] will be the next thing. [Participant 4]

Not only does Google Scholar provide participants with access to a huge database spanning both academic and grey literatures, but it also provides functionality that is highly valued, including the search functionality:

> Plus you can do lots of logic on Google Scholar. It's a really good way of excluding material as well as including it. If I want to search for something on terms of science communication, I can exclude just communication, which I don't want to know about. I like the logic structure in Google Scholar. [Participant 29]

Other functionality helped participants stay abreast of the latest, and in this case highly cited, research in their field:

> I can't read everything, when you are doing a search particularly online, like Google Scholar…if you looking particular topic and you think, that's paper interesting and you go to the abstract, on the right side of the screen it will tell you who has cited it, and who does cite it, so you know where that's going and you might find more recent, so that helps to extend the search…it's right there in the finger tips, you just go to Google Scholar, I've bookmarked Google Scholar so I go straight there, find my hit and have a look at the abstract and it tells me who has been citing it. [Participant 7]

5.2.4.2 Using social networking sites

For many participants in this study, social networking tools were not just a means to stay in touch with colleagues, but also a means of finding information. The following statement reveals how the participants were able to keep up to date with the latest research by monitoring social networks for key scholars. This included monitoring social network sites for new content from authors in their field:

> Because I have been sticking with the same topic for a while, I sort of know who the key researchers are, so I just keep an eye on their publications via things like ResearchGate. Or I will do a specific search for them and have a look at their institutional page, and just check if they published anything recently. [Participant 13]

It also involved actively searching social network sites for content related to their research when they had not been able to find anything relevant through their library's search functionality:

> If I can't find something through the [University Library] website I'll also often go looking on academia.edu or other sites like that, or even people's personal sites to see if they've put up a pre-publication version. [Participant 14]

5.2.4.3 Current awareness

Academics used a range of strategies to keep up-to-date with research in their field, from using social network sites (as noted above), to subscribing to RSS feeds, to regularly checking journal websites.

Some participants developed their own list of journals to follow and check on a regular basis. The following quotation illustrates this point, and also reveals that the participant was aware that their understanding or knowledge of where to find high quality information for their needs would evolve over time:

> You kind of build that up I think. I'm, technically speaking, still an early career researcher. I'm only three years post-PhD, so I'm not as knowledgeable as some people. I think it's just something that you build up really. You start with the major journals in your field. Then you follow the references in those papers, and that leads you to a different journal. You build up a sense of what's where. I know for example, if I want to look up something that's to do with research about the Internet, I'll use First Monday, which is a journal of Internet research. If I want something about public engagement with research, I'll use Science Communications. You build up that skill and that sense of what's going to be where over time I think. I personally have a reminder on my computer so that once a month its says, “check journals”. I have a list of journals. They're the major journals in my field and a few that I'm interested in. I just click on the recently published. It's quick to look once a month at 2 or 3 papers than to try and plough through a backlog. That way I find it keeps me abreast of what's happening in relatively little time. [Participant 29]

Other participants used RSS feeds for key journals so that they did not have to manually check journal websites.

> I use RSS feed from journals to keep myself updated, I just don’t manage to follow everything but as you can see I view

> thousands of these scientific journals and few thousands of thing that I have to read and decided if I'm interested or not because I just cannot catch up with information because it is too fast, and my interest is wide, so I use this kind of way to try to stay current. [Participant 3]

5.2.4.4 Finding open access information

Searching for open access information was not something that participants intentionally did. However, participants recognized that open access information would frequently emerge from the searches they conducted, and was easy to retrieve. This point is illustrated via the following quotations:

> It's not so much that I specifically choose to use open access information. It's more that these days an increasing number of journals will publish their articles in an open access format. And some journals are completely open access. Like [Open Access journal] is a completely open access journal. So, when you get journal results in the Medline searches, some of them just happen to be open access. [Participant 21]

> I don't necessarily look actually for open access articles, but if I do find an article that's of interest, it's very easy to get the full text because it is open access. I find that now that a lot of research is published in the open access journals, so it's very convenient. The databases will be able to have a link straight to the full text, full text articles. My experience is that it's quite convenient to use and able to find the full text without any real issues or having to do any in-depth search of the Internet or ask the librarian to find the full text. It's just all there with one click. [Participant 28]

5.2.4.5 Distinguishing between open and closed access

Many participants revealed they found it difficult and even unnecessary to distinguish between open and closed access resources. This was mostly because they were logged into the university's system when searching for

information. This meant that the full text would be available to them even if it was closed access, so long as their library had a subscription. This following quotation highlights this point:

> When I am searching through the university computer, and we've got subscriptions to the journals, it's not always possible to tell when something is open access or not. [Participant 20]

This had implications for the way researchers experienced open access as part of their information literacy because it essentially blurred the boundaries between open and closed access resources. The researcher did not need to distinguish between open and closed access, and even found it difficult to do so. It could be argued that open access became invisible because the participants did not know when a resource was available to them as a result of it being open access, and when it was available because the library subscribed to it.

A number of participants spoke about not needing to think about, or intentionally search for, open access information because of their work context. The following quotation illustrates this point. It also shows that the participant was aware that not all researchers had the same workplace advantages in accessing information:

> We're very lucky here because [participant's institution] library would pay most of the journal subscription, and they pay for the big one really impressive one like the [Subscription-based journal], and we have a good subscription with Elsevier, so typically when I find something about [research topic] that I want to download we have the subscription to it anyway so it doesn't matter to me at this point in my career whether it is open access or not, but if I'm in a small university that wasn't pay subscription to these publishing it would be matter a lot more, I've been looking at journals at the other day and because I haven't log on at the [Participant's institution] so I was an external source and it was charge me 40 pound for one article, so that almost $80 these days, so if I was in small university I couldn't afford

> that, so that's why it's great you making your research more available to I guess poor university that researchers struggling more on financial sort of thing, these people are really…they're trying to do research and they're really struggling because lot of instances lack of infrastructure whether it be lab space but also just accessing the publications, so I don't worry about that, I don't go only looking at open access publications, if it is something in an open access or a commercial journal it doesn't bother me I can download both. [Participant 6]

As the quote above illustrates, many participants demonstrated awareness that not all researchers had access to the same range of information. The importance and value of open access information was acknowledged by several participants, as evidenced by the following quotations:

> They do seem to publish a lot more articles, so maybe in that sense, they're actually more comprehensive in terms of there are more articles available in all different areas as opposed to the subscription journals, which tend to have a very, very clear and precise scope and focus, whereas open access tend to have a little bit more grit in their type of articles that they'll publish. [Participant 28]

> Having those open sources once comes straight to me, it saves my time looking for information in my field. I don't have to think about it. [Participant 8]

5.2.4.6 Assessing quality

The overall quality of information made available via open access channels was raised as a concern by several participants.

> If I'm just doing a search on Google Scholar or wherever and an interesting paper comes up that's been published in an open access journal, I'm usually a little bit cautious, to be honest. I'll usually read it, but it depends on what open access

> journal it is. So there are some that I think are less rigorous than others, so I will sometimes avoid things that have been published in there. [Participant 13]

In particular, several participants expressed concerns about quality because authors may have paid a fee to get their article published. There was some distrust of the open access journals that require payment to publish, as illustrated by the following quotations:

> I could tell that some problems that, because they just want to charge a fee by the authors, so in that case, they can be quite easy to accept a paper, and in that case, it probably won't be good studies. [Participant 27]

> They are the ones effectively that are predatory that generally have a title which is similar to reputable journals. But they basically accept any article for payment, and the quality of those particular articles are substandard. [Participant 23]

Building on this point, several participants noted the need to be vigilant and check for further detail relating to the information they found via open access, as evidenced by the following participant's comment who was wary of unfamiliar journals:

> I'm pretty sure I've clicked on articles that are freely available, certainly media articles like *The Conversation*[2] for example. I've used those sources, so yes I have accessed what I would consider open access resources, but most of the time it's actually using, it has to be a reputable journal, and some are open access. This information is freely available there from bogus journals and stuff, so part of it is about trying to make sure that it's actually an appropriate academic source….Yeah, certainly I would be looking at the journals and

[2] The Conversation is an independent source of news and views, sourced from the academic and research community and delivered direct to the public. The Conversation is available from https://theconversation.com/au/

> if it's a journal I'm not familiar with I'd probably look into it a bit more. [Participant 24]

Familiarity with journals can result in participants trusting those journals, with two participants noting that they would use information found in a journal if they had prior knowledge of the journal, its publication processes (especially peer review), and/or the authors who publish in it:

> I find that it's a really good journal. I find that the articles published on it are well written, and they're from reputable scientists. So, they're people who are well renowned in the field, and more published articles are in that journal. And, you know that they work is good. [Participant 21]

> I think there are quite a few dodgy ones out there now, but I think if I know it's a reputable one, and the [Open Access journal publisher] ones are probably the only ones I'm particularly familiar with that are open-access that I would be confident that they got to a reliable process. [Participant 12]

One participant suggested that quality was not black and white, with many closed access journals having issues in regard to maintaining quality:

> I don't personally think there's a quality difference between "Open Access" and "not Open Access", like not that I've found. I'm sure there are, there are crap Open Access journals, but there are crap journals that aren't Open Access too…. So I don't know that that's… to me it's not like all Open Access are good or bad, so I think it's just like every other journal that you have to be careful, you know. [Participant 10]

Building on this point, one participant noted that they used the same process for determining the quality of information in both open and closed access sources. They also noted that the focus of their critique was the specific article not the whole journal in which it is published, revealing the participants nuanced understanding that quality was a multi-layered concept,

and that the quality of an article and a journal were not always mutually dependent:

> I would critique the article like I would with any other article around the research methods and things like that. If I know that it's a reputable journal, and on their web page they clearly articulate what their process is for review of their articles, whether it's blind, peer review or whatever, then I would consider that I'm more interested in the article itself, so if I read through the article and I can see it's methodologically flawed or there's problems with it, then I would probably critique that within whatever I'm writing, rather than critique the journal itself. As far as I'm concerned, if the journal doesn't pop up in my usual database searches, say through Scopus, then I would be worried about why it doesn't. [Participant 12]

Another participant focused their selection of information resources based on the credibility of the journal. The following participant's statement illustrates this point by making reference to how they pay attention to the articles published by journals as a guide to the overall quality of a journal. The statement also reveals the participant's awareness and understanding of the interconnections and interdependencies between journals and the corporations that provide financial support to the resource:

> I guess when you're looking at the journal, you're looking at… what sort of things are they publishing, what's their funding. There's a couple that I know in my field that is particularly funded by the drug companies, and so I wouldn't trust anything that they publish. [Participant 10]

Importantly, many participants regarded the review process as a guarantee of the reliability and the quality of a journal. Two participants commented that the extent to which they selected information resources was significantly dependent on the rigor of the journal's peer review process. This point was illustrated by the following two quotations:

> The research articles that I use seem to be in peer reviewed journals. So they're open access, but they're still peer reviewed. It's the peer revision that makes them reliable and makes sure that they've been checked out. [Participant 21]

> It depends on the material. Sometimes it's papers that will have been peer reviewed and have been through the whole process. Again, you've got that slight of value there or at least rigor. Not necessarily value. Rigour would be a better word. [Participant 29]

The role of colleagues in helping to determine journal quality was highlighted by one participant. The following statement reveals how the participant was supported by colleagues to locate quality resources and to establish a process for locating good quality journals. The statement points to the collaborative or social nature of identifying appropriate information:

> My PhD supervisor was in the information systems. So you've had processes that they help you find or understand what is the reputable journals in your field. And sometimes, you find additional ones, and you discuss it with colleagues. But I think that's kind of the process. [Participant 25]

5.2.4.7 The role of open access information

The role of open access information was discussed by a small number of the study's participants. For one participant, information obtained via open access served only as a background and setting the context to their work:

> Some of the open access stuff is a little bit too general. It's good for background information and that sort of thing, and can give a lot of contexts and can drive some of the research questions that we dive in a bit further into trying to answer. It can set the scene and let us know how we're going and that sort of thing. [Participant 16]

Another participant observed that open access information was useful for obtaining access to information they may not otherwise have access to:

> Absolutely. Because like I said, sometimes I can't find things that are published in regular journals but the green open access or the author's final version is available to me, where my institution might not have a subscription to that particular channel. So yes, it's particularly valuable to me, the green open access. [Participant 25]

Finally, another participant noted that the limited range of open access journals in their field of study resulted in open access information having reduced benefit to them:

> Open access didn't really help me because… there weren't some key areas for me. [Participant 9]

The role of open access in participants' experience of finding information varies according to a number of factors, including discipline.

Social media enabled a culture of sharing, with many participants commenting on how social technologies made it easy to ask for, and distribute, papers. The following quotation illustrates this point:

> I use ResearchGate as another source, and it's not fully open access, but a lot of people are willing to share behind the scenes. Even all about Elsevier publications are not allowed to be posted in ResearchGate, but we can certainly privately email things between authors, and I find that brilliant because you can just quickly send. With ResearchGate, once it's uploaded once, you can share it. It stays behind the scenes, but you can share it. You don't have to keep uploading it for people. It's a good little sneak around…It's open and I've discovered people are very, very willing to share and it's just a matter... I also, I guess, as another open access source is that I email a lot of authors. I don't mind to email them and say, "Could you share this paper? I can't access it." They are always willing to email a copy. I do the same. If anybody asks, I always make an effort to share back because it's frustrating when you can't get something you need. [Participant 30]

5.2.4.8 Conclusion

Finding information was a core aspect of researchers' experience of open access as part of their information literacy. Importantly, as participants generally did not distinguish between open and closed access when searching for information, their experience of this theme had elements that are likely to be consistent with their experience of finding information generally. Most participants did not actively search for open access information, and many participants discussed the fact that they did not need, or were not able, to distinguish between open and closed access resources. This reveals something important about participants' information literacy: while they use open access information to learn, they do not seek it out specifically, and they may not be able to identify it. They were, however, cognisant of the fact that they were likely to find open access information when conducting their searches; they just may not be able to identify it.

5.3 Relationship between themes

The four themes presented above are related in different ways. Some themes provide foundations for others, some facilitate experiences in other themes, some occur through experiences of other themes, and some hinder experiences in other themes. Table 6 (overleaf) presents a broad overview of the relationships between the themes.

Themes	Short description	Relationships to other themes
Understanding Open Access (Theme 1)	Participants develop their understanding about open access and how it influences their behaviour on finding and publishing practices	*Understanding Open Access* is the core theme that influences all other themes. Influences: • Institutional Repositories • Publishing and Dissemination • Finding Information
Institutional Repositories (Theme 2)	Participants engage with institutional repositories for variety of purposes, that include finding information and promoting research	Influenced by: • Understanding Open Access • Finding Information • Publishing and Dissemination
Publishing and Dissemination (Theme 3)	Participants select where to publish by their identification of the journals, and how open access played a role in influencing their selection to publish.	Influences: • Institutional Repositories Influenced by: • Understanding Open Access • Finding Information
Finding Information (Theme 4)	Participants identify many and varied ways of finding and selecting information resources, including information from open access sources.	Influences: • Institutional Repositories • Publishing and Dissemination Influenced by: • Understanding Open access

Table 6. A broad overview of the relationships between the themes

Understanding Open Access is the core theme in this study's findings. It influences all other themes. It is the way participants understand open access and thus creates a foundation for their experiences of other themes. The way a participant understands open access informs, for example, the way they engage with it when *Publishing* and *Finding Information*.

The theme *Institutional Repository* depicts an aspect of the participants' experience of open access that is influenced by all other themes. Participants engage with an *Institutional Repository* as a result of their understanding of open access and a sense of what it can do for them. *Publishing and Dissemination* practices also impact on the way a participant

engages with the institutional repository. For example, a participant who publishes an article in a closed access journal may deposit it in the institutional repository to ensure it is widely available. In another example, participants who are aware of the role and function of the institutional repository may use it when they are *Finding Information*, particularly publications of colleagues or particular types of publications, like theses.

The theme *Publishing* is also influenced by other themes, especially the theme *Understanding Open Access*. Participants' understanding of open access, particularly its benefits, has influenced them to select open access journals when publishing. *Finding Information* also impacts on the way participants select where to publish, for instance, for a participant who uses Google or Google Scholar for finding information, open access articles may be included in the search result.

The theme *Finding Information* both influences other themes and is influenced by them. For example, it influences participants to deposit in the *Institutional Repository* because they are aware that material deposited there will surface via Google and therefore be widely available.

The experiences that occur in these themes can relate to other themes. These have been portrayed in the table as "influences" and "influenced by". Some themes are influenced by other themes. For example, *Understanding Open Access* creates the conditions that enable the experience in some other themes to occur. In this way, it influences experiences in other themes. For example, *Understanding Open Access* influences *Institutional Repository*, because it is the understanding of open access that creates these experiences, as illustrated by this quote:

> I do that before it was required. And I do it. And now it's not required to put an open access version in. I still do that wherever the copyright agreement allows me to, because I think if your work is available, it's more likely to be read. [Participant 25]

5.4 Conclusion

This chapter has presented the findings that emanated from the study. It has described the participants' experience of open access as part of their

information literacy, via four themes. These themes were: Understanding Open Access, Institutional Repositories, Publishing and Dissemination, and Finding Information. The next chapter will provide a discussion on the findings in terms of the different contributions they offer to theory and practice.

Chapter 6: Discussions and conclusion

6.1 Introduction

This chapter discusses the contributions this study provides to both theory and practice. The chapter begins with an overview of the research outlining main findings from the work. The next section presents the contribution the study's findings make to existing knowledge by positioning the study within the existing empirically derived literature base. It then continues with a discussion of the contribution this study makes to practice. The chapter concludes by outlining the study's limitations and suggested directions for future research.

6.2 Research overview

This study responded to the following research question: "How do Australian researchers experience open access as part of their information literacy?"

Emanating from this question, the study's objectives were identified as follows:

1. To identify Australian researchers' experience of open access as part of their information literacy
2. To provide descriptions of how researchers experience open access as part of their information literacy
3. To develop greater understanding about how researchers experience open access as part of their information literacy

The phenomenon being explored in this study is *open access and information literacy*. In this study, open access refers to free and unrestricted online access to peer-reviewed research articles without financial and legal barrier (Budapest Open Access Initiative, 2002; Bailey, 2007; Willinsky, 2006; Suber, 2015). The study adopts a relational perspective to information literacy (Bruce, 1997) which focuses on the various ways people relate to information and interact with elements of their information worlds. Through this

perspective the study does not seek to measure information literacy but instead seeks to explore and understand the complex (i.e., behavioural and experiential) relationship between people and information and their information worlds.

Using semi-structured interviews and thematic analysis, the study identified four themes that represent how the Australian researchers who participated in the study experience open access as part of their information literacy. Together the four themes reveal how participants' experience of open access as part of their information literacy relates to their role as informed and informing researchers. The term "informed researcher" relates to the participants' roles as information users, while the term "informing researcher" relates to the participants' roles as information producers. Across the four themes, it is through participants' roles as informed and informing researchers that their information literacy related to open access is enacted. The themes are:

Theme 1 - Understanding Open Access

Understanding Open Access explains the participants' understanding of what open access is and how it can benefit them as informed and informing researchers. Understanding what open access is, including the nature of open access publications, ways of accessing them, and the economic models that drive them, is an important aspect of a researcher's information literacy. This understanding helps them to navigate the practice of publishing and informs resource selection when they are looking at the research literature. In this study, participants understood "open access" to be "freely available".

Theme 2 – Institutional repositories

The participants engage with institutional repositories as both informed and informing researchers. As informed researchers, they use information found in repositories as part of their research. Institutional repositories enable the participants to interact with other scholars in their field through the publications other researchers have deposited. As informing researchers, they deposit their work in institutional repositories to enable other academics,

industry, and society at large to access and benefit from their work. Participants report on how institutional repositories play a role in allowing them to communicate their research, and to build their profile as a researcher.

Theme 3 – Publishing and dissemination

As informing researchers, participants make decisions about where to publish their work (in closed or open access sources), they also make judgements on using social media to disseminate their research. Each participant revealed their own preferences and insights on selecting where to publish, and the role that open access played in sharing research.

Theme 4 – Finding information

As informed researchers, participants find, evaluate, and use information as part of their research. For example, finding information as part of a literature review, and the practice of keeping up to date with the latest research in their field. While they may not actively seek out open access information, participants do engage with open access resources as part of their research.

6.3 Contribution to knowledge

The study's findings contribute to knowledge in two research spaces: open access and information literacy. Specific details are provided below. A table is also provided for each discussion of how the study contributes to research.

6.3.1 Open access

A growing number of studies explore the open access phenomenon. This research has focussed on the issues of impact and user perceptions. Table 7 (overleaf) presents a comparison of the current study's findings with the existing knowledge base relevant to the field of open access, with particular focus on impact. The current empirical base reveals that publishing open access has a positive impact on citations, with open access articles being cited more than closed access articles (see Koler-Povh, Juznic, & Turk, 2013;

Cahudhuri & Thoira, 2010; Norris, Oppenheim, & Rowland, 2008; Dorta-Gonzales & Santana-Jiimenez, 2017; Tang, 2017). The current study did not directly investigate whether open access increases citations. However, the study does reveal that many of the participants interviewed believed that open access would increase their citations or help raise their external profile as a researcher. The participants also noted that disseminating via open access channels would allow them to share their work more easily to desired audiences, such as industry.

Author/s	Article titles	Key findings	This study's findings
Koler-Povh, Juznic, & Turk (2013)	Impact of open access on citation of scholarly publications in the field of civil engineering	• Open access publications have a positive impact on citations of articles, with open access articles being cited more than closed access articles.	• Many participants acknowledged that open access publishing was a way to increase their citation count or to increase their profile as a researcher. Open access also provided a means to share their work with those in desired communities such as industry who may not be able to obtain closed access channels.
Gaule & Maystre (2011)	Getting cited: Does open access help?	• Open access does not necessarily increase the number of citations but publication quality and journal readership does.	
Chaudhuri & Thohira (2010)	Usage of open-access journals: Findings from eleven top science and medical journals	• Open access journal examined had an increase in usage over years.	
Norris, Oppenheim, & Rowland (2008)	The citation advantage of open-access articles	• In the subject areas of ecology, applied mathematics, sociology, and economics there is a citation advantage in publishing open access instead of closed access. • Participants had challenges in locating open access journals via search engines.	
Dorta-Gonzales & Santana-Jimenez (2017)	Prevalence and citation advantage of gold open access in the subject areas of the Scopus database	• Open access journals from the multidisciplinary and medicine subject areas have high citations numbers.	
Tang, Bever, & Yu (2017)	Open access increases citations of papers in Ecology	• Open access articles from the field of ecology received more citations that	

		the non-open access articles every year from 2009 to 2013.	
Ottaviani (2016)	The post-embargo open access citation advantage: It exists (probably), it's modest (usually), and the rich get richer (of course)	• Good quality articles received more citations when they were open access.	
Chua et al. (2017)	The impact factor of an open access journal does not contribute to an article's citations	• Open access publications received more citations compared with non-open access publications.	
Sotudeh, Gahsempour, & Yahgtin (2015)	The citation advantage of author-pays model: The case of Springer and Elsevier OA journals	• The number of open access papers has been growing more rapidly compared to toll access papers.	

Table 7. A comparison of the current study's findings to the existing knowledge base on impact of open access and researchers

Table 8 (overleaf) presents a comparison of the current study's findings to the existing knowledge base, relevant to the field of open access and with particular focus on the user perspective. A number of key observations can be made.

Firstly, recent empirical work by Mammo and Ngulube (2015) found that academics had a positive attitude to open access. However, a number of studies focussed on specific disciplines had contrasting findings. For example, Chang (2017) found that library and information science researchers still tended to publish in closed access channels. Building on this, McDonald et al. (2016) found disciplinary differences in academics' attitudes to open access. They observed that academics in the business discipline were the most sceptical, whereas academics in the sciences and health disciplines were the most positive towards open access. In the middle ground, with mixed attitudes to open access, were academics in the arts and humanities disciplines. The current study's findings provide further support to the McDonald et al. (2016) findings, with participants revealing variation or a spectrum of feelings toward open access. Some were highly positive to open access indicating that it was their preferred dissemination approach, others were strongly against open access indicating that they would only ever publish in open access as a last resort. However, the vast majority of the participants in the current study expressed a mixed view, in that they recognised the value of open access but also expressed some degree of scepticism or concern.

Secondly, the findings from the current study support the work of Lwoga and Questier (2015), Mcdonald et al. (2016), and Massoro and Othman (2017) in that the participants expressed concerns about the quality of open access journals, especially with respect to the peer review process. In addition, concern was noted about the user pays approach common with some open access journals. These concerns directly influenced the participants' decisions on selecting where to publish and choosing which information sources to use. Like St Jean et al. (2011), the current study showed the important influence of management or employers as well as colleagues and supervisors in determining where to publish (e.g., open or closed access).

Thirdly, this study supports Mammo and Ngulube (2015), finding that researchers did not intentionally search for open access information. More specifically, the current study showed that participants felt it was difficult and even unnecessary to distinguish between open and closed access information sources. Participants in the current study revealed that they found it difficult to determine when an information item was found was open or closed, due to the access to subscription databases provided by their institution's library.

Finally, the current study supports the work by Stanton and Liew (2011), with participants acknowledging the value of institutional repositories but expressing confusion about copyright and determining what version of an item could be submitted, as well as frustration in the functional aspects of using institutional repositories. The current study's findings also support the work by Borrego (2017), with participants expressing a preference to upload their scholarly works in social networking sites such as ResearchGate and Academia.edu.au, as these sites were easier to use and provided better reach or connection to desired audiences. As in the work by St Jean et al. (2011), participants in the current study had many different reasons for using institutional repositories, including it was mandated by their employer and wanting to share with their scholarly peers. However, unlike the work by St Jean et al. (2011), the participants in the current study did not express any confusion between institutional repositories and commercial databases available via the library.

Author/s	Article titles	Key findings	This study's findings
Stanton & Liew (2011)	Open access thesis in institutional repositories: An exploratory study of the perceptions of doctoral students	• Levels of awareness of doctoral students for open access and institutional repositories was low even though there was an institutional mandate. • Understanding of the benefits of open access and using the institutional repository but concerned about copyright and making submissions.	• Reveals researchers understand open access and how open access can benefit them as informed and informing researchers but had concerns about copyright and loading correct materials to the institutional repository. Frustrations also expressed about the difficulty in loading items to the repository due to systems issues.
Mammo & Ngulube (2015)	Academics' use and attitudes towards open access in selected higher learning institution in Ethiopia	• Participants have a positive attitude to open access, indicating an interest to use it in the future. • The open access resources they found would come up in their searchers although they admitted to not intentionally looking for open access per se.	• Participants did not have a black and white view of open access (e.g., positive v's negative). Instead there was a spectrum of valuing open access through to concerns about quality. • Many participants acknowledged that they did not intentionally look for open access • Participants felt it was difficult and even unnecessary to distinguish between open and closed access information sources and found it difficult to determine when an information item found was open or closed due to the access to subscription

			databases provided by the library.
Lwoga & Questier (2015)	Open access behaviours and perceptions of health science faculty and roles of information professionals	• Participants were aware of open access but many did not disseminate their work through open access channels. • There were a number of barriers to open access including awareness, skills, author pay, copyright and plagiarism concerns.	• Participants in the study were all aware of open access. • Many participants indicated using open access to find and share scholarly information however concerns were raised about publishing fees and costs, copyright and the quality of the peer review process, and impact factor. • Participants indicated that they published open access to make their work more available to key communities that would not access their work in closed access options, as well as to increase their citation counts and to build their profile as a researcher. Some spoke of publishing open access solely to meet the requirements of funding agencies. • Many participants also spoke of the need to still publish in closed access channels due to reasons of career progression or because of concern for the quality of the peer review process in open access journals stopping them from pursuing this type of

			dissemination.
St Jean, Rieh, Yakel, & Markey (2011)	Unheard voices: Institutional repository end-users	• Participants understood institutional repositories at different levels. • Participants' reasons and method of using institutional repositories were diverse. • Confusion about the relationship between institutional repositories and library databases.	• Participants' reasons for using institutional repositories varied, from it being a mandated requirement from their employer to wanting to share their work with others. • Participants expressed frustration in using the institutional repository because of confusion with copyright and knowing which version of an item to deposit and because of the poor system functionality.
Massoro & Othman (2017)	Antecedents for actual usages intention of open access journals in agricultural research institutions in Tanzania	• The most influencing factors for publishing in open access journals are colleagues, management, subordinates, and heads of departments. • Predatory publishers were strong determinants for not using open access. • The mistrust among researchers about the content of open access journals.	• Participants noted that the requirements of funding agencies and their work context (e.g., support from employers) significantly influenced their choices around where to publish especially in terms of open access fees. It also influenced their depositing of items in institutional repositories (e.g., the need to be seen by their employer to be research active). • Many participants expressed concerns about predatory journals, noting that this was a potential detriment to publishing

			open access.
Chang (2017)	Comparative study of characteristics of authors between open access and non-open access journals in library and information science	• Findings indicate that traditional scholarly publishing (e.g., closed access) has not faced obvious threats from open access publishing. • Academics in the Library and Information Science (LIS) discipline still preferred to publish in closed access.	• Participants in this study expressed mixed behaviour in regard to open access and closed access publishing. A small number of participants were strongly in favour and indicated that they would only publish in this channel in the future. Many of the participants however in this study expressed some degree of scepticism of open access channels, for a small number open access journals were a "last resort" to publish.
McDonald, Gibson, Yates, & Stephenson (2016)	An exploration of faculty experiences with open access journals publishing at two Canadian Comprehensive Universities	• Academics in the science and health disciplines were the most positive about open access journal publishing. Academics in the arts, humanities, and social sciences had mixed feelings in their perceptions of open access. Academics in the business field were the least positive. • Academics expressed concern about the quality and costs of open access journals.	• Participants in the study expressed variations in opinions on open access, with some highly positive, some highly sceptical and many sitting in the middle with mixed views. • Almost all participants indicated concerns about the quality of open access publishing, especially the peer review process and the costs associated with open access channels.
Borrego (2017)	Institutional repositories versus ResearchGate:	• Academics in the study tended to upload their scholarly outputs to ResearchGate more than their institutional repositories.	• Academics used institutional repositories to share their scholarly outputs, however many indicated a preference to use

	The depositing habits of Spanish researchers	• Two primary reasons for this were lack of knowledge of the existence of or how to use their institutional repository, and the advantages of using ResearchGate were stronger.	social networking sites such as ResearchGate or Academia.au as these sites were easier to use and provided better reach to desired audiences.

Table 8. A comparison of the current study's findings to the existing knowledge base on the user perspective of open access and researchers

6.3.2 Information literacy

Table 9 (overleaf) presents a comparison of the current study's findings to the existing knowledge base relevant to the field of information literacy. A number of key observations can be made.

Firstly, as per previous empirical work (Nicholson, 2017; Williamson et al., 2007; Green & Macauley, 2007), the participants in this study had a strong preference for finding information via Google and Google Scholar. The current study however also revealed that the participants frequently found information through social networking sites such as Research Gate and Academic.au, RSS feeds, creating their own list of key journals, regularly checking journal websites and through institutional repositories.

Secondly, participants in the current study indicated a preference for using social networking sites over institutional repositories. This preference arose as the functionality for the former was easier to navigate and use. Issues of having to manage the complexities of copyright were also identified as a barrier to using the technology (e.g., not knowing which version of a publication could be made available). A similar finding was noted by Roos (2005) when researchers selected where to search for information. Ease of access and use were the most important conditions for researchers when deciding both where to search and which items to read. Nicholas (2017), and Borrego and Anglada (2016) found that the use of social media to find information was popular for early career researchers, and education and behavioural science researchers.

Thirdly, a study by Zoellner et al. (2015) noted that researchers would often rely on graduate students to compile bibliographies. Although the current study did not replicate this finding, it did reveal that participants would rely on others, such as work colleagues, supervisors, and scholars in their field to navigate their information world, for example selecting where to publish and where to find information. The institutional repository was identified as a core vehicle to keep up to date with their colleagues' latest publications – in this way it acted as a kind of bibliography, although compiled by scholarly peers on their own works, and not graduate students.

Fourthly, the findings from this study support and extend the work by Nel and Fourie (2016) who observed that the information needs of researchers were influenced by the research environment and expectations of research output. The participants in the current study noted that requirements of funding agencies, and their work context significantly influenced their choices of where to publish, especially in terms of open access. They also influenced their decision to deposit items in institutional repositories, for example the need to be "counted" or seen by the university to "research active".

Fifthly, the current study supports the findings of Borrego and Anglada (2016). They observed that scholarly journals were the preferred mechanism for disseminating research, with open access being a factor of marginal interest in determining where to publish. For several participants in the current study open access was seen as a last resort for publishing their research findings, a place to go only when other closed and frequently higher ranked journals had been tried. The participants in the study had varied ways of determining where to publish, including impact factor, cost, the quality of the peer review process and where notable other researchers in their field published. Whether or not a journal was open access did not necessarily factor into the decision-making process for many of the participants.

Author/s	Article title	Key findings	This study's findings
Vezzosi (2009)	Doctoral students' information behaviour: An exploratory study at the University of Parma (Italy)	• Doctorate students prefer access through the web and appreciate simple and easy search tools. • Doctorate students became more aware about the importance of bibliographic database and advanced search as they moved further into their research degree.	• The participants preferred using social networking sites such as ResearchGate and Academia.au instead of institutional repositories for depositing and sharing their scholarly works, and preferred to use Google Scholar when searching for information.
Heading, Siminson, Purcell, & Pears (2010)	Finding and Managing Information: Generic information literacy and management skills for postgraduate researchers	• Formal structured training provided students the opportunity to develop and undertake a search strategy, and to confidently use bibliography management software and online databases.	Although the study did not specifically focus on how the participants searched for information generally, or what training they had received, the study revealed that participants: • were aware of both open and closed information sources including subscription databases. • recognised that open and closed information sources could serve different purposes in their information needs. • adopted diverse strategies to keep up-to-date with research, from using social network sites, to subscribing to RSS feeds, to regularly checking journal websites, to searching institutional repositories.
Whitworth, McIndoe, & Whitworth (2011)	Teaching media and information literacy to postgraduate researchers	• Participation in intentionally designed formal courses allowed research students to develop their information literacy	

		understanding and capacity.	• felt it was difficult and even unnecessary to distinguish between open and closed access information sources.
Alvarez, Franca, Cuenca, Bastos, Ueno, Barros, & Guimaraes (2013)	Information literacy: Perceptions of Brazilian HIV/AIDS researchers	• Researchers were familiar with information search strategies but had limited knowledge of academic databases. • Academic librarians need to find new ways to develop researchers' IL skills.	• Participants acknowledged some use of databases but overwhelmingly there was a preference for using Google Scholar to find information • Participants felt it was difficult and even unnecessary to distinguish between open and closed access information sources and found it difficult to determine when an information item found was open or closed due to the access to subscription databases provided by the library.
Roos (2005)	Medical scientists' information practices in the research work context	• Researchers' main formal sources of information are journal articles. • Ease of access and use are the most important conditions for researchers when searching for information. • Researchers obtain help and support to find information from fellow researchers. • Researchers search and use information most at	• Journal articles were discussed by participants as the primary information source, although they also acknowledged their growing use of other information sources such as social media sources and institutional repositories (e.g., eprints). • Participants noted that work colleagues, supervisors and fellow scholars in their field provided advice on selecting where to publish and find information. The institutional repository and social networking were identified as core sources for staying connected with colleagues and their work.

		the beginning and the end of the research work.	
Zoellner, Hines, Keenan, & Samson (2015)	Faculty research and publication practices	• Information sources were found through active searching using library resources and services. • Participants also rely on graduate students to compile bibliographies. • Researchers' reading patterns have changed with a preference for e-journal articles over books.	• Google was used by participants to locate information the researchers were unable to locate in any other way, and there was a strong preference for using Google Scholar to find information. Other non-library resources such as social networking sites like ResearchGate were also used for information searching. • Some participants developed their own list of journals to follow and check on a regular basis.
Nel & Fourie (2016)	Information behaviour and expectations of veterinary researchers and their requirements for academic library services	• Information needs of researchers are influenced by the research environment and expectations of research output.	• Participants noted that the requirements of funding agencies and their work context (e.g., support from employers) significantly influenced their choices around where to publish, especially in terms of open access fees. These also influenced their depositing of items in institutional repositories (e.g., the need to be seen by their employer to be research active).
Borrego & Anglada (2016)	Faculty information behaviour in the electronic environment:	• Databases are the preferred starting point for bibliographic searches • Scholarly journals are	• Google Scholar was the preferred first step for locating information. • For some participants, open access was seen to be the last resort for publishing. • Participants had varied ways of

	Attitudes towards searching, publishing and libraries	the preferred mechanism for disseminating research, with open access being a factor of marginal interesting in determining where to publish. • Social media was a popular source for finding scholarly information.	determining where to publish including impact factor, quality of the peer review process, cost, and where notable researchers in their field published. Whether or not a journal was open access did not necessarily factor into the decision-making process for many of the participants.
Nicholas, Boukacem-Zeghmouri, Rodriguez-Bravo, Xu, Watkinson, Abrizah, Herman, & Swigon (2017)	Where and how early career researchers find scholarly information: How ECRs find scholarly information	• Google and Google Scholar were most frequently used in finding scholarly information. • Social media was identified as a popular source for finding scholarly information. • Researchers were aware of open access in providing free access to articles.	• Google and Google Scholar were the most frequently used tools for finding information by the participants. • Social media was used to stay in touch with colleagues and to find information.
Williamson, Bernath, Wright, & Sullivan (2007)	Research students in the electronic age: Impact of changing of information	• Google and databases provided by the university were used for finding information. • Quality of results,	• Although Google Scholar was the preferred way to find information the participants also noted using databases and social media (e.g., Research Gate).

	behaviour of information literacy need	speed, and familiarity were the reason to use the tools mentioned above.	
Green & Macauley (2007)	Doctoral students' engagement with information: An American-Australian perspective	• Research students prefer to use Google scholar when searching for information. • Information literacy was seen to be something they do when undertaking a literature review.	

Table 9. A comparison of the current study's findings to the existing knowledge base on the information literacy of researchers

6.3.3 Open Access and Information Literacy

Only two studies have investigated the intersection of open access and information literacy: Grgic (2016) and Zuccala (2010). Both studies adopted a behavioural perspective to information literacy, where information literacy was seen as skills for searching, evaluating and presenting information within the context of open access. By using a relational perspective to understand information literacy, the current study was able to tap into both the behavioural and the experiential aspects. For example, as in the work by Grgic (2016) and Zuccala (2010), the participants in the current study were becoming more and more familiar with the web as a source of information, and were using a wide variety of different tools to locate information including open access channels.

Building on the work by Grgic (2016) and Zuccala (2010), the current study also revealed how understandings, personal preferences, motivations and assumptions played a vital role in participants' information literacy experiences when engaging with open access. For example, many participants commented about having a feeling of confusion in regards to open access, and this feeling informed the participants' information literacy experience, such as deciding where to publish or using an institutional repository.

In this study, participants' experience of open access as part of their information literacy was revealed as relating to their roles as informed and informing researchers. The term 'informed researchers' relates to the participants' role as information user, while the term 'informing researcher' relates to the participants' role as information producers. It is through participants' roles as informed and informing researchers that their information literacy related to open access is enacted. Importantly, as a relational perspective to information literacy was adopted, both behavioural (i.e., skills, knowledge) and experiential (i.e., preferences, attitudes, feelings) aspects are present in both roles.

6.4 Contributions to practice

The findings of this study have revealed significant practical implications, including for university librarians, institutional repositories, university management, and higher education bodies.

6.4.1 University librarians

The study highlights the essential role of the librarian in universities in supporting academics. The findings of this study can inform academic librarians in refining the way they design and deliver support programs and resources for researchers. For example, referring to the Publishing theme, university academics can help researchers to develop skills and knowledge to identify and avoid predatory journals, they can also assist researchers in setting up and using social media as a way to raise their profile and the visibility of their scholarly work.

6.4.2 Institutional repositories

The findings of this study shed light on how researchers engage with open access via institutional repositories as part of their information literacy. The findings also reveal that for many researchers the functionality of the repository is a major barrier. This study can be used to inform the design of institutional repositories in Australian universities to better support the information literacy experiences of researchers. The study's insights can significantly influence institutional repository designers to adopt human centred design grounded in the lived experience of researchers. For example, referring to the Institutional Repository theme, institutional repository designers could incorporate functionality that allows more automated processes, thus saving the researchers' time in using the systems.

6.4.3 University management and higher education authorities

Finally, the study's findings demonstrate that publishing fees were a key element in researchers' decision-making. The potential cost of publishing open access deterred many researchers from using this avenue to disseminate their work. This insight can aid university management and higher educational bodies such as Universities Australia, or the Council of Australian University

Librarians, to refine strategies and initiatives to reduce costs for the researcher of engaging with open access. The universities, through the libraries, already pay a significant amount of money for journal subscriptions. University management and higher education bodies need to identify and pursue alternative options to work with or without publishers.

6.5 Limitations of the study

This study has a number of limitations that need to be acknowledged.

Firstly, the study's focus on Australian researchers means that the findings may not be applicable to researchers in other countries. Researchers outside Australia may have different experiences of open access as part of their information literacy, and as such may provide a different perspective.

Secondly, participants in this study are limited to researchers based in universities. This limitation means that the findings may not be applicable to researchers not based in the university context. Researchers from non-university settings may provide different perspectives and understandings related to open access and information literacy.

Thirdly, in keeping with the purposive sampling approach in qualitative methods, the sample size in this study does not provide a statistically representative sample of researchers in Australia.

Fourthly, due to the nature of the qualitative method, the findings are specific to the group of researchers who participated in the study.

Finally, this study applied a particular qualitative research approach. Different qualitative research approaches may reveal alternative insights related to the phenomenon being explored.

6.6 Future directions of research

The findings of this research suggest a number of directions for future research, which are outlined below.

This research has explored Australian researchers' experience of open access as part of their information literacy. The participants were Australian

university based researchers. Future research could build upon these findings by extending the study to explore how other researchers, not affiliated with universities (e.g., researchers in commercial settings), experience the phenomenon. This would test the generalizability of the study's findings, and reveal if there were differences between different work contexts.

The findings from this study could also be expanded by investigating specific university disciplines in greater detail, for example studies focussed solely on exploring researchers' experiences from the Humanities or Law disciplines. This would help determine if different experiences of the phenomenon exist in different discipline areas, and reveal if different support is required for researchers in different disciplines to engage with open access as part of their information literacy.

Future research could also extend this study by exploring researchers from other countries, for example a developing country like Indonesia where the government is trying to enhance the visibility of Indonesian academics' outputs (Farida, Tjakraatmadja, Firman, & Basuki, 2015). Other countries will have different political, economic, and technological factors that influence the lived experiences of their researchers.

In terms of the theoretical perspective that informed this study, this study provides an example of research into academics' experience of open access as part of their information literacy from a relational perspective. There are some opportunities to explore the open access context from the behavioural perspective of information literacy where it describes and understands information literacy as a collection of skills and behaviours (Bruce, 1997) or from the socio-cultural perspective that emphasizes the relationship between individuals and their communities (Limberg, Sundin, & Talja, 2012).

The findings from this study have revealed how academics experience open access as part of their information literacy. Future research could explore the perspective of other key stakeholders, for example, university librarians who design and deliver services and resources to support researchers in their experience of open access as part of their information literacy. There are a small number of studies that have begun to investigate librarians and

open access, for example Alam (2014) who explored the transformation of the academic librarian's role in the open access environment, and Keane (2012) and Potvin (2013) who investigated how academic libraries prepared librarians to respond to the open access movement, in respect to their services to faculties and students. More empirical work is needed exploring academic librarians' understanding of open access as part of the information literacy experiences of researchers.

6.7 Conclusion

By applying a qualitative approach, this research reveals four different themes that depict aspects of Australian researchers' experience of open access related to their information literacy. The themes signify that open access is shaping how Australian researchers experience information literacy, influencing how Australian researchers are, and can be, informed and informing researchers. As informed and informing researchers, participants have their own individual understandings of what open access is, its benefits to them, and its benefits to the academy and society at large. Participants engage with institutional repositories as both informed and informing researchers. As informed researchers, they use information found in repositories as part of their research. As informing researchers, they deposit their work in the institutional repository to enable other academics, industry and society at large to access and benefit from their work.

As informing researchers, participants make decisions about where to publish their work, publish their work in closed and open access sources, and use social media to disseminate the research. As informed researchers, participants find, evaluate and use information as part of their research. This includes, for example, finding information as part of a literature review, and the practice of keeping up-to-date with the latest research in their field.

The findings from this study can guide librarians and other higher education professionals to provide support for researchers in Australian universities as they design and deliver services, resources, and support for open access and information literacy.

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Appendices

Appendix A: Recruitment Flyer

Queensland University of Technology Brisbane Australia	**PARTICIPATE IN RESEARCH** **Information for Prospective Participants**

The following research activity has been reviewed via QUT arrangements for the conduct of research involving human participation.

If you choose to participate, you will be provided with more detailed participant information, including who you can contact if you have any concerns.

Understanding How Researchers Experience Open Access as Part of Their Information Literacy

Research team contacts

Principal Researcher:	Heriyanto	PhD student
Associate Researchers:	Adjunct Professor Helen Partridge	Principal Supervisor
	Dr Kate Davis	Associate Supervisor

Information Systems School, Science and Engineering Faculty, Queensland University of Technology (QUT)

What is the purpose of the research?

The purpose of this research is to develop an understanding of researchers' experience open access as part of their information literacy.

Are you looking for people like me?

The research team is looking for Australian Universities academic researchers (sessional, contract and full time) who have used open access sources and content during the last 12month.

What will you ask me to do?

Your participation will involve taking part in a 30 to 60 minutes audio recorded interview at a location suitable to you and/or using Skype if you are not living in Brisbane. You will also be asked a few short questions (e.g. school, research topic) and to complete a written consent form.

The following questions are representative of the ones you will be asked during the interview:

- What research have you been working on over the last 12-month?
- Can you tell me about how you find information for your research?
- Can you tell me about the information sources you use for your research?
- How would you define open access?
- Can you tell me about a time when you used open access sources to find information for your research?
- Have you ever published in open access channels?

Are there any risks for me in taking part?
The research team does not believe there are any risks beyond normal day-to-day living associated with your participation in this research. It should be noted that if you do agree to participate you can withdraw from participation at any time during the project without comment or penalty.
Are there any benefits for me in taking part?
It is expected that this project will not benefit you directly. However, it will help to develop an understanding of how researchers experience open access as part of their information literacy. It may also benefit Australian universities who will be able to use the findings from the study to inform the ways they encourage and support researchers' engagement with open access and the development of institutional repositories thus helping to ensure the library services genuinely meet the information literacy needs of the researchers.
Will I be compensated for my time?
We would very much appreciate your participation in this research. Although it is unlikely this project will benefit you directly, it may contribute to a better understanding of how researchers experience open access as part of their information world. Your involvement in this research is entirely voluntary.
I am interested – what should I do next?
If you would like to participate in this study, please contact the researcher listed below for details of the next step: **Heriyanto** heriyanto@hdr.qut.edu.au You will be provided with further information to ensure that your decision and consent to participate is fully informed.
Thank You! **QUT Ethics Approval Number: 1500000045**

Appendix B: Participant Information Sheet

Queensland University of Technology Brisbane Australia	PARTICIPANT INFORMATION FOR QUT RESEARCH PROJECT - Interview -

Understanding How Researchers Experience Open Access as Part of Their Information Literacy

QUT Ethics Approval Number 1500000045

RESEARCH TEAM

Principal Researcher:	Heriyanto	Doctor of Philosophy student
Associate Researchers:	Adjunct Professor Helen Partridge	Principal Supervisor
	Dr Kate Davis	Associate Supervisor

School of Information Systems, Science and Engineering Faculty

Queensland University of Technology (QUT)

DESCRIPTION

This project is being undertaken as part of Doctor of Philosophy for Heriyanto.

The purpose of this project is to investigate how researchers experience open access as part of their information literacy.

You are invited to participate in this project because you are current academic researchers (sessional, contract and full time) and have used open access information during the last 12months.

PARTICIPATION

Your participation will involve an audio-recorded interview at a location suitable to you and/or using Skype if you are not living in Brisbane. The interview will take approximately 60 minutes of your time.

The following questions are representative of the ones you will be asked during the interview:

- What research have you been working on over the last 12-month?
- Can you tell me about how you find information for your research?
- Can you tell me about the information sources you use for your research?
- How would you define open access?
- Can you tell me about a time when you used open access sources to find information for your research?
- Have you ever published in an open access channels?

Your participation in this project is entirely voluntary. If you agree to participate you do not have to complete any question(s) you are uncomfortable answering. Your decision to participate or not participate will in no way impact upon your current or future relationship with QUT (e.g. research you undertaking) or with members of the research team. If you do agree to participate you can withdraw from the project at any time without comment or penalty.

EXPECTED BENEFITS

It is expected that this project will not benefit you directly. Your participation in this research may benefit Australian universities who will be able to use the findings from the study to inform the ways they encourage and support the researchers engagement with open access.

RISKS

There are minimal risks associated with your participation in this project. It is important to note that the focus of the research is exploring how you use open access information to learn about your research.

PRIVACY AND CONFIDENTIALITY

All comments and responses will be treated confidentially unless required by law. Any data collected as part of this project will be stored securely as per QUT's Management of research data policy.

Your interviews will be audio recorded:

- It is not possible to participate in the research without being audio recorded.
- You will not have the opportunity to verify your comments and responses prior to final inclusion.
- The recordings will be transcribed and the audio recordings will be retained until the project is completed, at which point they will be destroyed.
- The recordings and transcriptions will only be available to the research team.
- Your name will not appear in the transcription.

CONSENT TO PARTICIPATE

We would like to ask you to sign a written consent form (enclosed) to confirm your agreement to participate.

QUESTIONS / FURTHER INFORMATION ABOUT THE PROJECT

If have any questions or require further information please contact one of the researchers listed below.

Heriyanto	heriyanto@hdr.qut.edu.au	
Helen Partridge	h.partridge@qut.edu.au	07 3138 9047
Kate Davis	kate.davis@qut.edu.au	07 3138 9387

CONCERNS / COMPLAINTS REGARDING THE CONDUCT OF THE PROJECT

QUT is committed to research integrity and the ethical conduct of research projects. However, if you do have any concerns or complaints about the ethical conduct of the project you may contact the QUT Research Ethics Advisory Team on 07 3138 5123 or email ethicscontact@qut.edu.au. The QUT Research Ethics Advisory Team is not connected with the research project and can facilitate a resolution to your concern in an impartial manner.

Thank you for helping with this research project. Please keep this sheet for your information.

Appendix C: Consent form

Queensland University of Technology Brisbane Australia	CONSENT FORM FOR QUT RESEARCH PROJECT - Interview -

Understanding How Researchers Experience Open Access as Part of Their Information Literacy

QUT Ethics Approval Number 1500000045

RESEARCH TEAM CONTACTS

Heriyanto	heriyanto@hdr.qut.edu.au	
Helen Partridge	h.partridge@qut.edu.au	07 3138 9047
Kate Davis	kate.davis@qut.edu.au	07 3138 9047

STATEMENT OF CONSENT

By signing below, you are indicating that you:

- Have read and understood the information document regarding this project.
- Have had any questions answered to your satisfaction.
- Understand that if you have any additional questions you can contact the research team.
- Understand that you are free to withdraw at any time without comment or penalty.
- Understand that if you have concerns about the ethical conduct of the project you can contact the Research Ethics Advisory Team on 07 3138 5123 or email ethicscontact@qut.edu.au.
- Understand that the project will include an audio recording.
- Agree to participate in the project.

Name .....

Signature .....

Date .....

Please return this sheet to the investigator.

Appendix D: Interview – discussion guide

Understanding How Researchers Use Open Access Information to Learn: A Case Study of Queensland University of Technology

Discussion Guide – Duration: 30 to 60 minutes

1. Introduction (3-5 mins)
   a) Greeting
   b) Introduction to the project overall
   c) Purpose of the interview
   d) Confidentiality
   e) Consent process
   f) Individual opinion and experience (no right or wrong answer)
   g) Focus on the way researchers use open access information to learn about their research
   h) Audio recorded

2. Prompt Questions: (20-50 mins)
   a) Can you tell me what your research topic is?
      Probe questions:
      - Are you a research student or academic researcher?
      - What topic/s have you been working on over the last 12 months?
      - What is your research method/s?
      - Where are you up to with the research? If student, what stage of candidature are you at?
   b) Can you tell me about how you discover information for your research?
      Probe questions:
      - Where do you look for information?
      - How do you go about looking for information?

c) Can you tell me about the information sources you use for your research?

Probe questions:

- What types of sources do you use?
- What types of sources do you use for particular aspects of your research?
- What do you like most about these sources?
- What do you like least about these sources?

d) How would you define “open access”?

Probe questions:

- What do you think open access encompasses?
- What do you think isn’t encompassed by the term “open access”?

e) Can you tell me about a time when you used other open access sources to find information for your research?

Probe questions:

- Why did you use open access?
- What kinds of open access source did you use?
- How did you find that source?
- What did you like most about using that source?
- What did you like least about using that source?
- How do you keep informed from that source?

f) Can you tell me about a time when you have used QUT ePrints to find information for your research?

Probe questions:

- What kinds of information have you used from ePrints for your research?
- Why did you use ePrints to get this information?
- What did you like most about using ePrints?
- What did you like least about using ePrints?

General probe questions

a. Could you explain that further?
b. What do you mean by that?
c. Why is that important?
d. Could you please give me an example?

3. Summary (2 mins)

a. Question & Answer Time
b. Thank participant

Appendix E: Pilot study

Before the study was fully undertaken, the decision to implement a pilot study was made whose primary aim would be to test the feasibility of the proposed study. Specifically, when conducting a qualitative study, a pilot study is useful to help the researcher to assess the acceptability of the interview or the interview recruitment protocol, or both (Kim, 2010). Moreover, a pilot study is beneficial to the researcher to self-evaluate their capability and readiness when conducting a qualitative study (Beebe, 2007).

A pilot study was conducted by recruiting and interviewing researchers from the Information Systems (IS) School, Science and Engineering Faculty, QUT. It aimed to refine the data collection techniques that would be used for the main study. By the end of the pilot study, the researcher was able to evaluate and develop relevant questions to ask the participants. Moreover, it was useful refine the data collection to best capture the phenomenon of the study (Yin, 2009).

The preparation for the pilot included applying for ethics approval and writing a recruitment email intended for the IS school e-list. This is the e-list that is used by IS school academics. The expectation was to get 4 researchers from the IS school who had used open access sources during the previous 12 months of their research. However, the recruitment took longer than expected. As the recruitment message did not get anyone to participate, even after a second email sent two weeks after the first, the recruitment was mentioned verbally to academics through informal conversations confirming whether they had received the email. This resulted in four researchers participating in the pilot study.

Each interview ran 30-45 minutes and mainly asked about the participants' experience in using open access sources. In order to hear their story systematically, five general questions were prepared for the interviews. They were:

a. What research have you been working on over the last 12-month?
b. Can you tell me about how you find information for your research?

c. How would you define open access?
d. Can you tell me about a time when you used open access sources to find information for your research?
e. Can you tell me about a time when you published in open access sources?

During the pilot, the researcher noticed that he very much relied on the questions that were prepared. When the researcher finished transcribing, the researcher found there were some responses from the participants that the researcher should have followed up for more detail. The researcher had led the participants to answer the prepared questions, but the researcher still could not acquire meaty data. There were some responses that should have been explored deeper. The experience of conducting interviews encouraged the researcher to learn about how to conduct effective interviews.

A fundamental principle of in-depth interviewing is to listen. It is more about trying to hear the meaning of what participant is saying, understanding where there is a subtext that needs to be explored and hearing the nuances in the participant's account. In keeping with the nature of the interview, the follow-up questions could be determined by the participant's answer. Therefore, a list of thematic probe questions might help in preparing the follow-up questions. The researcher also needed to listen carefully to remember the points that needed to be followed up at a later stage. One response from an interviewee may trigger two or three points to probe. A swift decision needed to be taken about the immediate issue to be followed up. From the interview transcripts, it seems important to make a short note to return to the other issues raised, either once we have dealt with the immediate issue or later in the interview when we are dealing with a relevant topic.

One example is drawn from a transcript:

- Can I take you back to something you said earlier about open access article's quality?
- You said earlier Google search gives you the most recent papers, can you tell me further about that?

The next thing that was important to note during the interview was the difference in tone of voice and body language of the participants. That can

happen when the participants try to convey their state of mind through their tone of voice, manner or body language. For example, when they feel doubtful about a view, or conversely, they look very sure about what they are saying, this may be a signal for the interviewer as a researcher to explore further. However, it might be essential to keep them talking, and later ask whether they have other views or experiences. The right probe question for the conversation was:

- You sound very certain about a mandatory policy for researcher archiving their work at the institutional repository, what makes you so certain?

Additionally, it is easy to make the assumption from how the participants have responded about a particular course of action. However, reading the interview transcripts, it was clear that it would have taken on a deeper and richer meaning if I had asked for little more explanation. What is important is to turn the assumption into a question.

Another important thing that was noted after the second pilot interview was the structure of the questions. Asking a question about how the participant defined open access in the beginning of the interview made the participant look a bit uneasy. This can be seen from a pause in the participant's response, and the responses they provided.

In the next interview, the question about how they understood open access was delayed until they had explained their experiences in using and publishing in open access sources. The question was then placed at the end of the interview, before the interviewer closed the conversation. In this event, the participants responded better and the answer was more flowing than in the previous interviews.

Appendix F: List of coding groups

Coding group	Short description
Commitment to open access	The quote indicates the participant is committed to open access as a cause, or is committed to publishing open access
Commitment to openness	A broader commitment to openness - may be indicated by a statement about a belief in openness generally, or specifically they may talk about both open access and open data
Defining open access	The participant's personal understanding of open access - whether it's right or wrong
Deposit processes	Data that refers to aspects of the process of depositing in an institutional repository generally
Deposit processes / barriers	Specifically quotes that deal with negative aspects or barriers related to the process of depositing in an institutional repository
Disadvantages of putting unpublished work online	This is about posting unpublished works online, in an institutional repository or on a personal or team website - specifically the disadvantages of it
Economics	This is about the economic aspects of publishing models
Finding information resources	Quotes about finding information resources generally; this is about finding resources for them to read and cite
Finding open access material	Specifically about looking for open access resources or the process of finding open access resources as part of broader research
Funder requirement for open access	Use for data that indicates the funder requires the research outputs to be published open access
Identifying IR benefit	Quotes that identify one or more benefits of institutional repository; this is slightly different from motivations for depositing; it's talking about the benefits they have seen or indicating that they have not yet seen benefits
Identifying open access benefit	Quotes that identify one or more benefits of open access; again, different from motivations for publishing open access. This is about the benefits they have seen or indicating that they have not yet seen
Identifying open access journals	Quotes about how participants identify open access resources; also, quotes that indicate they don't need to identify them because their library access is so good that it's irrelevant
Keeping up-to-date	Quotes about keeping up to date with new research outputs
Making work accessible	Quotes that indicate a desire to or experiences related to making work accessible, including to non-academic audiences
Monitoring citations of own work	Participants monitor / track the performance of their own publications
Motivations	Motivations related to openness generally
Motivations for depositing	Motivations for depositing in the IR
Motivations for not publishing open access	Reasons why participants don't publish open access

Motivations for open data	Reasons why participants make their data open
Motivations for publishing open access	Reasons why participants publish open access
Motivations for social networking	Reasons why participants use social networking for research related purposes
Motivations for using IR	Motivations for using the institutional repository - not for depositing, but for using it in other ways
Navigating copyright issues	This is about how participants navigate copyright issues - relates to illegal sharing; also working out what the participants can and can't deposit in the institutional repository; how the participants can share on social networking
Open data	Anything related to open data, other than motivations for open data
Paying publication fees	Anything to do with paying publication fees
Peer review process	Peer review process generally
Peer review process (OA)	Specifically peer review process for open access publications
Perceptions of open access	People's perceptions of open access - not scepticism
Personal branding (building profile)	Quote related to building a personal profile or a research identity
Promoting research	Promoting research in various ways - including through social media
Publication process	Quote about publication processes - specifically non-open access publications
Publishing process (OA)	Quote about the process of publishing in an open access publication (not an institutional repository)
Repository usability	Quote about the functionality or usability of institutional repositories. This is more about usability / functionality of the front end, e.g., for discovery, than for depositing. Use deposit processes / barriers for usability in terms of uploading papers
Repository usage	Usage of IRs - both own institution and other institutions - on the front end (e.g., for information discovery)
Sceptical about open access peer review process	Scepticism about open access peer review process
Sceptical with open access	General scepticism about open access
Sceptical with open access journal	Scepticism about open access journals, including a particular journal
Selecting information resources	About the process of selecting information resources; including evaluating or assessing them; how they select resources to use - not how they choose where to publish. This is about selecting information resources to use
Selecting where to publish	This is about how participants choose where they are going to publish
Social network usage	Anything to do with social network usage including academic social networks and mainstream social media
Using open access resources	About using open access resources - not finding or selecting